UNITED STATES DEPARTMENT OF AGRICULTURE

# SPECIFICATIONS

# EDINBURG-KERRVILLE BLDG 4 DIP VAT REDESIGN – A-E PREDESIGN AND DESIGN SERVICES

# PN: 12805B23D0001

**MOORE FIELD EDINBURG, TX**

**Prepared For:**
**U.S. Department Of Agriculture**
**USDA APHIS**
**5601 Sunnyside Avenue**
**RM 3-2102**
**Beltsville, Maryland 20705**

**Contract No.**
**12805B23D0001**

**Task Order No.**
**1232SA25F03843**

**Prepared By:**
**The Johnson-McAdams Firm, P.A.**
**Greenwood, Mississippi**

**August 31, 2026**

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SECTION 01 00 01

WARRANTY OF CONSTRUCTION
**01/20**

PART 1 GENERAL

1.1 WARRANTY OF CONSTRUCTION

1.1.1 General

In addition to any other warranties set out elsewhere in this contract, the Contractor warrants that work performed under this contract conforms to the contract requirements and is free of any defect of equipment, material or design furnished, or workmanship performed by the Contractor or any of his subcontractors or suppliers at any tier. Such warranty shall continue for a period of one year from the date of final acceptance of the project. Under this warranty, the Contractor shall remedy at his own expense any such failure to conform or any such defect. In addition, the Contractor shall remedy at his own expense any damage to Government owned or controlled real or personal property, when that damage is the result of the Contractor's failure to conform to contract requirements or any such defect of equipment, material, workmanship, or design. The Contractor shall also restore any work damaged in fulfilling the terms of this clause. The Contractor's warranty with respect to work repaired or replaced hereunder will run for one year from the date of such repair or replacement.

1.1.2 Notification and Compliance

The Government shall notify the Contractor in writing within a reasonable time after the discovery of any failure, defect, or damage. Should the Contractor fail to remedy any failure, defect, or damage described in paragraph 1.1.1 above within a reasonable time after receipt of notice thereof, the Government shall have the right to replace or repair and have the cost billed to the Contractor.

1.2 EXPRESSED WARRANTY

In addition to the other rights and remedies provided by this clause, all subcontractors', manufacturers', and suppliers' warranties expressed or implied, respecting any work and materials shall, at the direction of the Government, be enforced by the Contractor for the benefit of the Government. In such case, if the Contractor's warranty under paragraph 1.1.1 above has expired, any suit directed by the Government to enforce a subcontractor's, manufacturer's, or supplier's warranty shall be at the expense of the Government. The Contractor shall obtain any warranties which the subcontractors, manufacturers, or suppliers would give in normal commercial practice. If so directed by the Contracting Officer, the Contractor shall require any such warranties to be executed in writing to the Government.

1.3 LATENT DEFECTS

The Warranty specified herein shall not limit the Government's rights with respect to latent defects, gross mistake, or fraud.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 00 11

GENERAL PARAGRAPHS
**06/20**

PART 1 GENERAL

1.1 DESCRIPTION OF WORK

1.1.1 Work To Be Done

The work to be done consists of furnishing all labor, materials, and equipment and performing all work in strict accordance with these specifications for Edinburg-Kerrville BLDG 4 Dip Vat Redesign in Edinburg, Texas. Bidder shall provide Base Bid pricing and Alternate pricing as defined in the contract documents.

1.1.2 Location

The site of work is at the USDA Cattle Fever Tick Research Laboratory (CFTRL) in Edinburg, Texas.

1.1.3 Principal Features of Work

Base Bid: Demolish existing roofing back to rigid frame structure, demolish existing canopy at west end of structure complete including foundation (fill void left from foundation removal with select fill and compact) , demolish rail fencing and gates complete including concrete foundations (fill holes with select fill and compact), demolish CMU block structure complete including foundation(fill void left from foundation removal with select fill and compact), demolish existing sump pit to extents indicated (backfill remaining structure with select fill and compact), demolish existing water piping to watering troughs complete to extents indicated, demolish existing electrical power and lighting systems complete to extents indicated. Provide extension to rigid frame as indicated, install 8" purlins as indicated, provide new exposed fastener metal roof panels over reflective thermal insulation, provide metal fence railing and gates for corrals as indicated, provide concrete walking surface as indicated, provide concrete alteration to the existing dip vat as indicated, provide reinforced concrete walking surface with rough broomed finish as indicated, provide plumbing to watering troughs as indicated, provide power and lighting as indicated.

Additive Alternate #1: Provide systems as indicated above to create additional cattle corral pens 9, 10, and 11 as indicated.

Additive Alternate #2: Additive Alternate #2 provides galvanization for all fencing, gate, and bollard steel.

1.2 SITE VISIT

USDA will conduct a single pre-bid site inspection for all interested contractors that will be coordinated with David De La PazKnipling. Contractor must be able to provide valid US or state photo ID and must be with USDA escort for access.

Contact:

U.S. Department of Agriculture, ARS

Attn: David De La Paz
Knipling-Bushland U.S. Livestock Insects Research Laboratory
2700 Fredericksburg Road Kerrville, TX 78028
Email: David.delapaz@usda.gov
Phone: (830) 300-8655 (Office)

1.3 COORDINATION OF WORK

1. Work space and space available for storing materials shall be subject to the approval of the Contracting Officer. Do not store materials and equipment in other than assigned areas.

2. Execute work so as to interfere as little as possible with normal functioning of the facility, including operations of utility services and any existing equipment, and with work being done by others.

3. Minimize interference of construction activities with the flow of traffic by keeping roads, walks and entrances to grounds, to parking areas and to occupied areas of buildings clear of construction materials, debris, construction equipment and vehicles except where noted on plans or in specifications. Provide unobstructed access to areas required to remain in operation.

4. The Contractor shall complete all work within 240 consecutive calendar days after Notice to Proceed. Within 14 calendar days after Contract Award, the Contractor shall provide the Government with a detailed schedule of dates on which the Contractor plans to accomplish the work. In addition, the Contractor shall notify the Contracting Officer 20 days in advance of starting work. A planned work schedule shall be submitted to the Contracting Officer for approval. The schedule shall include a detailed description of the methods and equipment to be used for each operation and the sequence of operations. This schedule shall be updated monthly and submitted to the Contracting Officer for approval.

5. The Contractor shall take all measures and provide all material necessary for protecting existing equipment, property, and ongoing operations in affected areas of construction against dust, noise and debris.

6. Power sources for the Contractor's operations will be available and supplied by the Government to the Contractor as long as its use does not interfere with the operations of the Government's facilities. Water sources will be available and supplied by the Government to the Contractor as long as its use does not interfere with the operations of any building. The Contractor is responsible for extending utility sources to his work locations. Such extensions shall be removed after work is completed, to the satisfaction of the Contracting Officer.

7. All references in the specifications to the Contracting Officer and Contracting Officer's Representative shall constitute referrals initially to the Contracting Officer's Representative.

8. Contractor shall provide sanitary facilities for use by his employees and the employees of his subcontractors working on job site.

1.4 DRAWINGS

All dimensions shown of existing work and all dimensions required for work that is to be connected with existing work shall be verified by the Contractor by actual field measurements. Any work at variance with that specified or shown on the drawings, shall not be performed by the Contractor until the Contractor has received approval in writing from the Contracting Officer.

1.5 MATERIAL SAFETY DATA SHEETS (MSDS)

Material Safety Data Sheets must be submitted to the Contracting Officer prior to the commencement of any work with any substance that may be considered hazardous.

1.6 UTILITIES OR STRUCTURAL ITEMS CONFLICTING WITH PROPOSED CONSTRUCTION

In the event the Contractor finds an area where existing utilities or structural items conflict with proposed construction, the Contracting Officer shall be notified immediately. The Contractor shall then propose a method of solving the problem. Work shall not resume in the problem area until the Contractor has received written approval from the Contracting Officer. There shall be no additional cost to the Government.

1.7 ACCEPTANCE OF WORK

1. Final inspection will not be made until the contract work is completed. The Contractor shall notify in writing the Contracting Officer fifteen working days prior to the date on which the work will be ready for final inspection.

2 Final inspection and acceptance of the work shown by the drawings and specifications forming a part of this contract shall not be binding or conclusive on the Government if it shall be shown that (1) the Contractor has willfully or through collusion with persons or firms engaged in the performance of the contract supplied inferior materials, equipment, or workmanship, or (2) the Contractor has otherwise departed from the terms of the contract.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

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SECTION 01 00 21

CONTRACTOR QUALITY CONTROL
**01/20**

PART 1 GENERAL

1.1 GENERAL

This section covers the quality control inspection, sampling, and testing of all materials, services, and workmanship required to be performed by the contract drawings, specifications, and other contract requirements. The Contractor shall perform all quality control inspection, sampling, and testing required by this contract and shall review and certify that each submittal is correct and in strict conformance with the contract drawings and specifications for all shop drawings and lists of materials, fixtures, and equipment. All work performed under this contract shall be in strict accordance with procedures and requirements detailed on the plans and specifications.

1.1.1 Certificate of Compliance

Upon completion of the contract work and prior to final acceptance, the Contractor shall submit to the Contracting Officer a certificate of compliance which states that all materials, equipment, design, and workmanship incorporated in the work are in strict compliance with contract drawings and specifications except for approved deviations. The certificates of compliance shall be signed and dated by both the manufacturer's representative and the Contractor. Based upon this certificate of compliance and other contract requirements, the Contractor shall be responsible for correcting at his expense any defects, omissions, and deviations (other than those previously approved) which are detected during the contract period or during the one-year warranty period as defined in Section 01 00 01, WARRANTY OF CONSTRUCTION.

1.2 INSPECTION

The Contractor shall inspect all materials, services, and workmanship to assure conformance with contract requirements.

1.3 REVIEW AND APPROVAL OF SHOP DRAWINGS, SUBMITTALS, MATERIALS, FIXTURES, AND EQUIPMENT

1. Prior to submission to the Government for approval, the Contractor shall review and certify that all shop drawings, submittals, lists of materials, fixtures, and equipment as called for under the various headings of these specifications are correct and in strict compliance with the contract drawings and specifications. These drawings, submittals and lists shall be accurate, complete, and adequately detailed. The top copy of data comprising each item listed on the transmittal form shall be identified as having received the quality control inspector's approval by being so stamped, signed and dated. Approval of shop drawings, submittals, materials, fixtures, and equipment by the Contracting Officer will not relieve the Contractor of the responsibility for any errors which may exist and the Contractor shall be responsible for the satisfactory construction of all work. All samples of materials submitted as required by these specifications shall be properly identified and labeled for ready identification, and upon being approved, stored at the site of the

work for jobsite use until all work has been accepted by the Contracting Officer. The Contractor shall provide four (4) sets of all shop drawings.

2. All proposed deviations required by the Contractor shall be noted in the "Remarks" column of the Form. The Contractor shall set forth in writing the reason for any deviations and annotate such deviations on the shop drawing. The Government reserves the right to rescind inadvertent approval of shop drawings containing unnoted deviations.

1.4 DAILY RECORDS

The Contractor shall maintain a daily record of quality control measures performed for each shift of Contractor or Subcontractor operations on an appropriate format. These records shall provide factual evidence that continuous quality control inspections have been performed, including but not limited to the following: job progress, problems encountered; problems resolved; type and number of inspections; results of inspections or tests including all computations; nature of defects; causes for rejection; safety violations; proposed remedial action; and corrective action taken. These records shall cover both conforming and defective items and shall include a statement that all supplies and materials incorporated in the work are in full compliance with the terms of the contract. The Contractor shall maintain a current record of all inspections and shall furnish to the Contracting Officer on a daily basis, legible copies of all inspection records for his permanent retention. The daily records of inspections shall cover all work placement subsequent to the previous report and shall be verified by the Contractor's designated representative.

1.5 RECURRING DEFICIENCIES

If recurring deficiencies in an item or items being inspected by the Contractor indicate that the inspection system is not providing adequate quality control, the Contractor shall take such corrective measures as deemed necessary by the Contracting Officer.

1.6 CONFLICTS

In the event of a difference of opinion between the Contractor and the Contracting Officer, the Contracting Officer shall govern.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 01 00

GENERAL REQUIREMENTS
**February 29, 2024**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

| | |
|---|---|
| ASTM E84 | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials |

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

| | |
|---|---|
| NFPA 10 | (2018; ERTA 1-2 2018) Standard for Portable Fire Extinguishers |
| NFPA 30 | (2024) Flammable and Combustible Liquids Code |
| NFPA 51B | (2024) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work |
| NFPA 70 | (2023; ERTA 4 2023) National Electrical Code |
| NFPA 101 | (2021; TIA 21-1) Life Safety Code |
| NFPA 241 | (2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations |

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

| | |
|---|---|
| 29 CFR 1926 | Safety and Health Regulations for Construction |

1.2 GENERAL INTENTION

A. Contractor shall completely prepare the site for building operations, including demolition and removal of existing structures, and furnish labor and materials and perform work as required by the contract documents.

B. All employees of the general contractor and subcontractors shall comply with ARS's site-specific security management program, as identified by the ARS Location Manager, at all times on ARS property, and shall be restricted from unauthorized access.

C. Prior to commencing work:

1. The General Contractor shall provide proof that an OSHA certified "competent person" (CP) 29 CFR 1926.20(b)(2) will maintain a presence

at the work site whenever the general or subcontractors are present.

2. The General Contractor and Tradesmen shall receive an On-site Safety briefing from the location Facilities Manager or designated representative.

D. Training:

1. All employees of the general contractor or subcontractors shall have the 10-hour OSHA certified Construction Safety course and /or other relevant competency training. Supervisors shall have completed the 30-hour OSHA training.

2. Submit training records of all such employees to the Contracting Officer for approval before the start of work.

1.3 PROJECT COORDINATION

A. Construction Project Schedule: See contract, FAR 52.236-15 Schedules for Construction Contracts. The Contractor will not begin work on site until the construction project schedule has been approved by the Contracting Officer. The Contractor shall submit one electronic copy of the updated construction project schedule with each progress payment request.

B. Contract Inspector's Daily Report: Contractor shall submit to the Contracting Officer a daily report listing the names of all those present on the job site each day, along with their Job Classification, hours of work and a brief description of the work performed by each on that day. The Contract Inspector's Daily Report is to be submitted at the beginning of work each day for the previous day's work and shall be maintained throughout the term of the contract.

C. Formal Meetings: Any meeting (deemed formal by the contractor, the Contracting Officer Representative (COR) or the Contracting Officer will have the minutes of said meeting prepared by the contractor with copies provided to the COR and Contracting Officer within twenty-four (24) hours of meeting time. The minutes will include Date, Name of Attendees (with contact info), Old Business, and New Business subjects (or narratives). Regardless of who requests a meeting (the COR or the Contractor or the Contracting Officer) the contractor may be required to provided hard copy pictures of the issues to be discussed. Each issue will be linked to a contract specification section number and/or subject area by page number. Issues discussed will also include an action/completion date.

1.4 SUBMITTALS - GENERAL

A. All submittals referenced by the procurement documents including but not limited to Plans, Specifications, Statements of Work (SOW), Supplemental Information, etc. shall be submitted electronically to the Contracting Officer (CO), Construction Officer Representative (COR) and Project Architect (A/E).

B. All material sample submittals referenced by the procurement documents including but not limited to Plans, Specifications, Statements of Work (SOW), Supplemental Information, etc. shall be provided to the Contracting Officer (CO), Construction Officer Representative (COR) and Location Manager/Facilities Manager (LM/FM).

C. Contractor shall provide documentation that submitted materials comply with current Buy America Act (BAA) requirements.

D. Contractor shall utilize products listed in the USDA "BioPreferred Program" to the greatest extent possible.

E. All items submitted shall be accompanied by the SUBMITTAL LOG appended to the GENERAL SPECIFICATIONS.

1. The Contractor shall maintain one (1) log for all submittals and shall update the information as required by Construction Progress.

1.5 SAFETY SUBMITTALS

A. Contractor shall submit a "site specific" Safety Plan to the Contracting Officer that describes how the contractor is to comply with all applicable OSHA standards and NFPA codes pertaining to construction work to include, but not limited to: General Safety and Health Provisions, Occupational Health and Environmental Controls, Personal Protective and Life Saving Equipment, Hand and Power Tools, Welding and Cutting, Electrical, Scaffolds, Fall Protection, Excavations, Demolition, Stairways and Ladders, Toxic and Hazardous Substances, Confined Space Entry, Lockout/Tagout, Respiratory Protection, Hearing Protection. Safety Plan shall identify Contractor and sub-contractor employees who are authorized/qualified to perform electrical work. No work shall proceed until written approval of the Safety Plan by the Contracting Officer. Safety Plan shall describe how the Contractor and/or sub- contractors are to secure equipment, supplies, tools, and chemical products. Contractor and/or sub- contractor equipment, supplies, tools, and chemical products are to be always under the control of the Contractor, equipment, supplies tools, and chemical products found not to be under the control of the Contractor or sub-contractor are subject to confiscation and will be reported to the Contracting Officer.

B. The Contractor shall inspect the entire construction site daily, including days without construction work if there are NFPA 101 Life Safety Code deficiencies existing while construction work is shut down, the results of that inspection shall be documented on the attachment "DAILY INSPECTION FORM", and the form is to be signed and submitted to the Contracting Officer.

C. Contractor shall provide continuous monitoring for noise, dust, chemical vapor generation, volatile organic compounds, and vibration. At no time shall noise exposure to any ARS staff exceed 75 dB. Contractor shall provide to the COR with copies to the Contracting Officer a written weekly update of the test results in addition to the proposed work for the subsequent two weeks (a two week "look- ahead")involving any construction activity that will produce loud noises, heavy vibrations, strong smells, or significant dust inside ARS staff occupied space.

D. Contractor and sub-contractors shall notify in writing the Contracting Officer at least one week prior to executing high risk work (this may be included in the two week "look-ahead" update). High risk work includes, but is not limited to, tasks requiring Lockout/Tag Out, live electrical work, hot work, work at heights, trenching/ shoring, crane operations, and confined space entry. Contractor's Competent Person shall provide the COR with a copy to the Contracting Officer written weekly documented safety inspection results.

E. Contractor shall provide to the Contracting Officer for approval, copies of Safety Data Sheets (SDS) for **EVERY CHEMICAL WITHOUT EXCEPTION** prior to bringing such chemical on the work site.

1.6 CONSTRUCTION SECURITY REQUIREMENTS

A. Security Plan:

1. The General Contractor shall provide a site-specific security plan that defines both physical and administrative security procedures that will remain effective for the entire duration of the project.

2. The General Contractor is responsible for assuring that all sub-contractors working on the project and their employees also comply with these regulations.

B. Contractor Personnel:

1. The required Contractor Personnel information and the security application procedures vary depending upon the project worksite location. Each Task Order Request for Proposal (RFP) will contain the contact information for the Location Monitor, who will be able to provide the Contractor Personnel information and security application procedures required for the specific site.

C. Security Procedures:

1. Contractor's employees shall not enter the project site without being in conformance with location's security plan. They may also be subject to inspection of their personal effects when entering or leaving the project site.

2. For working outside the "regular hours" as defined in the contract, The Contractor shall give a ten (10) day notice to the Contracting Officer so that security arrangements can be provided for the employees. This notice is separate from any notices required for utility shutdown described later in this section.

3. No photography of ARS premises is allowed without written permission of the Contracting Officer.

4. ARS reserves the right to close or shut down the project site and order the Contractor's employees off the premises in the event of a national emergency or Government shutdown. The Contractor may return to the site only with the written approval of the Contracting Officer.

D. Document Control:

1. The Contractor is responsible for safekeeping of all drawings, project manuals and other project information. This information shall be shared only with those with a specific need to accomplish the project.

2. These documents shall not be removed or transmitted from the project site without the written approval of Contracting Officer.

3. All paper waste or electronic media such as CD's and diskettes shall be shredded and destroyed in a manner acceptable to the ARS.

E. Motor Vehicle Restrictions:

1. Vehicle authorization request shall be required for any vehicle entering the site and such request shall be submitted 24 hours before the date and time of access. Access shall be restricted to picking up and dropping off materials and supplies and Contractor must be on site to receive all deliveries.

2. Separate permits shall be issued for Contractor and its employees for parking in designated areas only.

1.7 FIRE SAFETY

A. Applicable Publications: Publications listed below form part of this Article to extent referenced. Publications are referenced in text by basic designations only.

1. American Society for Testing and Materials (ASTM):

ASTM E84 Surface Burning Characteristics of Building Materials

2. National Fire Protection Association (NFPA):

NFPA 10 Standard for Portable Fire Extinguishers

NFPA 30 Flammable and Combustible Liquids Code

NFPA 51B Standard for Fire Prevention During Welding, Cutting and Other Hot Work

NFPA 70 National Electrical Code

NFPA 241 Standard for Safeguarding Construction, Alteration, and Demolition Operations

3. Occupational Safety and Health Administration (OSHA):

29 CFR 1926 Safety and Health Regulations for Construction

B. Fire Safety Plan: Establish and maintain a fire protection program in accordance with 29 CFR 1926. Prior to start of work, prepare a plan detailing project-specific fire safety measures, including periodic status reports, and submit to the Contracting Officer for review for compliance with contract requirements. Prior to any worker for the contractor or subcontractors beginning work, they shall undergo a safety briefing provided by the contractor's competent person per OSHA requirements. This briefing shall include information on the construction limits, ARS safety guidelines, means of egress, break areas, work hours, locations of restrooms, use of ARS equipment, etc. Documentation shall be provided to the Contracting Officer that individuals have undergone contractor's safety briefing.

C. Site and Building Access: Maintain free and unobstructed access to facility emergency services and for fire, police, and other emergency response forces in accordance with NFPA 241.

D. Separate temporary facilities, such as trailers, storage sheds, and dumpsters, from existing buildings and new construction by distances in accordance with NFPA 241. For small facilities with less than 20 feet

exposing overall length, separate by 10 feet.

E. Temporary Heating and Electrical: Install, use, and maintain installations in accordance with 29 CFR 1926, NFPA 241 and NFPA 70.

F. Means of Egress: Do not block exiting for occupied buildings, including paths from exits to roads. Minimize disruptions and coordinate with Contracting Officer.

G. Egress Routes for Construction Workers: Maintain free and unobstructed egress. Inspect daily. Report findings and corrective actions weekly to the Contracting Officer.

H. Fire Extinguishers: Provide and maintain extinguishers in construction areas and temporary storage areas in accordance with 29 CFR 1926, NFPA 241 and NFPA 10.

I. Flammable and Combustible Liquids: Store, dispense and use liquids in accordance with 29 CFR 1926, NFPA 241 and NFPA 30.

J. Existing Fire Protection: Do not impair automatic sprinklers, smoke and heat detection, and fire alarm systems, except for portions immediately under construction, and temporarily for connections. Provide fire watch for impairments more than 4 hours in a 24-hour period. Request interruptions 48 hours in advance and coordinate with the Location Manager and Contracting Officer. All existing or temporary fire protection systems (fire alarms, sprinklers) located in construction areas shall be tested as coordinated with the Contracting Officer. Parameters for the testing and results of any tests performed shall be recorded by the General Contractor and copies provided to the Contracting Officer.

K. Smoke Detectors: Prevent accidental operation. Remove temporary covers at end of work operations each day. Coordinate with the Contracting Officer.

L. Hot Work: Perform and safeguard hot work operations in accordance with NFPA 241 and NFPA 51B. Coordinate with the Contracting Officer. Designate contractor's responsible project- site fire prevention program manager to coordinate permit hot work.

M. Fire Hazard Prevention and Safety Inspections: Inspect entire construction area daily. Coordinate with, and report findings and corrective actions weekly to the COR with a copy to the Contracting Officer.

N. Smoking: Smoking is prohibited in and adjacent to construction areas inside existing buildings and additions under construction. In separate and detached buildings under construction, smoking is prohibited except in designated smoking rest areas.

O. Dispose of waste and debris in accordance with NFPA 241. Remove from buildings daily.

P. Perform other construction, alteration, and demolition operations in accordance with 29 CFR 1926.

1.8 OPERATIONS AND STORAGE AREAS

A. The Contractor shall confine all operations (including storage of

materials) on Government premises to areas authorized or approved by the Contracting Officer. The Contractor shall hold and save the Government, its officers, and agents, free and harmless from liability of any nature occasioned by the Contractor's performance.

B. The Contractor shall, under regulations prescribed by the Contracting Officer, use only established roadways, or use temporary roadways constructed by the Contractor when and as authorized by the Contracting Officer. When materials are transported in prosecuting the work, vehicles shall not be loaded beyond the loading capacity recommended by the manufacturer of the vehicle or prescribed by any Federal, State, or local law or regulation. When it is necessary to cross curbs or sidewalks, the Contractor shall protect them from damage. The Contractor shall repair or pay for the repair of any damaged curbs, sidewalks, or roads.

C. Working space and space available for storing materials shall be as determined by the Contracting Officer.

D. Workers are subject to rules of the Location applicable to their conduct which shall be provided by the Contracting Officer.

E. Execute work to interfere as little as possible with normal functioning of Location as a whole, including operations of utility services, fire protection systems and any existing equipment, and with work being done by others. Use of equipment and tools that transmit vibrations and noises through the building structure are not permitted in buildings that are occupied during construction except as permitted by the Contracting Officer where required by limited working space.

1. Do not store materials and equipment in other than assigned areas.

2. Schedule delivery of materials and equipment to immediate construction working areas within buildings in use by the ARS in quantities sufficient for not more than two (2) workdays.

3. Where access by ARS personnel to vacated portions of buildings is not required, storage of Contractor's materials and equipment may be permitted subject to fire and safety requirements.

F. Contractor shall take all measures and provide all material necessary for protecting existing equipment and property in affected areas of construction against dust and debris, so that equipment and affected areas to be used in the ARS operations will not be hindered. Contractor shall permit access to ARS personnel through construction areas which serve as routes of access to such affected areas and equipment. Coordinate alteration work in areas occupied by ARS so that operations will continue during the construction period.

G. Utilities Services: Existing utility services for ARS shall be maintained at all times. Provide temporary facilities, labor, materials, equipment, connections, and utilities to assure uninterrupted services. Where necessary to cut existing water, steam, gases, sewer or air pipes, or conduits, wires, cables, etc. of utility services or of fire protection systems and communications systems (including telephone), they shall be cut and capped at suitable places where shown; or, in absence of such indication, where directed by the Contracting Officer.

1. No utility service such as water, gas, steam, sewers or electricity, or fire protection systems and communications systems may

be interrupted without prior written approval of the Contracting Officer. Electrical work shall be accomplished with all affected circuits or equipment de-energized. When an electrical outage cannot be accomplished, work on any energized circuits or equipment shall not commence without the Contracting Officer's prior knowledge and written approval.

2. Contractor shall submit a request to interrupt any such services to the Contracting Officer, in writing, Two weeks advanced notice and again 48 hours in advance of proposed interruption. Request shall state reason, date, exact time of, and approximate duration of such interruption.

3. Contractor will be advised (in writing) of approval of request, or of which other date and/or time such interruption will cause least inconvenience to operations of ARS. Interruption time approved by ARS may occur at other than Contractor's normal working hours.

4. In case of a contract construction emergency, service will be interrupted on approval of the Contracting Officer. Such approval will be confirmed in writing as soon as practical.

H. To minimize interference of construction activities with flow of ARS traffic, comply with the following:

1. Keep roads, walks and entrances to grounds, to parking and to occupied areas of buildings clear of construction materials, debris and standing construction equipment and vehicles.

I. Coordinate the work for this contract with other construction operations as directed by Contracting Officer. This includes the scheduling of traffic and the use of roadways, as specified in paragraph 1.11, USE OF ROADWAYS.

1.9 ALTERATIONS

A. Survey: Before any work is started, the Contractor shall make a thorough survey with the Contracting Officer or their designated representative, of areas of the building in which alterations occur and areas which are anticipated routes of access, and furnish a report, signed, to the COR with a copy to the Contracting Officer.

1. Shall note any discrepancies between drawings and existing conditions at site.

2. Shall designate areas for working space, materials storage and routes of access to areas within buildings where alterations occur, and which have been agreed upon by Contractor and Contracting Officer.

1.10 DISPOSAL AND RETENTION

A. Materials and equipment accruing from work removed and from demolition of buildings or structures, or parts thereof, shall be disposed of as follows:

1. Items that remain property of the Government shall be removed or dislodged from present locations in such a manner as to prevent damage which would be detrimental to re- installation and reuse. Store such items in locations as directed by Contracting Officer.

2. Items not reserved shall become property of the Contractor and be removed by Contractor from ARS property.

3. Items of portable equipment and furnishings located in rooms and spaces in which work is to be done under this contract shall remain the property of the Government. When rooms and spaces are vacated by the ARS during the alteration period, such items which are NOT required by drawings and specifications to be either relocated or reused will be removed by the Government in advance of work to avoid interfering with Contractor's operation.

B. Upon completion of contract, deliver work complete and undamaged. Existing work (walls, ceilings, partitions, floors, mechanical and electrical work, lawns, paving, roads, walks, etc.) disturbed or removed as a result of performing required new work, shall be patched, repaired, reinstalled, or replaced with new work, and refinished and left in as good condition as existed before commencing work.

C. At Contractor's own expense, Contractor shall immediately restore to service and repair any damage caused by Contractor's workers to existing piping and conduits, wires, cables, etc., of utility services or of fire protection systems and communications systems (including telephone) which are indicated on drawings and which are not scheduled for discontinuance or abandonment.

1.11 USE OF ROADWAYS

A. For hauling, use only established public roads and roads on ARS property and, when authorized by the Contracting Officer, such temporary roads which are necessary in the performance of contract work. Temporary roads shall be constructed by the Contractor at Contractor's expense.

1.12 AVAILABILITY AND USE OF UTILITY SERVICES

A. See contract, FAR Clause 52.236-14, Availability and Use of Utility Services.

1.13 TESTS

A. Conduct final tests required in various sections of specifications in presence of an authorized representative of the Contracting Officer. Contractor shall furnish all labor, materials, equipment, instruments, and forms, to conduct and record such tests.

1.14 CONSTRUCTION WASTE MANAGEMENT

A. Responsibilities. The contractor shall employ processes that ensure the generation of as little waste as possible and shall avoid the generation of waste due to the following:

1. Over-packaging.

2. Error.

3. Poor planning, layout.

4. Over ordering.

5. Breakage.

6. Mishandling.

7. Contamination.

8. General weather damage, excluding extreme acts of nature (i.e. tornados, hurricanes, flash floods, blizzards, etc.).

B. Procedures.

1. Of the inevitable waste that is generated, as many of the waste materials as economically feasible shall be reused, salvaged or recycled.

2. Waste disposal in landfills shall be minimized to the greatest extent possible.

3. Waste Diversion Goals.

i Demolition, Major Remodeling: Minimum 10% of total project waste shall be diverted from landfill.

ii Interior Remodeling: Minimum 10% of total project waste shall be diverted from landfill.

iii. The following waste categories, at a minimum, shall be diverted from landfill:

a) Green waste (biodegradable landscaping materials).

b) Soil.

c) Inert material (concrete, asphalt, masonry).

d) Clean dimensional wood, palette wood.

e) Engineered wood products: plywood, particle board, I-joists, etc.

f) Cardboard, paper, packaging.

g) Asphalt roofing materials.

h) Insulation.

i) Gypsum board.

j) Carpet and pad.

k) Paint.

l) Plastics: ABS, PVC.

m) Beverage containers.

C. Description of Work.

1. Includes:

i. Waste Management Plan development and implementation.

ii. Meetings to discuss goals, issues and training for the Waste Management Plan.

iii. Techniques to minimize waste generation.

iv. Sorting and separation of waste materials.

v. Reuse of salvaged materials on site.

vi. Salvage of existing materials and items for reuse or resale.

vii. Recycling of materials that cannot be reused or sold.

viii. Record keeping of receipts and records of salvaged, recycled or land filled materials.

2. Related Elements:

i. Alternates.

ii. Construction Waste Management.

iii. Site Demolition.

iv. Site Clearing.

v. Slope Protection/Erosion Control.

vi. Asphalt Concrete.

vii. Crushed Stone Paving.

viii. Portland Cement Concrete Paving.

ix. Valve Boxes.

x. Storm Sewers.

xi. Chain Link Fences and Gates.

xii. Walk, Road and Parking Appurtenances.

xiii. Miscellaneous Landscaping Materials.

xiv. Concrete, Concrete Formwork, and Concrete Reinforcement.

xv. Cast-in-Place Concrete.

xvi. Unit Masonry.

xvii. Structural Steel.

xviii. Steel Roof Deck/Steel Floor Deck.

xix. Cold Formed Metal Framing.

xx. Metal Fabrications.

xxi. Rough and Finish Carpentry.

xxii. Engineered Structural Wood.

xxiii. Plastic Lumber.

xxiv. Building Insulation.

xxv. Modified Bitumen Roofing.

xxvi. Metal Doors.

xxvii. Wood and Plastic Doors and Frames.

xxviii. Metal Support Systems.

xxix. Gypsum Wallboard.

xxx. Acoustical Treatment.

xxxi. Resilient Flooring.

xxxii. Tile and Carpet.

xxxiii. Painting.

xxxiv. Toilet Compartments.

xxxv. Louvers and Vents.

xxxvi. Signage and Graphics.

xxxvii. Ductwork and Ductwork Accessories

D. Definitions.

1. Class III Landfill: A landfill that accepts non-hazardous resources such as household, commercial and industrial waste resulting from construction, remodeling, repair and demolition operations.

2. Clean: Untreated and unpainted; uncontaminated with adhesives, oils, solvents, mastics and like products.

3. Construction and Demolition Waste: Includes all non-hazardous resources resulting from construction, remodeling, alterations, repair and demolition operations.

4. Dismantle: The process of parting out a building in such a way as to preserve the usefulness of its materials and components.

5. Disposal: Acceptance of solid wastes at a legally operating facility for the purpose of land filling (includes Class III landfills and inert fills).

6. Inert Backfill Site: A location, other than inert fill or other disposal facility, to which inert materials are taken for the purpose of filling an excavation, shoring or other soil engineering operation.

7. Inert Fill: A facility that can legally accept inert waste, such as asphalt and concrete exclusively for the purpose of disposal.

8. Inert Solids/Inert Waste: Non-liquid solid resources including, but not limited to, soil and concrete that does not contain hazardous waste or soluble pollutants at concentrations in excess of water-quality objectives established by a regional water board and does not contain significant quantities of decomposable solid resources.

9. Mixed Debris: Loads that include commingled recyclable and non-recyclable materials generated at the construction site.

10. Mixed Debris Recycling Facility: A solid resource processing facility that accepts loads of mixed construction and demolition debris for the purpose of recovering re-usable and recyclable materials and disposing non-recyclable materials.

11. Permitted Waste Hauler: A company that holds a valid permit to collect and transport solid wastes from individuals or businesses for the purpose of recycling or disposal.

12. Recycling: The process of sorting, cleansing, treating, and reconstituting materials for the purpose of using the altered form in the manufacture of a new product. Recycling does not include burning, incinerating or thermally destroying solid waste.

    i. On-site Recycling. Materials that are sorted and processed on site for use in an altered state in the work, i.e., concrete crushed for use as a sub-base in paving.

    ii. Off-site Recycling. Materials hauled to a location and used in an altered form in the manufacture of new products.

13. Recycling Facility: An operation that can legally accept materials for the purpose of processing the materials into an altered form for the manufacture of new products. Depending on the types of materials accepted and operating procedures, a recycling facility may or may not be required to have a solid waste facility permit or be regulated by the local enforcement agency.

14. Re-Use: Materials that are recovered for use in the same form, on-site or off-site.

15. Return: To give back reusable items or unused products to vendors for credit.

16. Salvage: To remove waste materials from the site for resale or re-use by a third party.

17. Source-Separated Materials: Materials that are sorted by type at the site for the purpose of reuse and recycling.

18. Solid Waste: Materials that have been designated as non-recyclable and are discarded for the purposes of disposal.

19. Transfer Station: A facility that can legally accept solid waste for the purpose of temporarily storing the materials for re-loading onto

other trucks and transporting them to a landfill for disposal or recovering some materials for re-use or recycling.

E. Guides: No preference is given to the recycles listed below; they are listed for the convenience of the contractor.

1. Dirt/clean fill.

2. Green/landscaping waste.

3. Concrete, asphaltic concrete.

4. Cardboard, paper, packaging.

5. Clean dimensional wood, palette wood.

6. Usable palettes.

7. Metals from banding, ductwork, piping, rebar, roofing, other trim, steel, iron, galvanized sheet steel, stainless steel, aluminum, copper, zinc, lead, brass, and bronze.

8. Carpet and pad.

9. Gypsum board.

10. Paint.

11. Insulation.

12. Asphalt shingles.

13. Beverage containers.

F. Submittals.

1. Waste Management Plan. Prior to any waste removal, the Contractor shall submit their Waste Management Plan to the ARS. The Plan shall contain the following:

i. Analysis of the estimated job site waste to be generated, including types and quantities.

ii. Proposed alternatives to land filling. Contractor shall prepare a list of each material proposed to be salvaged, re-used, or recycled during the course of the project.

iii. Methods handling of materials to be recycled.

2. On Site:

i. Materials separation

ii. Materials storage

iii. Materials protection, where applicable

3. Off site:

i Provide name of mixed debris recycling facility; include list of materials to be recycled.

ii. Procedures. A description of the means to be employed in recycling the above materials consistent with requirements for acceptance by designated facilities.

iii. Landfill Options. The name of the landfill(s) where trash will be disposed of.

a) Meetings. Contractor shall conduct Construction Waste Management meetings. Meetings shall include the Subcontractor, the Project Manager and representatives as designated by the Contracting Officer. At a minimum, waste management goals and issues shall be discussed at pre-bid meetings, pre-construction meetings and regular job-site meetings.

b) Transportation. A description of the means of transportation of the recyclable materials (whether materials will be site-separated and self-hauled to designated facilities, or whether mixed materials will be collected by a waste hauler and removed from the site) and destination of materials.

c) Waste Management Plan Implementation.

(i) Manager. The contractor shall designate an on-site party (or parties) responsible for instructing workers and subcontractors and overseeing and documenting results of the Waste Management Plan for the project.

(ii) Distribution. The contractor shall distribute copies of the Waste Management Plan to the Contracting Officer and Location Manager.

(iii) Instruction. The contractor shall provide on-site instruction of appropriate separation, handling, recycling, salvage, reuse and return methods to be used by all parties at appropriate stages of the project.

(iv) Separation Facilities. The contractor shall lay out and label a specific area to facilitate separation of materials for reuse, salvage, recycling, and return. Recycling and waste bin areas are to be kept neat and clean and clearly marked in order to avoid contamination or mixing of materials.

(v) Hazardous Wastes. Hazardous wastes shall be separated, stored, and disposed of according to local, state and federal regulations.

4. Reports: The Contractor shall submit (monthly, quarterly, at end of job) a Waste Management Progress Report. The report shall contain the amount (in tons or cubic yards) of material land filled from the project, the identity of the landfill, the total amount of tipping fees paid at the landfill and the total disposal cost. Include legible copies of manifests, weight tickets, receipts and invoices. Manifests shall be from recycle and/or disposal site operators that can legally accept the materials for the purpose of reuse, recycling, or disposal.

5. For each material recycled, reused or salvaged from the project,

provide the following:

i. Amount (in tons or cubic yards).

ii. Date removed from the job site.

iii. Receiving party.

iv. Transportation cost.

v. Amount of any money paid or received for the recycled or salvaged material. Net total cost or savings of salvage or recycling each material. Attach manifests, weight tickets, receipts, and/or invoices. Indicate the project information, including project title, name of company completing form, and beginning and ending dates of period covered by summary form.

1.15 WARRANTY OF CONSTRUCTION

A. GENERAL: In addition to any other warranties set out elsewhere in this contract, the Contractor warrants that work performed under this contract conforms to the contract requirements and is free of any defect of equipment, material or design furnished, or workmanship performed by the Contractor or any of his subcontractors or suppliers at any tier. Such warranty shall continue for a period of one year from the date of final acceptance of the project. Under this warranty, the Contractor shall remedy at his own expense any such failure to conform or any such defect. In addition, the Contractor shall remedy at his own expense any damage to Government owned or controlled real or personal property, when that damage is the result of the Contractor's failure to conform to contract requirements or any such defect of equipment, material, workmanship, or design. The Contractor shall also restore any work damaged in fulfilling the terms of this clause. The Contractor's warranty with respect to work repaired or replaced hereunder will run for one year from the date of such repair or replacement.

B. NOTIFICATION AND COMPLIANCE: The Government shall notify the Contractor in writing within a reasonable time after the discovery of any failure, defect, or damage. Should the Contractor fail to remedy any failure, defect, or damage described in paragraph above within a reasonable time after receipt of notice thereof, the Government shall have the right to replace or repair and have the cost billed to the Contractor.

C. EXPRESSED WARRANTY: In addition to the other rights and remedies provided by this clause, all subcontractors', manufacturers', and suppliers' warranties expressed or implied, respecting any work and materials shall, at the direction of the Government be enforced by the Contractor for the benefit of the Government. In such case, if the Contractor's warranty under paragraph A above has expired, any suit directed by the Government to enforce a subcontractor's, manufacturer's, or supplier's warranty shall be at the expense of the Government. The Contractor shall obtain any warranties which the subcontractors, manufacturers, or suppliers would give in normal commercial practice. If so directed by the Contracting Officer, the Contractor shall require any such warranties to be executed in writing to the Government.

D. LATENT DEFECTS: The Warranty specified herein shall not limit the Government's rights with respect to latent defects, gross mistake, or fraud.

## 1.16 BIOBASED PREFERRED PRODUCTS

A. The Contractor must comply with Sections 9002 of the Farm Security and Rural Investment Act of 2002; the Food Conservation and Energy Act of 2008; and the Agricultural Act of 2014; and the Federal Acquisition Regulation to provide biobased products. The Contractor shall also comply with the clause at FAR 52.223-2, Affirmative Procurement of Biobased Products Under Service and Construction Contracts contained in Section I. The Contractor shall also comply with the provision at FAR 52.223-1, Biobased Product Certification contained in Section K.

B. The Contractor shall utilize products and material made from biobased materials (e.g., biobased erosion control materials, biobased composite panels, biobased roof coatings, biobased insulating foam, biobased interior paints and coatings, and biobased wood and concrete sealers). All supplies and materials shall be of a type and quality that conform to applicable Federal specifications and standards. All supplies and materials to be used in the performance of work described herein are subject to the approval of the Contracting Officer Representative (COR).

C. The following is an example list of products that may be used in this minor construction contract for which biobased products are available and the designated biobased content. The list is not all inclusive. It is desirable that vendors be able to supply the greatest number of biobased products listed meeting the health and environmental specifications.

Adhesive and Mastic Removers-58 percent
Asphalt and Tar Removers-80 percent
Asphalt Restorers-68 percent 2
Carpet-7 percent
Carpet and Upholstery Cleaners-54 percent and Spot Removers-7 percent
Composite Panels - Acoustical-37 percent; Interior Panels-55 percent; Plastic Lumber-23 percent; Structural Interior Panels-89 percent; Structural Wall Panels-94 percent and Countertops and Solid Surface-89 percent
Concrete and Asphalt Cleaners-70 percent
Concrete and Asphalt Release Fluids-87 percent
Dust Suppressants-85 percent
Erosion Control Materials-77 percent
Floor Coverings (Non Carpet)-91 percent
General Purpose Cleaners-54 percent
Hand Soaps and Cleaners-64 percent
Industrial Cleaners-41 percent
Insulating Foam-7 percent
Leather, Vinyl, and Rubber Care Products-55 percent
Paint- Interior Paints and Coatings, Latex-20 percent and Oil-based-67 percent
Paint Removers-41 percent
Roof Coatings-20 percent
Wastewater Systems Coatings-47 percent
Water Tank Coatings-59 percent
Wood and Concrete Sealers - Membrane Concrete Sealers-11 percent and Penetrating Liquids-79 percent
Wood and Concrete Stains-39 percent

D. Biobased products that are designated for preferential procurement under USDA's BioPreferred Program must meet the required minimum biobased content as stated in USDA's Final Rules available at www.biopreferred.gov.

The Contractor shall provide data for their biobased products such as biobased content. In addition to the biobased products designated by the USDA BioPreferred Program, the Contractor is encouraged to use other commercially available biobased products such as biobased products certified with a label. The Contractor shall submit with the initial proposal a complete list of biobased products, indicating the name of the manufacturer, cost of each material, and the intended use of each of the materials that are to be used in carrying out the requirements of this contract.

E. Within thirty (30) days of contract award, the Contractor shall provide a complete list of USDA designated biobased products that it intends to use or supply during performance of this minor construction contract, including products used or supplied by subcontractors. For each product, the Contractor shall identify the product and brand/manufacturer. For each designated product, the Contractor shall also identify the biobased content. The Contractor shall provide detail on how the Contractor intends to keep abreast of the development and increasing availability of designated and non-designated biobased products and how any new or improved products will be incorporated on an ongoing basis into contract performance. The list shall be approved by the COR in writing. The Contractor shall not commence work until this list is approved by the COR in writing. The list must be reviewed and updated annually, and as required by the COR.

## PART 2 PRODUCTS

Not used.

## PART 3 EXECUTION

Not used.

### 3.1 USDA FORMS

Hot Work Permit

Daily Inspection Form

HOT WORK PERMIT
NEEDED WHEN ANY SPARK OR HEAT PRODUCING EQUIPMENT IS TO BE USED

Project Name:
Name of Contractor's Firm:
Contractor Contact Name:
Contractor Contact Telephone Number
Date:
Building/Location:
Work To Be Done:

_____

Any Special Precautions:

Fire Watch Required: ___ Yes ___ No
The location where the work is to be performed has been examined, necessary precautions have been taken, and permission is granted for this work.

Signed
(Contractor Individual Responsible for Authorizing Hot Work)
Permit Expires: _____ (Date)

Time Hot Work Started: _____ Time Hot Work Completed:

FINAL CHECK-UP
Work area and all adjacent areas to which sparks and heat might have spread (including floors above and below and on opposite sides of walls) were inspected 30 minutes after the work was completed and were found fire safe.

Signed_____ Date _____ Contractor's Fire Watch)

(Page 1 of 2)

ATTENTION
Before approving any cutting and welding permit, the contractor's authorized representative or their appointee shall inspect the work area and confirm that pre-cautions have been taken to prevent fire in accordance with NFPA Standard No. 51B.

PRECAUTIONS

1. Sprinklers are in service where installed

2. Cutting and welding equipment in good repair

3. Within 35 feet; floors swept clean of combustible, no combustible material or flammable liquids, all wall and floor openings covered, and covers suspended beneath work to collect sparks

4. When working on enclosed equipment and in confined space, equipment and area is free of flammable vapors

5. Fire watch provided during and 30 minutes after operation (60 minutes for torch applied roofing operations)

6. Portable fire extinguisher with adequate rating available in the immediate vicinity

7. Standpipe system in service where installed

8. Protection of any sprinkler heads when hot work is in close proximity

9. Smoking prohibited in immediate vicinity

10. Non-combustible shields provided when hot work is done near combustible walls, partitions, floors, roofs

11. Prohibition of hot work on pipes contacting combustible walls

12. Personnel trained in use of equipment including portable fire extinguishers and sounding a fire alarm

13. Final check-up conducted after 30 minutes

DAILY INSPECTION FORM

INSTRUCTIONS: This form is to be utilized when hazards are posed by NFPA 101 deficiencies or construction activities are in progress. Remediation must be implemented upon project start and continuously enforced through project completion to provide a level of life safety comparable to that described in the Life Safety Code. Submit completed forms to the Contracting Officer.

| | |
|---|---|
| Project Name: | Project Number: |
| Construction & Impact Description: | Construction Location:<br><br>Affected Areas: |
| Project COR | Project Start Date |
| Project Competent Person (CP) | Estimated Duration |
| | Completion Date |
| Contractor | |
| GC Supervisor | Telephone: |
| Contractor CP | |

| Inspection Period:<br>Responses: | SUN | MON | TUE | WED | THR | FRI | SAT |
|---|---|---|---|---|---|---|---|
| 1. Are exits readily accessible and provide unobstructed egress? | | | | | | | |
| 2. Have alternate exits been established if required due to inaccessibility of existing exits? | | | | | | | |
| 3. If alternate exits have been established, are personnel in the area informed and aware of their relocation? | | | | | | | |
| 4. Are the existing and relocated exits clearly marked and able to be seen in the event of a fire or emergency? | | | | | | | |
| 5. Are evacuation routes posted with follow-up inspections required by construction impact changes in escape routes? | | | | | | | |
| 6. Are written procedures and guidelines posted in the immediate and adjacent areas for what to do and who to call in the event of a fire or emergency? | | | | | | | |

| Inspection Period:<br>Responses: | SUN | MON | TUE | WED | THR | FRI | SAT |
|---|---|---|---|---|---|---|---|
| 7. Are personnel in immediate/adjacent areas aware and informed in procedures and guidelines to follow in the event of fire or emergency? | | | | | | | |
| 8. Is there free and unobstructed access to services for emergency personnel (eg, fire, medical, security)? | | | | | | | |
| 9. Are fire alarm (eg, pull station), detection (eg, smoke/heat), suppression (eg, sprinkler, extinguisher) systems in working order, protected and unobstructed with locations identified? | | | | | | | |
| 10. If the fire alarm, detection, suppression systems are impaired or temporarily non-functional, has a fire watch for the area, as required or necessary, been trained and established? | | | | | | | |
| 11. If the fire alarm, detection, suppression systems are impaired, have measures been taken to provide temporary equivalent equipment and/or systems for adequate protection? Note date for equivalent measures. | | | | | | | |
| 12. If the fire alarm, detection, suppression systems are impaired, are equivalent equipment/systems inspected and tested at least monthly? | | | | | | | |
| 13. If temporary fire alarm, detection, suppression systems are installed, are personnel in the area aware and trained on how to operate or utilize them in the event of fire or emergency? | | | | | | | |
| 14. Has the "No Smoking" policy been posted, implemented and enforced in the construction area? | | | | | | | |
| 15. Are temporary partitions built to be fire/smoke tight with fire retardant noncombustible material and inspected daily for integrity? | | | | | | | |

| Inspection Period:<br>Responses: | SUN | MON | TUE | WED | THR | FRI | SAT |
|---|---|---|---|---|---|---|---|
| 16. Is construction site access restricted to authorized personnel only including warning signs and secured at the end of each day? | | | | | | | |
| 17. Is construction area hazard surveillance conducted daily? | | | | | | | |
| 18. Is construction area storage, waste, debris and excess materials being daily managed properly to reduce fire or safety hazards? | | | | | | | |
| 19. Are construction activities and materials prosecuted, handled, stored, and secured in an orderly and safe manner? | | | | | | | |
| 20. Is the generation, spread and exposure of construction dust, fumes, noise, odor, smoke controlled with appropriate fume, odor, vapor ventilation provided to control noxious, infectious, toxic exposure and store/protect flammable/combustible products? | | | | | | | |
| 21. Has a GC Safety Manager been designated with routine site safety meetings conducted to ensure awareness of Life Safety Code? | | | | | | | |
| 22. Is personnel protective equipment (e.g., safety glasses, ear plugs, hard hats) required and being used? | | | | | | | |
| 23. If there are hand/safety rails, scaffolding or ladders required, are they in place, in good condition and being used in a safe manner? | | | | | | | |
| 24. Are the construction site (buildings and exterior grounds) hazards (e.g., fall/trip) guarded and free of potential safety violations? | | | | | | | |
| 25. Do electrical panels, temporary wiring, extension cords (3 wire grounded type), tools, and equipment appear to be installed, utilized, and functioning in a safe manner? | | | | | | | |
| 26. If there are temporary electrical outlets provided, do they have ground fault protection at the receptacle/panel? | | | | | | | |

| Inspection Period:<br>Responses: | SUN | MON | TUE | WED | THR | FRI | SAT |
|---|---|---|---|---|---|---|---|
| 27. If hazardous equipment/systems need to be de-energized, are applicable "Lockout/Tagout" procedures being followed? | | | | | | | |
| 28. Are utility services (e.g., electrical, steam, water, waste, gas) properly secured at the end of each day? | | | | | | | |
| 29. If there is any hot work (welding, soldering, cutting) being performed within the construction site, have additional fire safety precautions been taken and necessary equipment provided? | | | | | | | |
| 30. If there is any hot work (welding, soldering, cutting) being performed on the construction site, has Contracting Officer been notified? Has the Hot Work Permit been approved? | | | | | | | |
| 31. If hazardous products are present, are they limited to the amount needed and used daily? | | | | | | | |
| 32. Are hazardous products disposed according to EPA requirements? | | | | | | | |
| 33. Are all hazardous products present or being used (e.g., flammable, combustible, corrosive, noxious) labeled with MSDS information readily available? | | | | | | | |
| 34. If infection control is required, are the appropriate policies and procedures known and being followed? | | | | | | | |
| 35. Are all safety incidents documented and reported to the Contracting Officer? | | | | | | | |
| | | | | | | | |
| Contractor CP Initials Performing Daily Inspections: | | | | | | | |

Inspection Comments/Findings: (PROVIDE DETAILED EXPLAINATION OF EXCEPTIONS/DEFICIENTIES)

Signature/Date:

Project CP GC Safety Manager

-- End of Section --

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SECTION 01 11 00

SUMMARY OF WORK
**08/15, CHG 2: 08/21**

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Utility Locate Report

1.2 OCCUPANCY OF PREMISES

Building will not be occupied during performance of work under this Contract.

Before work is started, arrange with the Contracting Officer a sequence of procedure, means of access, space for storage of materials and equipment, and use of approaches, corridors, and stairways.

1.3 EXISTING WORK

In addition to FAR 52.236-9 Protection of Existing Vegetation, Structures, Equipment, Utilities, and Improvements:

a. Remove or alter existing work in such a manner as to prevent injury or damage to any portions of the existing work which remain.

b. Repair or replace portions of existing work which have been altered during construction operations to match existing or adjoining work, as approved by the Contracting Officer. At the completion of operations, existing work must be in a condition equal to or better than that which existed before new work started.

1.4 LOCATION OF UNDERGROUND UTILITIES

Obtain digging permits prior to start of excavation, and comply with Installation requirements for locating and marking underground utilities. Verify existing utility locations indicated on contract drawings, within area of work.

Identify and mark all utilities prior to construction Scan the construction site with Ground Penetrating Radar (GPR), electromagnetic, or sonic equipment, and mark the surface of the ground or paved surface where existing underground utilities are discovered. Verify the elevations of existing piping, utilities,and any type of underground obstruction not indicated, or specified to be removed, that is indicated or discovered during scanning, in locations to be traversed by piping, ducts, and other work to be conducted or installed. Verify elevations before installing new work closer than nearest manhole or other structure at which an adjustment in grade can be made. Submit a Utility Locate Report to include

a plan map and utility drawing of location using GPR, electromagnetic, or sonic equipment to the Government for review prior to Construction.

1.4.1 Notification Prior to Excavation

Notify the Installation Department at least 30 calendar days prior to starting excavation work.

1.5 SALVAGE MATERIAL AND EQUIPMENT

All materials to be demolished and removed that are composed primarily of metal shall be delivered and unloaded at the DRMO Lot Recycling Center. Other items designated by the Contracting Officer to be salvaged remain the property of the Government. Segregate, itemize, deliver and off-load the salvaged property at the Government designated location.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

U.S. DEPARTMENT OF AGRICULTURE • AGRICULTURAL RESEARCH SERVICE

# CONSTRUCTION PROGRESS AND PAYMENT SCHEDULE

**IMPORTANT:** Read instructions on reverse before completing this form.

| CONTRACT NO. | AMOUNT | CALENDAR DAYS | STARTING DATE | COMPLETION DATE |
|---|---|---|---|---|
| | | | | |

| PROJECT | CONTRACTOR'S NAME & ADDRESS (Include Zip code) |
|---|---|
| LOCATION | |

| DESCRIPTION OF BRANCH OF WORK | PROGRESS SCHEDULE | | | | SCHEDULE OF PAYMENTS *(Use in conjunction with ARS Form 372)* | | |
|---|---|---|---|---|---|---|---|
| | Value | Percent of work | DATE | | Completed this period | Previously Completed | Total Completed |
| | | | Start | Complete | | | |
| | A | B | C | D | E | F | G |
| 1. | | % | | | | | |
| 2. | | % | | | | | |
| 3. | | % | | | | | |
| 4. | | % | | | | | |
| 5. | | % | | | | | |
| 6. | | % | | | | | |
| 7. | | % | | | | | |
| 8. | | % | | | | | |
| 9. | | % | | | | | |
| 10. | | % | | | | | |
| 11. | | % | | | | | |
| 12. | | % | | | | | |
| 13. | | % | | | | | |
| 14. | | % | | | | | |
| 15. | | % | | | | | |
| 16. | | % | | | | | |
| 17. | | % | | | | | |
| 18. | | % | | | | | |
| 19. | | % | | | | | |
| 20. | | % | | | | | |
| 21. | | % | | | | | |
| 22. | | % | | | | | |
| 23. | | % | | | | | |
| **Following items to be used for Change Orders, Amendments, and other.** | | | | | | | |
| 24. | | % | | | | | |
| 25. | | % | | | | | |
| 26. | | % | | | | | |
| 27. | | % | | | | | |
| 28. | | % | | | | | |
| 29. | | % | | | | | |
| 30. | | % | | | | | |
| 31. | | % | | | | | |
| **TOTAL** | | | | | | | |

REMARKS

| CONTRACTOR'S SIGNATURE | DATE | CONTRACTING OFFICER'S APPROVAL | DATE |
|---|---|---|---|
| | | | |



This form was electronically produced by USDA/ARS/OCIO/EASB.

## INSTRUCTIONS FOR COMPLETION OF FORM ARS - 371

The construction Progress - Payment Schedule shall be submitted within **14 calendar days after the date of receipt of Notice to Proceed.**

1. Complete blocks entitled: Contract Number, Amount, Calendar Days, Starting Date, Completion Date, Project, Location, and Contractor's Name and Address.

2. Complete Column A**, Value,** indicating a complete breakdown of each branch of work in dollar value.

3. Complete Column B**, Percent of Work,** showing the percentage of each branch of work in relation to the whole project.

4. Complete Columns C and D to show the approximate dates each branch of work will begin and be completed. Final completion date in Column D must not exceed the completion date at the top of the form.

5. Contractor shall sign the form at the bottom and submit to the Contracting Officer for his approval. An approved copy will be returned to the Contractor for his file.

**This form is required even though partial payments are not requested.** If partial payments are requested, a copy of the approved Form ARS-371 must accompany each Payment Request (Form ARS-372), and Columns E, F, and G must be filled in according to the work completed.

# CONTRACTOR'S REQUEST FOR PAYMENT TRANSMITTAL

| CONTRACT NO. | PARTIAL PAYMENT NO. | FINAL PAYMENT ☐ |
|---|---|---|
| PROJECT | REQUISITION NO. | |
| LOCATION | FOR PERIOD BEGINNING | ENDING |

| ITEM | AMOUNT | |
|---|---|---|
| 1. Amount of original contract | $ | |
| 2. Change orders and/or amendments | $ | |
| 3. Total adjusted contract prices | $ | |
| 4. Value of work completed to end of period | | $ |
| 5. Value of material stored at the site *(Itemize below)* | | $ |
| 6. Total value of work completed and stored material *(Line 4 plus Line 5).* | | $ |
| 7. Less _____% retainage | | $ |
| 8. **Total due contractor thru end of this period** | | $ |
| 9. Less previous requests | | $ |
| 10. **Net amount due contractor this payment** | | $ |

MATERIAL STORED AT THE SITE *(See Item 5 above)*

| Description | Value |
|---|---|
| | |

## CONTRACTOR'S CERTIFICATION OF PAYMENT

By signing this request for payment, I certify that I have made payment from the proceeds of prior payments, and that I will make timely payment from the proceeds of this payment, of amounts due my subcontractors and suppliers in accordance with my contractual arrangements with them.

| SIGNATURE OF CONTRACTOR'S AUTHORIZED REPRESENTATIVE | DATE SIGNED |
|---|---|
| RECOMMENDED FOR PAYMENT *(Signature of EPM, COR, CM or A.E. as appropriate)* | DATE SIGNED |
| CONCUR *(Signature of EPM or COR as appropriate)* | DATE SIGNED |

# MATERIAL APPROVAL SUBMITTAL

| TO: USDA/ARS CO<br>CC: A&E, COR | FROM: (Contractor) | PROJECT TITLE: | CONTRACT NUMBER: |
|---|---|---|---|

| (To be completed by the Contractor) | | | Government Use Only | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| SUBMITTAL NUMBER: | PREVIOUS SUBMITTAL NUMBER (only if items listed were submitted previously): | | A&E Recommendation | | | COR/EPM Recommendation | | | Contracting Officer Approval/Disapproval | | |
| Line No. | Specification Sect. Para No., Draw No. | Description of Material (Include type, model, mfg) | APP | DISAP | See Comments on Reverse | APP | DISAP | See Comments on Reverse | APP | DISAP | See Comments on Reverse |
| | | | | | | | | | | | |
| | | | | | | | | | | | |
| | | | | | | | | | | | |
| | | | | | | | | | | | |
| | | | | | | | | | | | |

**If an item is disapproved, a resubmittal is due within _____calendar days of the Contracting Officer's signature below.**

| Contractor (Ktr) | A&E | COR/EPM | Contracting Officer |
|---|---|---|---|
| TYPE or PRINT NAME & TITLE | TYPE or PRINT NAME & TITLE | TYPE or PRINT NAME & TITLE | TYPE or PRINT NAME & TITLE |
| SIGNATURE: | SIGNATURE: | SIGNATURE: | SIGNATURE: |
| DATE: | DATE: | DATE: | DATE: |
| By completing this form, the above signed contractor certifies that the material complies with all specifications of the subject contract. | DATE RECEIVED FROM KTR: | DATE RECEIVED FROM A&E: | DATE RECEIVED FROM COR: |

| CONTRACT #: | SUBMITTAL #: |
|---|---|

## COMMENTS

**Instructions:** Enter the initials of the individual making the comment, the Line Number of the item for which the comment is being made, and the comment.

| |
|---|
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| |

SECTION 01 33 00

SUBMITTAL PROCEDURES
**08/18**

PART 1 GENERAL

1.1 SHOP DRAWINGS

The Contractor shall furnish the Contracting Officer six copies with such promptness as to cause no delay in this work or the work of any other contractors, one sepia copy of the shop drawings as required for the various subheadings for the Contracting Officer's approval. Shop drawings of any other subcontractor should be reviewed and approved by each subcontractor where their contract work comes in contact with or interferes with that of another subcontractor.

The Contracting Officer's approval, however, shall not relieve the Contractor from responsibility for errors in shop drawings. Shop drawings are the instruments of the Contractor, not the Contracting Officer; the Drawings and Specifications shall always be final.

1.2 SAMPLES

Within and not later than thirty (30) days from the date of the General Contract, the General Contractor shall have secured and submitted to the Contracting Officer for his consideration, two samples of each kind and grade of materials and finish as noted under the various headings of the specifications. These samples shall be finished completely and shall accurately represent the work as it is proposed to be installed. The Contracting Officer will give immediate consideration to such samples and shall notify the General Contractor within ten (10) days after receipt of either approval or rejection. If such samples are rejected, the General Contractor shall, without delay, submit corrected samples for approval, and shall continue to do so until such approval is given the Contracting Officer. All approvals will be given by the Contracting Officer to the General Contractor in writing.

It shall be the Contractor's responsibility to obtain materials at such time as to cause no delay to the project. No allowance for additional time nor substitution of approved materials will be allowed for Contractor's negligence in late ordering of materials.

1.3 PRODUCT DATA

Submit six copies of preprinted material such as illustrations, standard schedules, performance charts, instructions, brochures, diagrams, manufacturer's descriptive literature, catalog data, and other data to illustrate portion of work.

1.4 O & M MANUALS

O & M Manuals shall be submitted in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

1.5 GOVERNMENT ACTIONS POSSIBLE

Submittals will be returned to the Contractor with one of the following notations:

a. Submittals marked "approved" or "approved as submitted" authorize Contractor to proceed with work covered.

b. Submittals marked "approved" as noted or "approval except as noted; resubmission not required" authorize Contractor to proceed with work as noted provided the Contractor takes no exception to the notations.

c. Submittals marked "revise" and "resubmittal" or "disapproved" indicate submittal is incomplete or does not comply with design concept or requirements of the contract documents and shall be resubmitted with appropriate changes. No work shall proceed for this item until resubmittal is approved.

1.6 SUBMITTAL REGISTER

Contractor shall submit items as listed in each specification section.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

# SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
Edinburg-Kerrville BLDG 4 Dip Vat Redesign

CONTRACTOR

| ACTIVITY NO | TRANSMITTAL NO | SPEC SECT | DESCRIPTION ITEM SUBMITTED | PARAG # RAPH | CLASSIFICATION | GOVT OR A/E REVWR | CONTRACTOR: SCHEDULE DATES | | | CONTRACTOR ACTION | | | APPROVING AUTHORITY | | | | | MAILED TO CONTR/ DATE RCD FRM APPR AUTH | REMARKS |
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| | | | | | | | SUBMIT | APPROVAL NEEDED BY | MATERIAL NEEDED BY | ACTION CODE | DATE OF ACTION | DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR | DATE FWD TO OTHER REVIEWER | DATE RCD FROM OTH REVIEWER | ACTION CODE | DATE OF ACTION | | |
| (a) | (b) | (c) | (d) | (e) | (f) | | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | | (q) | (r) |
| | | 01 11 00 | SD-01 Preconstruction Submittals | | | | | | | | | | | | | | | | |
| | | | Utility Locate Report | 1.4 | | | | | | | | | | | | | | | |
| | | 01 35 26 | SD-01 Preconstruction Submittals | | | | | | | | | | | | | | | | |
| | | | Site Specific Safety Plan/Accident | 1.7 | G | | | | | | | | | | | | | | |
| | | | Prevention Plan (SSSP/APP) | | | | | | | | | | | | | | | | |
| | | | SD-06 Test Reports | | | | | | | | | | | | | | | | |
| | | | Notifications and Reports | 1.10 | | | | | | | | | | | | | | | |
| | | | SD-07 Certificates | | | | | | | | | | | | | | | | |
| | | | Contractor Safety Self-Evaluation | 1.4 | | | | | | | | | | | | | | | |
| | | | Checklist | | | | | | | | | | | | | | | | |
| | | | Crane Operators/Riggers | 1.6.1.4 | | | | | | | | | | | | | | | |
| | | | Standard Lift Plan | 1.7.2.2 | G | | | | | | | | | | | | | | |
| | | 01 45 00 | SD-01 Preconstruction Submittals | | | | | | | | | | | | | | | | |
| | | | Contractor Quality Control (CQC) | 1.4.2 | G | | | | | | | | | | | | | | |
| | | | Plan | | | | | | | | | | | | | | | | |
| | | | SD-06 Test Reports | | | | | | | | | | | | | | | | |
| | | | Test Reports | 1.8 | | | | | | | | | | | | | | | |
| | | | SD-07 Certificates | | | | | | | | | | | | | | | | |
| | | | QC Manager | 1.4.2 | | | | | | | | | | | | | | | |
| | | | SD-11 Closeout Submittals | | | | | | | | | | | | | | | | |
| | | | Testing And Deficiency Records | 1.11.1 | | | | | | | | | | | | | | | |
| | | 01 78 23 | SD-10 Operation and Maintenance | | | | | | | | | | | | | | | | |
| | | | Data | | | | | | | | | | | | | | | | |
| | | | Prefinal | 3.2.1 | G | | | | | | | | | | | | | | |
| | | | Final | 3.2.2 | G | | | | | | | | | | | | | | |
| | | | SD-11 Closeout Submittals | | | | | | | | | | | | | | | | |

# SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
Edinburg-Kerrville BLDG 4 Dip Vat Redesign

CONTRACTOR

| ACTIVITY NO | TRANSMITTAL NO | SPEC SECT | DESCRIPTION ITEM SUBMITTED | PARAG # RAPH | CLASSIFICATION | GOVT OR A/E REVWR | CONTRACTOR: SCHEDULE DATES | | | CONTRACTOR ACTION | | | APPROVING AUTHORITY | | | | | MAILED TO CONTR/ DATE RCD FRM APPR AUTH | REMARKS |
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| | | 01 78 23 | Record Drawings And Utility Information | 1.3.6 | G | | | | | | | | | | | | | |
| | | | Training Outline | 3.1.2 | G | | | | | | | | | | | | | |
| | | | Completion Record | 3.1.2 | G | | | | | | | | | | | | | |
| | | 02 41 00 | SD-01 Preconstruction Submittals | | | | | | | | | | | | | | | |
| | | | Demolition Plan | 1.2.1 | G | | | | | | | | | | | | | |
| | | | Deconstruction Plan | 1.2.1 | G | | | | | | | | | | | | | |
| | | | Existing Conditions | 1.9 | | | | | | | | | | | | | | |
| | | | SD-07 Certificates | | | | | | | | | | | | | | | |
| | | | Notification | 1.7 | G | | | | | | | | | | | | | |
| | | 03 30 53 | SD-02 Shop Drawings | | | | | | | | | | | | | | | |
| | | | Installation Drawings | 1.4 | G | | | | | | | | | | | | | |
| | | | SD-03 Product Data | | | | | | | | | | | | | | | |
| | | | Air-Entraining Admixture | 2.2.3.1 | | | | | | | | | | | | | | |
| | | | Expansion Joint Filler Strips, Premolded | 2.2.6 | | | | | | | | | | | | | | |
| | | | Joint Sealants - Field Molded Sealants | 2.2.7 | | | | | | | | | | | | | | |
| | | | Conveying and Placing Concrete | 3.2 | | | | | | | | | | | | | | |
| | | | Mix Design Data | 2.3 | G | | | | | | | | | | | | | |
| | | | Ready-Mix Concrete | 2.3 | | | | | | | | | | | | | | |
| | | | Mechanical Reinforcing Bar Connectors | 2.2.5 | | | | | | | | | | | | | | |
| | | | SD-06 Test Reports | | | | | | | | | | | | | | | |
| | | | Aggregates | 2.2.2 | | | | | | | | | | | | | | |
| | | | Concrete Mixture Proportions | 2.1.3 | G | | | | | | | | | | | | | |

# SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
Edinburg-Kerrville BLDG 4 Dip Vat Redesign

CONTRACTOR

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| | | 03 30 53 | Compressive Strength Testing | 3.8.3 | G | | | | | | | | | | | | | | |
| | | | Slump | 3.8.3 | G | | | | | | | | | | | | | | |
| | | | Air Content | 3.8.3 | | | | | | | | | | | | | | | |
| | | | Water | 2.2.4 | | | | | | | | | | | | | | | |
| | | | SD-07 Certificates | | | | | | | | | | | | | | | | |
| | | | Cementitious Materials | 2.2.1 | | | | | | | | | | | | | | | |
| | | | Aggregates | 2.2.2 | | | | | | | | | | | | | | | |
| | | 05 50 13 | SD-02 Shop Drawings | | | | | | | | | | | | | | | | |
| | | | CORRAL RAILINGS | 2.3 | G | | | | | | | | | | | | | | |
| | | | GATES | 2.4 | G | | | | | | | | | | | | | | |
| | | | BOLLARDS | 2.2 | G | | | | | | | | | | | | | | |
| | | | Anchors | 1.5.1 | G | | | | | | | | | | | | | | |
| | | | MISCELLANEOUS | 1.5 | G | | | | | | | | | | | | | | |
| | | | SD-03 Product Data | | | | | | | | | | | | | | | | |
| | | | Pipe | 2.1.1 | G | | | | | | | | | | | | | | |
| | | | Hinges | 1.5.1 | G | | | | | | | | | | | | | | |
| | | | Latches | 1.5.1 | G | | | | | | | | | | | | | | |
| | | | Galvanizing | 1.5.1 | G | | | | | | | | | | | | | | |
| | | | Galvanizing | 2.7 | G | | | | | | | | | | | | | | |
| | | | Hardware | 2.4 | G | | | | | | | | | | | | | | |
| | | | SD-07 Certificates | | | | | | | | | | | | | | | | |
| | | | GALVANIZING | 1.5.1 | G | | | | | | | | | | | | | | |
| | | | GALVANIZING | 2.7 | G | | | | | | | | | | | | | | |
| | | | WELDERS | 1.3 | G | | | | | | | | | | | | | | |
| | | | Welding | 1.3 | G | | | | | | | | | | | | | | |
| | | | Mill | 2.1.1 | G | | | | | | | | | | | | | | |

# SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
Edinburg-Kerrville BLDG 4 Dip Vat Redesign

CONTRACTOR

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| (a) | (b) | (c) | (d) | (e) | (f) | | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | | (q) | (r) |
| | | 13 34 19 | SD-01 Preconstruction Submittals | | | | | | | | | | | | | | | | |
| | | | Existing-Condition Survey | 1.9 | G | | | | | | | | | | | | | | |
| | | | Verification Report | 1.9 | G | | | | | | | | | | | | | | |
| | | | Erection Plan | 1.9 | G | | | | | | | | | | | | | | |
| | | | SD-02 Shop Drawings | | | | | | | | | | | | | | | | |
| | | | Existing-To-New Interfaces | 1.9 | G | | | | | | | | | | | | | | |
| | | | Rigid Frames | 1.9 | G | | | | | | | | | | | | | | |
| | | | Columns | 1.9 | G | | | | | | | | | | | | | | |
| | | | Rafters | 1.9 | | | | | | | | | | | | | | | |
| | | | Purlin Extensions | 1.9 | G | | | | | | | | | | | | | | |
| | | | Splices | 1.9 | G | | | | | | | | | | | | | | |
| | | | Bracing | 1.9 | G | | | | | | | | | | | | | | |
| | | | Roof Panels | 1.9 | G | | | | | | | | | | | | | | |
| | | | Fasteners | 1.9 | G | | | | | | | | | | | | | | |
| | | | Fasteners | 1.9 | G | | | | | | | | | | | | | | |
| | | | Flashings | 1.9 | G | | | | | | | | | | | | | | |
| | | | Trim | 1.9 | G | | | | | | | | | | | | | | |
| | | | Closures | 1.9 | G | | | | | | | | | | | | | | |
| | | | Insulation Layout | 1.9 | G | | | | | | | | | | | | | | |
| | | | SD-03 Product Data | | | | | | | | | | | | | | | | |
| | | | Structural Materials | 1.9 | G | | | | | | | | | | | | | | |
| | | | Red-Oxide Primer | 1.9 | G | | | | | | | | | | | | | | |
| | | | Exposed-Fastener Roof Panels | 1.9 | G | | | | | | | | | | | | | | |
| | | | Sealants | 1.9 | G | | | | | | | | | | | | | | |
| | | | Prodex Total 5M Plus Insulation | 1.9 | G | | | | | | | | | | | | | | |
| | | | SD-05 Design Data | | | | | | | | | | | | | | | | |

# SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
Edinburg-Kerrville BLDG 4 Dip Vat Redesign

CONTRACTOR

| ACTIVITY NO | TRANSMITTAL NO | SPEC SECT | DESCRIPTION ITEM SUBMITTED | PARAG # RAPH | CLASSIFICATION | GOVT OR A/E REVWR | CONTRACTOR: SCHEDULE DATES | | | CONTRACTOR ACTION | | | APPROVING AUTHORITY | | | | | MAILED TO CONTR/ DATE RCD FRM APPR AUTH | REMARKS |
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| | | 13 34 19 | Signed And Sealed Structural | 1.9 | G | | | | | | | | | | | | | | |
| | | | Calculations | | | | | | | | | | | | | | | | |
| | | | SD-07 Certificates | | | | | | | | | | | | | | | | |
| | | | Welding Qualifications | 1.9 | G | | | | | | | | | | | | | | |
| | | | Mill Certificates | 1.9 | G | | | | | | | | | | | | | | |
| | | | Manufacturer/Erector | 1.9 | G | | | | | | | | | | | | | | |
| | | | Qualifications | | | | | | | | | | | | | | | | |
| | | | SD-11 Closeout Submittals | | | | | | | | | | | | | | | | |
| | | | Record Drawings | 1.9 | G | | | | | | | | | | | | | | |
| | | | Finish Repair Records | 1.9 | G | | | | | | | | | | | | | | |
| | | | Warranties | 1.9 | G | | | | | | | | | | | | | | |
| | | | Final Inspection Report | 1.9 | G | | | | | | | | | | | | | | |
| | | 22 00 00 | SD-02 Shop Drawings | | | | | | | | | | | | | | | | |
| | | | Plumbing System | 3.4.1 | G | | | | | | | | | | | | | | |
| | | | SD-03 Product Data | | | | | | | | | | | | | | | | |
| | | | Recycled Content for Steel Pipe | 2.1 | S | | | | | | | | | | | | | | |
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# SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
Edinburg-Kerrville BLDG 4 Dip Vat Redesign

CONTRACTOR

| ACTIVITY NO | TRANSMITTAL NO | SPEC SECT | DESCRIPTION ITEM SUBMITTED | PARAG # RAPH | CLASSIFICATION | GOVT OR A/E REVWR | CONTRACTOR: SCHEDULE DATES | | | CONTRACTOR ACTION | | APPROVING AUTHORITY | | | | | MAILED TO CONTR/ DATE RCD FRM APPR AUTH | REMARKS |
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| | | | SD-07 Certificates | | | | | | | | | | | | | | | |
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| | | | Data | | | | | | | | | | | | | | | |
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SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS
**01/20**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

| | |
|---|---|
| ASSP A10.34 | (2001; R 2012) Protection of the Public on or Adjacent to Construction Sites |
| ASSP Z359.0 | (2012) Definitions and Nomenclature Used for Fall Protection and Fall Arrest |

ASTM INTERNATIONAL (ASTM)

| | |
|---|---|
| ASTM F855 | (2015) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment |

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

| | |
|---|---|
| IEEE 1048 | (2003) Guide for Protective Grounding of Power Lines |
| IEEE C2 | (2017; Errata 1-2 2017; INT 1 2017) National Electrical Safety Code |

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

| | |
|---|---|
| NFPA 241 | (2019) Standard for Safeguarding Construction, Alteration, and Demolition Operations |
| NFPA 70 | (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code |
| NFPA 70E | (2021) Standard for Electrical Safety in the Workplace |

U.S. ARMY CORPS OF ENGINEERS (USACE)

| | |
|---|---|
| EM 385-1-1 | (2014) Safety and Health Requirements Manual |

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

| | |
|---|---|
| 29 CFR 1910 | Occupational Safety and Health Standards |
| 29 CFR 1910.146 | Permit-required Confined Spaces |

| | |
|---|---|
| 29 CFR 1910.147 | The Control of Hazardous Energy (Lock Out/Tag Out) |
| 29 CFR 1915 | Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment |
| 29 CFR 1926 | Safety and Health Regulations for Construction |
| 29 CFR 1926.16 | Rules of Construction |
| CPL 2.100 | (1995) Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR 1910.146 |

## 1.2 DEFINITIONS

### 1.2.1 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are dangerous to personnel, and who has authorization to take prompt corrective measures with regards to such hazards.

### 1.2.2 Competent Person, Confined Space

The CP, Confined Space, is a person meeting the competent person requirements as defined EM 385-1-1 Appendix Q, with thorough knowledge of OSHA’s Confined Space Standard, 29 CFR 1910.146, and designated in writing to be responsible for the immediate supervision, implementation and monitoring of the confined space program, who through training, knowledge and experience in confined space entry is capable of identifying, evaluating and addressing existing and potential confined space hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

### 1.2.3 Competent Person, Cranes and Rigging

The CP, Cranes and Rigging, as defined in EM 385-1-1 Appendix Q, is a person meeting the competent person, who has been designated in writing to be responsible for the immediate supervision, implementation and monitoring of the Crane and Rigging Program, who through training, knowledge and experience in crane and rigging is capable of identifying, evaluating and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

### 1.2.4 Competent Person, Excavation/Trenching

A CP, Excavation/Trenching, is a person meeting the competent person requirements as defined in EM 385-1-1 Appendix Q and 29 CFR 1926, who has been designated in writing to be responsible for the immediate supervision, implementation and monitoring of the excavation/trenching program, who through training, knowledge and experience in excavation/trenching is capable of identifying, evaluating and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.5 Competent Person, Fall Protection

The CP, Fall Protection, is a person meeting the competent person requirements as defined in EM 385-1-1 Appendix Q and in accordance with ASSP Z359.0, who has been designated in writing by the employer to be responsible for immediate supervising, implementing and monitoring of the fall protection program, who through training, knowledge and experience in fall protection and rescue systems and equipment, is capable of identifying, evaluating and addressing existing and potential fall hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.6 Competent Person, Scaffolding

The CP, Scaffolding is a person meeting the competent person requirements in EM 385-1-1 Appendix Q and designated in writing by the employer to be responsible for immediate supervising, implementing and monitoring of the scaffolding program. The CP for Scaffolding has enough training, knowledge and experience in scaffolding to correctly identify, evaluate and address existing and potential hazards and also has the authority to take prompt corrective measures with regard to these hazards. CP qualifications must be documented and include experience on the specific scaffolding systems/types being used, assessment of the base material that the scaffold will be erected upon, load calculations for materials and personnel, and erection and dismantling. The CP for scaffolding must have a documented, minimum of 8-hours of scaffold training to include training on the specific type of scaffold being used (e.g. mast-climbing, adjustable, tubular frame), in accordance with EM 385-1-1 Section 22.B.02.

1.2.7 Competent Person (CP) Trainer

A competent person trainer as defined in EM 385-1-1 Appendix Q, who is qualified in the material presented and who possesses a working knowledge of applicable technical regulations, standards, equipment and systems related to the subject matter on which they are training Competent Persons. A competent person trainer must be familiar with the typical hazards and the equipment used in the industry they are instructing. The training provided by the competent person trainer must be appropriate to that specific industry. The competent person trainer must evaluate the knowledge and skills of the competent persons as part of the training process.

1.2.8 High Risk Activities

High Risk Activities are activities that involve work at heights, crane and rigging, excavations and trenching, scaffolding, electrical work, and confined space entry.

1.2.9 High Visibility Accident

A High Visibility Accident is any mishap which may generate publicity or high visibility.

1.2.10 Load Handling Equipment (LHE)

LHE is a term used to describe cranes, hoists and all other hoisting equipment (hoisting equipment means equipment, including crane, derricks, hoists and power operated equipment used with rigging to raise, lower or

horizontally move a load).

1.2.11 Medical Treatment

Medical Treatment is treatment administered by a physician or by registered professional personnel under the standing orders of a physician. Medical treatment does not include first aid treatment even through provided by a physician or registered personnel.

1.2.12 Near Miss

A Near Miss is a mishap resulting in no personal injury and zero property damage, but given a shift in time or position, damage or injury may have occurred (e.g., a worker falls off a scaffold and is not injured; a crane swings around to move the load and narrowly misses a parked vehicle).

1.2.13 Operating Envelope

The Operating Envelope is the area surrounding any crane or load handling equipment. Inside this "envelope" is the crane, the operator, riggers and crane walkers, other personnel involved in the operation, rigging gear between the hook, the load, the crane's supporting structure (i.e. ground or rail), the load's rigging path, the lift and rigging procedure.

1.2.14 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.2.15 Qualified Person, Fall Protection (QP for FP)

A QP for FP is a person meeting the requirements of EM 385-1-1 Appendix Q, and ASSP Z359.0, with a recognized degree or professional certificate and with extensive knowledge, training and experience in the fall protection and rescue field who is capable of designing, analyzing, and evaluating and specifying fall protection and rescue systems.

1.2.16 Recordable Injuries or Illnesses

Recordable Injuries or Illnesses are any work-related injury or illness that results in:

a. Death, regardless of the time between the injury and death, or the length of the illness;

b. Days away from work (any time lost after day of injury/illness onset);

c. Restricted work;

d. Transfer to another job;

e. Medical treatment beyond first aid;

f. Loss of consciousness; or

g. A significant injury or illness diagnosed by a physician or other licensed health care professional, even if it did not result in (a) through (f) above.

1.2.17 Load Handling Equipment (LHE) Accident or Load Handling Equipment Mishap

A LHE accident occurs when any one or more of the eight elements in the operating envelope fails to perform correctly during operation, including operation during maintenance or testing resulting in personnel injury or death; material or equipment damage; dropped load; derailment; two-blocking; overload; or collision, including unplanned contact between the load, crane, or other objects. A dropped load, derailment, two-blocking, overload and collision are considered accidents, even though no material damage or injury occurs. A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped boom, dropped load, or roll over). Document any mishap that meets the criteria described in the Contractor Significant Incident Report (CSIR).

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Site Specific Safety Plan/Accident Prevention Plan (SSSP/APP); G

SD-06 Test Reports

Notifications and Reports

SD-07 Certificates

Contractor Safety Self-Evaluation Checklist

Crane Operators/Riggers

Standard Lift Plan; G

1.4 CONTRACTOR SAFETY SELF-EVALUATION CHECKLIST

Contracting Officer will provide a "Contractor Safety Self-Evaluation checklist" to the Contractor at the pre-construction conference. Complete the checklist monthly and submit with each request for payment voucher. An acceptable score of 90 or greater is required. Failure to submit the completed safety self-evaluation checklist or achieve a score of at least 90 may result in retention of up to 10 percent of the voucher.

1.5 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this contract, comply with the most recent edition of USACE EM 385-1-1, and the following federal, state, and local laws, ordinances, criteria, rules and regulations. Submit matters of interpretation of standards to

the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

## 1.6 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

### 1.6.1 Personnel Qualifications

#### 1.6.1.1 Site Safety and Health Officer (SSHO)

Provide an SSHO that meets the requirements of EM 385-1-1 Section 1. The SSHO must ensure that the requirements of 29 CFR 1926.16 are met for the project. Provide a Safety oversight team that includes a minimum of one (1) person at each project site to function as the Site Safety and Health Officer (SSHO). The SSHO or an equally-qualified Alternate SSHO must be at the work site at all times to implement and administer the Contractor's safety program and government-accepted Accident Prevention Plan. The SSHO and Alternate SSHO must have the required training, experience, and qualifications in accordance with EM 385-1-1 Section 01.A.17, and all associated sub-paragraphs.

If the SSHO is off-site for a period longer than 24 hours, an equally-qualified alternate SSHO must be provided and must fulfill the same roles and responsibilities as the primary SSHO. When the SSHO is temporarily (up to 24 hours) off-site, a Designated Representative (DR), as identified in the AHA may be used in lieu of an Alternate SSHO, and must be on the project site at all times when work is being performed. Note that the DR is a collateral duty safety position, with safety duties in addition to their full time occupation.

#### 1.6.1.2 Contractor Quality Control (QC) Manager:

The Contractor Quality Control Manager can be the SSHO on this project.

#### 1.6.1.3 Competent Person Qualifications

Provide Competent Persons in accordance with EM 385-1-1, Appendix Q and herein. Competent Persons for high risk activities include confined space, cranes and rigging, excavation/trenching, fall protection, and electrical work. The CP for these activities must be designated in writing, and meet the requirements for the specific activity (i.e. competent person, fall protection).

The Competent Person identified in the Contractor's Safety and Health Program and accepted Accident Prevention Plan, must be on-site at all times when the work that presents the hazards associated with their professional expertise is being performed. Provide the credentials of the Competent Persons(s) to the Contracting Officer for information in consultation with the Safety Office.

##### 1.6.1.3.1 Competent Person for Confined Space Entry

Provide a Confined Space (CP) Competent Person who meets the requirements of EM 385-1-1, Appendix Q, and herein. The CP for Confined Space Entry must supervise the entry into each confined space.

1.6.1.4 Crane Operators/Riggers

Provide Operators meeting the requirements in EM 385-1-1, Section 15.B for Riggers and Section 16.B for Crane Operators. In addition, for mobile cranes with Original Equipment Manufacturer (OEM) rated capacities of 50,000 pounds or greater, designate crane operators qualified by a source that qualifies crane operators (i.e., union, a government agency, or an organization that tests and qualifies crane operators). Provide proof of current qualification.

1.6.2 Personnel Duties

1.6.2.1 Duties of the Site Safety and Health Officer (SSHO)

The SSHO must:

a. Conduct daily safety and health inspections and maintain a written log which includes area/operation inspected, date of inspection, identified hazards, recommended corrective actions, estimated and actual dates of corrections. Attach safety inspection logs to the Contractors' daily production report.

b. Conduct mishap investigations and complete required accident reports. Report mishaps and near misses.

c. Use OSHA's Form 300 to log work-related injuries and illnesses occurring on the project site for Prime Contractors and subcontractors. Post and maintain the Form 300 on the site Safety Bulletin Board.

d. Maintain applicable safety reference material on the job site.

e. Attend the pre-construction conference, pre-work meetings including preparatory meetings, and periodic in-progress meetings.

f. Review the APP and AHAs for compliance with EM 385-1-1, and approve, sign, implement and enforce them.

g. Establish a Safety and Occupational Health (SOH) Deficiency Tracking System that lists and monitors outstanding deficiencies until resolution.

h. Ensure subcontractor compliance with safety and health requirements.

i. Maintain a list of hazardous chemicals on site and their material Safety Data Sheets (SDS).

j. Maintain a weekly list of high hazard activities involving energy, equipment, excavation, entry into confined space, and elevation, and be prepared to discuss details during QC Meetings.

k. Provide and keep a record of site safety orientation and indoctrination for Contractor employees, subcontractor employees, and site visitors.

Superintendent, QC Manager, and SSHO are subject to dismissal if the above duties are not being effectively carried out. If Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the

above duties are again being effectively carried out.

1.6.3 Meetings

1.6.3.1 Preconstruction Conference

a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction conference. This includes the project superintendent, Site Safety and Occupational Health officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).

b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the contract. This list of proposed AHAs will be reviewed at the conference and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays.

c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin work until an APP is established that is acceptable to the Contracting Officer.

1.6.3.2 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors on the project location. The SSHO, supervisors, foremen, or CDSOs must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.7 SITE SPECIFIC SAFETY PLAN/ACCIDENT PREVENTION PLAN (SSSP/APP)

A qualified person must prepare the written site-specific SSSP/APP. Prepare the SSSP/APP in accordance with the format and requirements of EM 385-1-1, Appendix A, and as supplemented herein. Cover all paragraph and subparagraph elements in EM 385-1-1, Appendix A. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling authority" for all work site safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of

the contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed by an officer of the firm (Prime Contractor senior person), the individual preparing the APP, the on-site superintendent, the designated SSHO, the Contractor Quality Control Manager, and any designated Certified Safety Professional (CSP) or Certified Health Physicist (CIH). The SSHO must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Submit the APP to the Contracting Officer 15 calendar days prior to the date of the preconstruction conference for acceptance. Work cannot proceed without an accepted SSSP/APP. Once reviewed and accepted by the Contracting Officer, the SSSP/APP and attachments will be enforced as part of the contract. Disregarding the provisions of this contract or the accepted SSSP/APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and amend the SSSP/APP, as necessary, throughout the life of the contract. Changes to the accepted SSSP/APP must be made with the knowledge and concurrence of the Contracting Officer, project superintendent, SSHO and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original SSSP/APP as they are discovered. Should any severe hazard exposure (i.e. imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ASSP A10.34), and the environment.

1.7.1 Names and Qualifications

Provide plans in accordance with the requirements outlined in Appendix A of EM 385-1-1, including the following:

a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.

b. Qualifications of competent and of qualified persons. As a minimum, designate and submit qualifications of competent persons for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance.

1.7.2 Plans

Provide plans in the SSSP/APP in accordance with the requirements outlined in Appendix A of EM 385-1-1, including the following:

1.7.2.1 Confined Space Entry Plan

Develop a confined or enclosed space entry plan in accordance with EM 385-1-1, applicable OSHA standards 29 CFR 1910, 29 CFR 1915, and 29 CFR 1926, OSHA Directive CPL 2.100, and any other federal, state and local regulatory requirements identified in this contract. Identify the qualified person's name and qualifications, training, and experience. Delineate the qualified person's authority to direct work stoppage in the event of hazardous conditions. Include procedure for rescue by contractor personnel and the coordination with emergency responders. (If there is no confined space work, include a statement that no confined space work exists and none will be created.)

1.7.2.2 Standard Lift Plan (SLP)

Plan lifts to avoid situations where the operator cannot maintain safe control of the lift. Prepare a written SLP in accordance with EM 385-1-1, Section 16.A.03, using Form 16-2 for every lift or series of lifts (if duty cycle or routine lifts are being performed). The SLP must be developed, reviewed and accepted by all personnel involved in the lift in conjunction with the associated AHA. Signature on the AHA constitutes acceptance of the plan. Maintain the SLP on the LHE for the current lift(s) being made. Maintain historical SLPs for a minimum of 3 months.

1.7.2.3 Safety and Health Plan

Identify the safety and health aspects, and prepare and submit a safety and Health Plan.

1.8 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.9 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment. Government has no responsibility to provide emergency medical treatment.

1.10 NOTIFICATIONS and REPORTS

1.10.1 Mishap Notification

Notify the Contracting Officer as soon as practical, but no more than twenty-four hours, after any mishaps, including recordable accidents, incidents, and near misses, as defined in EM 385-1-1 Appendix Q, any report of injury, illness, load handling equipment (LHE) or rigging mishaps, or any property damage. The Contractor is responsible for obtaining appropriate medical and emergency assistance and for notifying fire, law enforcement, and regulatory agencies. Immediate reporting is required for electrical mishaps, to include Arc Flash; shock; uncontrolled release of hazardous energy (includes electrical and non-electrical); load handling equipment or rigging; fall from height (any level other than same surface); and underwater diving. These mishaps must be investigated in depth to identify all causes and to recommend hazard control measures.

Within notification include Contractor name; contract title; type of contract; name of activity, installation or location where accident

occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any mishap.

CONFINED SPACE ENTRY REQUIREMENTS.

Confined space entry must comply with Section 34 of EM 385-1-1, OSHA 29 CFR 1926, OSHA 29 CFR 1910, OSHA 29 CFR 1910.146, and OSHA Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory PPE includes:

a. Hard Hat

b. Long Pants

c. Appropriate Safety Shoes

d. Appropriate Class Reflective Vests

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. An employee check-in/check-out communication procedure must be developed to ensure employee safety.

3.1.2 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear

density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. Notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

### 3.1.3 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e. 29 CFR Part 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during construction operations, stop that portion of work and notify the Contracting Officer immediately. Within 14 calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification pursuant to FAR 52.243-4, "Changes" and FAR 52.236-2, "Differing Site Conditions."

## 3.2 PRE-OUTAGE COORDINATION MEETING

Apply for utility outages at least 3 days in advance. As a minimum, the request must include the location of the outage, utilities being affected, duration of outage and any necessary sketches. Special requirements for electrical outage requests are contained elsewhere in this specification section. Once approved, and prior to beginning work on the utility system requiring shut down, attend a pre-outage coordination meeting with the Contracting Officer to review the scope of work and the lock-out/tag-out procedures for worker protection. No work will be performed on energized electrical circuits unless proof is provided that no other means exist.

## 3.3 WORK PLATFORMS

### 3.3.1 Scaffolding

Provide employees with a safe means of access to the work area on the scaffold. Climbing of any scaffold braces or supports not specifically designed for access is prohibited. Comply with the following requirements:

a. Scaffold platforms greater than 20 feet in height must be accessed by use of a scaffold stair system.

b. Ladders commonly provided by scaffold system manufacturers are prohibited for accessing scaffold platforms greater than 20 feet maximum in height.

c. An adequate gate is required.

d. Employees performing scaffold erection and dismantling must be qualified.

e. Scaffold must be capable of supporting at least four times the maximum intended load or without appropriate fall protection as delineated in

the accepted fall protection and prevention plan.

f. Stationary scaffolds must be attached to structural building components to safeguard against tipping forward or backward.

g. Special care must be given to ensure scaffold systems are not overloaded.

h. Side brackets used to extend scaffold platforms on self-supported scaffold systems for the storage of material are prohibited. The first tie-in must be at the height equal to 4 times the width of the smallest dimension of the scaffold base.

i. Scaffolding other than suspended types must bear on base plates upon wood mudsills (2 in x 10 in x 8 in minimum) or other adequate firm foundation.

j. Scaffold or work platform erectors must have fall protection during the erection and dismantling of scaffolding or work platforms that are more than six feet.

k. Delineate fall protection requirements when working above six feet or above dangerous operations in the Fall Protection and Prevention (FP&P) Plan and Activity Hazard Analysis (AHA) for the phase of work.

## 3.4 EQUIPMENT

### 3.4.1 Material Handling Equipment (MHE)

a. Material handling equipment such as forklifts must not be modified with work platform attachments for supporting employees unless specifically delineated in the manufacturer's printed operating instructions. Material handling equipment fitted with personnel work platform attachments are prohibited from traveling or positioning while personnel are working on the platform.

b. The use of hooks on equipment for lifting of material must be in accordance with manufacturer's printed instructions. Material Handling Equipment Operators must be trained in accordance with OSHA 29 CFR 1910, Subpart N.

c. Operators of forklifts or power industrial trucks must be licensed in accordance with OSHA.

### 3.4.2 Machinery and Mechanized Equipment

a. Proof of qualifications for operator must be kept on the project site for review.

b. Manufacture specifications or owner's manual for the equipment must be on-site and reviewed for additional safety precautions or requirements that are sometimes not identified by OSHA or USACE EM 385-1-1. Incorporate such additional safety precautions or requirements into the AHAs.

## 3.5 EXCAVATIONS

Soil classification must be performed by a competent person in accordance with 29 CFR 1926 and EM 385-1-1.

3.5.1 Utility Locations

Provide a third party, independent, private utility locating company to positively identify underground utilities in the work area in addition to any station locating service and coordinated with the station utility department.

3.5.2 Utility Location Verification

Physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within three feet of the underground system.

3.5.3 Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces

Utilities located within and under concrete slabs or pier structures, bridges, parking areas, and the like, are extremely difficult to identify. Whenever contract work involves chipping, saw cutting, or core drilling through concrete, bituminous asphalt or other impervious surfaces, the existing utility location must be coordinated with station utility departments in addition to location and depth verification by a third party, independent, private locating company. The third party, independent, private locating company must locate utility depth by use of Ground Penetrating Radar (GPR), X-ray, bore scope, or ultrasound prior to the start of demolition and construction. Outages to isolate utility systems must be used in circumstances where utilities are unable to be positively identified. The use of historical drawings does not alleviate the Contractor from meeting this requirement.

3.6 ELECTRICAL

Perform electrical work in accordance with EM 385-1-1, Appendix A, Sections 11 and 12.

3.6.1 Conduct of Electrical Work

As delineated in EM 385-1-1, electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases, obtain an energized work permit from the Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Attach temporary grounds in accordance with ASTM F855 and IEEE 1048. Perform all high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by NFPA 70, high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety shoes,

insulating gloves and electrical arc flash protection for personnel as required by NFPA 70E. Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and 29 CFR 1910.147.

### 3.6.2 Qualifications

Electrical work must be performed by QP personnel with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with State, Local requirements applicable to where work is being performed.

### 3.6.3 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

Check grounding circuits to ensure that the circuit between the ground and a grounded power conductor has a resistance low enough to permit sufficient current flow to allow the fuse or circuit breaker to interrupt the current.

### 3.6.4 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system and signed by the electrical CP or QP.

-- End of Section --

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SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS
**05/24**

PART 1 GENERAL

1.1 REFERENCES

Various publications are referenced in other sections of the specifications to establish requirements for the work. These references are identified in each section by document number, date and title. The document number used in the citation is the number assigned by the standards producing organization (e.g., ASTM B564 Standard Specification for Nickel Alloy Forgings). However, when the standards producing organization has not assigned a number to a document, an identifying number has been assigned for reference purposes.

1.2 ORDERING INFORMATION

The addresses of the standards publishing organizations whose documents are referenced in other sections of these specifications are listed below, and if the source of the publications is different from the address of the sponsoring organization, that information is also provided.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)
555 12th Street NW, Suite 1000
Washington, DC 20004
Ph: 202-624-5800
Fax: 202-624-5806
E-Mail: info@aashto.org
Internet: https://www.transportation.org/

AMERICAN CONCRETE INSTITUTE (ACI)
38800 Country Club Drive
Farmington Hills, MI 48331-3439
Ph: 248-848-3800
Fax: 248-848-3701
Internet: https://www.concrete.org/

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)
130 East Randolph, Suite 2000
Chicago, IL 60601
Ph: 312-670-2400
Fax: 312-670-5403
Steel Solutions Center: 866-275-2472
E-mail: solutions@aisc.org
Internet: https://www.aisc.org/

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Two Park Avenue
New York, NY 10016-5990
Ph: 800-843-2763
Fax: 973-882-1717
E-mail: customercare@asme.org
Internet: https://www.asme.org/

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)
520 N. Northwest Highway
Park Ridge, IL 60068
Ph: 847-699-2929
E-mail: customerservice@assp.org
Internet: https://www.assp.org/

AMERICAN WATER WORKS ASSOCIATION (AWWA)
6666 W. Quincy Avenue
Denver, CO 80235 USA
Ph: 303-794-7711 or 800-926-7337
Fax: 303-347-0804
Internet: https://www.awwa.org/

AMERICAN WELDING SOCIETY (AWS)
8669 NW 36 Street, #130
Miami, FL 33166-6672
Ph: 800-443-9353
Email: customercare@aws.org
Internet: https://www.aws.org/

ASTM INTERNATIONAL (ASTM)
100 Barr Harbor Drive, P.O. Box C700
West Conshohocken, PA 19428-2959
Ph: 610-832-9500
Fax: 610-832-9555
E-mail: service@astm.org
Internet: https://www.astm.org/

EUROPEAN UNION (EU)
European Commission
Rue de la Loi 200
1000 Bruxelle
Belgium
Ph: +32 2 299 96 96
Internet: https://commission.europa.eu/index_en

ILLUMINATING ENGINEERING SOCIETY (IES)
120 Wall Street, Floor 17
New York, NY 10005-4001
Ph: 212-248-5000
Fax: 212-248-5018
E-mail: membership@ies.org
Internet: https://www.ies.org/

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)
445 and 501 Hoes Lane
Piscataway, NJ 08854-4141
Ph: 732-981-0060 or 800-701-4333
Fax: 732-981-9667
E-mail: onlinesupport@ieee.org
Internet: https://www.ieee.org/

INTERNATIONAL CODE COUNCIL (ICC)
200 Massachusetts Ave, NW Suite 250
Washington, DC 20001
Ph: 800-786-4452 or 888-422-7233
Fax: 202-783-2348

E-mail: order@iccsafe.org
Internet: https://www.iccsafe.org/

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)
3050 Old Centre Ave. Suite 101
Portage, MI 49024
Ph: 269-488-6382 or 1-888-300-6382
Fax: 269-488-6383
Internet: https://www.netaworld.org/

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)
127 Park Street, NE
Vienna, VA 22180-4602
Ph: 703-281-6613
E-mail: info@msshq.org
Internet: http://msshq.org

METAL BUILDING MANUFACTURERS ASSOCIATION (MBMA)
1300 Sumner Avenue
Cleveland, OH 44115-2851
Ph: 216-241-7333
Fax: 216-241-0105
Internet: https://www.mbma.com/

NATIONAL ELECTRICAL CONTRACTORS ASSOCIATION (NECA)
1201 Pennsylvania Ave. NW, Suite 1200
Washington, DC 20004
Ph: 202-991-6300
Fax: 202-217-4171
Internet: https://www.necanet.org/

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)
1300 North 17th Street, Suite 900
Arlington, VA 22209
Ph: 703-841-3200
Email: communications@nema.org
Internet: https://www.nema.org

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
1 Batterymarch Park
Quincy, MA 02169-7471
Ph: 800-344-3555
Fax: 800-593-6372
Internet: https://www.nfpa.org

NSF INTERNATIONAL (NSF)
789 North Dixboro Road
P.O. Box 130140
Ann Arbor, MI 48105
Ph: 734-769-8010 or 800-NSF-MARK
Fax: 734-769-0109
E-mail: info@nsf.org
Internet: http://www.nsf.org

PLASTIC PIPE AND FITTINGS ASSOCIATION (PPFA)
800 Roosevelt Road
Building C, Suite 312
Glen Ellyn, IL 60137

Ph: 630-858-6540
Fax: 630-790-3095
Internet: https://www.ppfahome.org/

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)
400 Commonwealth Drive
Warrendale, PA 15096
Ph: 877-606-7323 or 724-776-4841
Fax: 724-776-0790
E-mail: customerservice@sae.org
Internet: https://www.sae.org/

U.S. ARMY CORPS OF ENGINEERS (USACE)
CRD-C DOCUMENTS available on Internet: https://www.wbdg.org/army
THE CAD/BIM TECHNOLOGY CENTER FOR FACILITIES, INFRASTRUCTURE, AND ENVIRONMENT (USACE)on Internet:
https://cadbimcenter.erdc.dren.mil/
Order Other Documents from:
Official Publications of the Headquarters, USACE
E-mail: hqpublications@usace.army.mil
Internet: http://www.publications.usace.army.mil/
or
https://www.hnc.usace.army.mil/Missions/Engineering-Directorate/TECHINFO/

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)
1200 Pennsylvania Avenue, N.W.
Washington, DC 20004
Ph: 202-564-4700
Internet: https://www.epa.gov
--- Some EPA documents are available only from:
National Technical Information Service (NTIS)
5301 Shawnee Road
Alexandria, VA 22312
Ph: 703-605-6060 or 1-800-363-2068
Fax: 703-605-6880
TDD: 703-487-4639
E-mail: info@ntis.gov
Internet: https://www.ntis.gov/

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)
8601 Adelphi Road
College Park, MD 20740-6001
Ph: 866-272-6272
Internet: https://www.archives.gov/
Order documents from:
Superintendent of Documents
U.S. Government Publishing Office (GPO)
732 N. Capitol Street, NW
Washington, DC 20401
Ph: 202-512-1800 or 866-512-1800
Bookstore: 202-512-0132
Internet: https://www.gpo.gov/

UL SOLUTIONS (UL)
333 Pfingsten Road
Northbrook, IL 60062
Ph: 877-854-3577 or 360-817-5500
E-mail: CustomerExperienceCenter@ul.com

Internet: https://www.ul.com/
UL Directories available through IHS at https://accuristech.com/

PART 2 PRODUCTS

Not used

PART 3 EXECUTION

Not used

-- End of Section --

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SECTION 01 45 00

QUALITY CONTROL
**08/23, CHG 3: 08/25**

PART 1 GENERAL

1.1 PAYMENT

Separate payment will not be made for providing and maintaining an effective Quality Control program. Include all associated costs in the applicable Bid Pricing Schedule item.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Contractor Quality Control (CQC) Plan; G

SD-06 Test Reports

Test Reports

Soil density tests, concrete tests, welding/inspection reports when required, galvanizing or coating certifications, and other tests required by the technical sections.

SD-07 Certificates

QC Manager

Qualification statement; testing laboratory accreditation; welder qualifications; material and coating certifications; Completion Certification

SD-11 Closeout Submittals

Testing And Deficiency Records

Final testing log, punch lists, and record drawings.

1.3 GENERAL REQUIREMENTS

Establish and maintain a Quality Control program that verifies materials, equipment, workmanship, fabrication, installation, and testing comply with the Contract Documents. Quality Control is the Contractor's responsibility and does not replace Government quality assurance or Contracting Officer acceptance.

The QC program may be scaled to the size and complexity of this project but must provide qualified supervision, inspections, documentation, testing, correction of deficiencies, and current record drawings.

## 1.4 QUALITY CONTROL (QC) PROGRAM REQUIREMENTS

Establish and maintain a QC program as described in this section.

### 1.4.1 Meetings

Attend one preconstruction QC coordination meeting before starting field work. Review the QC Plan, definable features of work, submittals, testing, reporting, safety coordination, project schedule, and authority of the QC Manager. Hold additional QC meetings when requested by the Contracting Officer or when significant quality problems require coordination.

### 1.4.2 Contractor Quality Control (CQC) Plan

Submit a concise project-specific QC Plan at least 10 calendar days before field construction. Do not begin affected work until the plan is accepted. Include:

QC organization and names, qualifications, duties, authority, and contact information for the QC Manager and alternate.

A list of subcontractors, testing agencies, fabricators, and qualified inspectors.

Definable features of work and the planned preparatory, initial, and follow-up control activities.

Submittal, testing, deficiency/rework, and record-drawing procedures.

Forms proposed for daily reports, control-phase checklists, testing log, and deficiency tracking.

Procedures for receiving, storing, protecting, and verifying materials.

### 1.4.3 Acceptance of the Quality Control (QC) and Design Quality Control (DQC) Plan

The Contracting Officer's acceptance of the Contractor QC Plan, or interim plan applicable to the particular feature of work to be started, and Design Quality Control Plan is required prior to the start of design and construction. Work outside of the accepted interim plan will not be permitted to begin until acceptance of a CQC Plan or another interim plan containing the additional work. The Contracting Officer reserves the right to require changes in the QC and DQC Plan and operations as necessary, including removal or addition of personnel, to ensure the specified quality of work. The Contracting Officer reserves the right to interview any member of the QC and DQC organization at any time to verify the submitted qualifications. All QC and DQC organization personnel are subject to acceptance by the Contracting Officer. The Contracting Officer may require the removal of any individual for non-compliance with quality requirements specified in the Contract.

### 1.4.4 Preliminary Construction Work Authorized Prior to Acceptance

The only construction work that is authorized to proceed prior to the acceptance of the QC Plan is mobilization of storage and office trailers, temporary utilities, and surveying with specific prior approval of the Contracting Officer.

### 1.4.5 Notification of Changes

Notify the Contracting Officer, in writing, of any proposed changes in the QC Plan or changes to the QC organization personnel. Proposed changes are subject to acceptance by the Contracting Officer.

## 1.5 QUALITY CONTROL (QC) ORGANIZATION

Designate one QC Manager with authority to stop nonconforming work, reject defective material, direct corrective action, and coordinate testing and inspections. For this small project, the QC Manager may also serve as Project Superintendent or Site Safety and Health Officer, provided the individual is present during construction activities and can perform each duty without conflict.

The QC Manager must have at least 3 years of construction experience, including experience with cast-in-place concrete and welded or fabricated steel work, and must demonstrate successful quality-control or supervisory responsibility on at least one comparable project. Submit a resume and references.

### 1.5.1 Alternate Quality Control (QC) Manager

Identify an alternate qualified to act during the QC Manager's absence. The alternate may be a superintendent or foreman meeting the experience requirements and must have authority to stop or reject nonconforming work

### 1.5.2 Testing and Inspection Personnel

Use independent testing agencies meeting the applicable ASTM qualification standard for soil and concrete testing. Use qualified personnel for welding inspection when such inspection is required. Testing laboratories and inspectors must not be owned or controlled by a subcontractor whose work they inspect unless approved in writing by the Contracting Officer.

## 1.6 SUBMITTAL AND DELIVERABLES REVIEW AND APPROVAL

The QC Manager must verify each submittal is complete, coordinated, and compliant before forwarding it to the Government. Identify deviations clearly. Do not fabricate or install work requiring an approved submittal until the required action has been received.

## 1.7 THREE PHASES OF CONTROL

Apply the three phases of control to each definable feature of work. At a minimum, use separate control phases for earthwork and post excavations; reinforcing and cast-in-place concrete; steel pipe posts, rails, gates, welding, and erection; galvanizing or field-applied protective coatings; gate hardware; and water piping or livestock-watering components when included.

### 1.7.1 Preparatory Phase

Before starting each definable feature, review applicable drawings, specifications, submittals, qualifications, materials, testing requirements, tolerances, safety provisions, and prerequisite work. Inspect delivered materials and the work area. Resolve conflicts before work begins. Document the meeting on a preparatory-phase checklist.

1.7.2 Initial Phase

At the start of each definable feature, inspect the first representative portion of work. Establish workmanship, dimensions, alignment, spacing, weld quality, coating repair, concrete placement, and other acceptance standards. Correct deficiencies before production continues. Document the inspection on an initial-phase checklist.

1.7.3 Follow-Up Phase

Inspect ongoing work as needed to maintain the established quality. Verify tests are performed, deficiencies are corrected, and concealed work is acceptable before covering or encasing it. Record follow-up inspections in the daily CQC report.

1.7.4 Additional Phases

Repeat preparatory and initial phases when work quality is unacceptable, personnel or methods change, work resumes after a substantial interruption, or the Contracting Officer directs.

1.7.5 Deficiency/Rework Items List

Maintain a current list of deficient or nonconforming work showing description, location, responsible party, date identified, corrective action, and date accepted as corrected. Do not conceal deficient work. Submit the updated list with progress-payment documentation and at closeout.

1.8 TESTING

Prepare a Testing Plan and Log listing each required test, specification reference, frequency, responsible agency, planned date, actual date, result, and report-submittal date. Coordinate the final frequencies with the technical specifications.

As applicable to the Contract Work, include soil density testing for compacted fill; concrete sampling, slump, air content when specified, temperature, and compressive-strength cylinders; inspection of reinforcing and embedded steel before concrete placement; verification of welder qualifications and visual weld inspection; and coating or galvanizing thickness/certification requirements.

Provide the Contracting Officer at least 2 working days' notice before scheduled field tests when practicable. Submit signed test reports within 5 working days after testing, or within 2 working days after laboratory results become available. Immediately report failing results and take corrective action.

Test reports must identify the project, date, location, specification requirement, procedure, actual result, acceptance criteria, pass/fail status, testing personnel, and laboratory.

1.9 COMPLETION INSPECTIONS

1.9.1 Contractor Punch-Out Inspection

Before requesting Government inspection, the QC Manager must inspect all

work and prepare a punch list. Correct the listed items, update record drawings and testing records, operate gates and hardware, verify drainage and final grading, and confirm coatings and repaired galvanized surfaces are complete.

1.9.2 Pre-Final Inspection

Delete the fixed 14-calendar-day final inspection notice requirement.

Notify the Contracting Officer at least 7 calendar days before the requested pre-final inspection. Correct pre-final deficiencies before requesting final acceptance. The QC Manager and Project Superintendent must attend unless excused by the Contracting Officer.

1.10 QUALITY CONTROL (QC) CERTIFICATIONS

Include the following certification in each CQC report: 'On behalf of the Contractor, I certify that this report is complete and correct and that the materials, equipment, and work described comply with the Contract Documents except as specifically noted.' At completion, the QC Manager must certify that the work has been completed, inspected, and tested and is in compliance with the Contract Documents except for items specifically accepted by the Contracting Officer.

1.11 DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER

Prepare one CQC report for each day that onsite or significant off-site Contract work is performed. No report is required for days when no work occurs unless a significant event, test result, delivery, deficiency, or Government direction must be documented. Submit reports electronically by the end of the next working day.

Each report must include, as applicable: date and weather; personnel and subcontractors onsite; work performed and location; materials received; equipment used; preparatory, initial, and follow-up inspections; tests and results; submittals reviewed; deficiencies and corrective actions; instructions or changes; photographs of concealed work and significant progress; safety issues affecting quality; and the QC certification.

1.11.1 Testing and Deficiency Records

Maintain current electronic copies of the Testing Plan and Log, test reports, control-phase checklists, deficiency/rework list, approved submittals, material certificates, meeting records, photographs, and punch lists. Make records available to the Contracting Officer upon request and submit final records at closeout.

1.11.2 Record Drawings

Keep record drawings current and show field deviations, post and foundation locations, underground piping when included, and approved changes.

1.12 NOTIFICATION ON NON-COMPLIANCE

The Contracting Officer will notify the Contractor of any detected non-compliance with the Contract. Take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, is deemed sufficient for the purpose of notification. If

the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of a claim for extension of time for excess costs or damages by the Contractor.

1.13 DELIVERY, STORAGE, AND HANDLING

Designate receiving/storage areas for incoming material to be delivered according to installation schedule and to be placed convenient to work area in order to minimize waste due to excessive materials handling and misapplication. Store and handle materials in a manner as to prevent loss from weather and other damage. Keep materials, products, and accessories covered and off the ground, and store in a dry, secure area. Prevent contact with material that may cause corrosion, discoloration, or staining. Protect all materials and installations from damage by the activities of other trades.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 50 00

PROJECT CLOSEOUT
**01/20**

PART 1 GENERAL

1.1 AS-BUILT DRAWINGS

The Contractor shall furnish red line as built drawings to the contracting officer that indicate all variations in locations, construction, equipment, etc. from that shown on the original drawings.

1.2 SUBSTANTIAL COMPLETION AND COMPLETION OF PUNCH LIST CORRECTIONS

Set date and notify Contracting Officer as to time construction will be ready for preliminary punch list and, upon completion of that list, set time and date for inspection of punch list corrections. The failure to include any items on such list does not alter the responsibility of the Contractor to complete all Work in accordance with the Contract Documents. When the Contracting Officer on the basis of an inspection, determines that the Work or designated portion thereof is sufficiently complete, in accordance with the Contract Documents, so that the Owner or separate contractors can occupy or utilize the Work or designated portion thereof for the use for which it is intended, he will prepare a Certificate of Substantial Completion which shall establish the Date of Substantial Completion, shall state the responsibilities of the Owner and the Contractor for security, maintenance, heat, utilities, damage to the Work, and insurance, and shall fix the time within which the Contractor shall complete the items listed therein or attached thereto. Warranties required by the Contract Documents shall commence on the Date of Substantial Completion of the Work or designated portion thereof unless otherwise provided in the Certificate of Substantial Completion. The Certificate of Substantial Completion shall be submitted to the Owner and the Contractor for their written acceptance of the responsibilities assigned to them in such Certificate.

1.3 FINAL INSPECTION

The Contractor shall notify the Contracting Officer and Owner in writing 10 days prior to the date that the project will be ready for final inspection. An inspection of the project will be made by the Contracting Officer and Contractor at a mutually agreed upon time. If the project is complete, in the opinion of the Contracting Officer, the Contracting Officer shall issue a certificate as set forth hereinafter. Minor unfinished items of work may be excluded from the certificate. The Contracting Officer and Contractor shall mutually agree on a date for the completion of these items of work. Should the Contractor fail to complete the work within the agreed time, the Owner shall have the option of having the unfinished work completed.

1.4 RELEASE OF LIENS AND CERTIFICATION ALL BILLS ARE PAID

The Contractor, before final acceptance is given by the Contracting Officer, shall furnish for himself and all sub contractors, in triplicate, duly certified by a notary, a full release of liens, damages, and a guarantee that all materials and labor are paid in accordance with the Contractual Documents.

## 1.5 GUARANTEE

The Contractor, before final acceptance by the Contracting Officer, shall furnish the Owner, in triplicate, a written notarized guarantee. All Contractors shall guarantee their work for a period of one year from the date of Substantial Completion as stated by the Contracting Officer. Neither the final Certificate for Payment, nor any provision in the Contract Documents shall relieve the contractors of their responsibility for negligence or defective materials or workmanship within the extent and period covered by law; and upon written notice he shall remedy any defects due thereto, and pay all expenses for any damages to other work resulting therefrom. Emergency defective workmanship and defective materials within the year's guarantee shall be corrected within a few days after receipt of a written order from the Contracting Officer. Any items that are not considered of an emergency nature by the Contracting Officer may be corrected prior to the expiration of the Guarantee period.

The wording for the Guarantee shall be as follows:

> "We hereby guarantee all work performed by us on the above captioned project to be free from defective materials and workmanship for a period of one year, or for such longer period of time as may be called for in the Contractual Documents for such portions of the work."

## 1.6 OTHER FINAL DOCUMENTS

Deliver to the Owner, through the Contracting Officer, all required bonds, warranties, individual guarantees, etc. that are required by the various following sections of these specifications.

## 1.7 ACCEPTANCE

Upon satisfactory submittal of the above documents the Contracting Officer shall issue to the Owner a certificate recommending final acceptance of the project. Minor unfinished items of work may be listed on the certificate and excepted from final acceptance. A sum sufficient to cover the cost of completing these unfinished items of work will be withheld by the Owner from the final payment. All other funds including retainage shall be due and payable upon receipt by the Owner of the Contracting Officer's certificate. Upon certification by the Contracting Officer that the items of work have been completed, all monies withheld by the Owner shall be due the Contractor.

# PART 2 PRODUCTS

Not used.

# PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 57 50

TEMPORARY ENVIRONMENTAL CONTROLS
**10/19**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

| | |
|---|---|
| 40 CFR 112 | Oil Pollution Prevention |
| 40 CFR 152 - 186 | Pesticide Programs |
| 40 CFR 261 | Identification and Listing of Hazardous Waste |
| 40 CFR 262 | Standards Applicable to Generators of Hazardous Waste |
| 40 CFR 263 | Standards Applicable to Transporters of Hazardous Waste |
| 40 CFR 264 | Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities |
| 40 CFR 403 | General Pretreatment Regulations for Existing and New Sources of Pollution |
| 49 CFR 171 | General Information, Regulations, and Definitions |
| 49 CFR 172 | Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements |

1.2 DEFINITIONS

1.2.1 Sediment

Sediment is soil and other debris that have eroded and have been transported by runoff water or wind.

1.2.2 Solid Waste

Solid waste is a solid, liquid, semi-solid or contained gaseous waste. A solid waste can be a hazardous waste, non-hazardous waste, or non-Resource Conservation and Recovery Act (RCRA) regulated waste. Types of solid waste typically generated at construction sites may include:

1.2.2.1 Rubbish

Combustible and noncombustible wastes such as paper, boxes, glass, crockery, metal, lumber, cans, and bones.

1.2.2.2 Debris

Combustible and noncombustible wastes such as ashes and waste materials resulting from construction or maintenance and repair work, leaves, and tree trimmings.

1.2.2.3 Chemical Waste

This includes salts, acids, alkalies, herbicides, pesticides, and organic chemicals.

1.2.3 Sanitary Wastes

1.2.3.1 Sewage

Waste characterized as domestic sanitary sewage.

1.2.3.2 Garbage

Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.

1.2.4 Hazardous Waste

Hazardous substances as defined in 40 CFR 261 or as defined by applicable state and local regulations.

1.2.5 Oily Waste

Petroleum products and bituminous materials.

1.2.6 Landscape Features

Trees, plants, shrubs and ground covers.

1.2.7 Hazardous Substances

As defined in EPA PL 96 510.

1.2.8 Hazardous Materials

As defined in DOT Regulation 49 CFR 171 and listed in 49 CFR 172.

1.3 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain, during the life of the contract, environmental protection as defined. Plan for and provide environmental protective measures to control pollution that develops during normal construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Comply with Federal, state, and local regulations pertaining to the environment, including water, air, solid waste, hazardous waste and substances, oily substances and noise pollution.

PART 2 PRODUCTS

2.1 NATURAL RESOURCES

Preserve the natural resources within the project boundaries and outside the limits of permanent work. Restore to an equivalent or improved condition upon completion of work. Confine construction activities to within the limits of the work indicated or specified.

2.1.1 Landscape Features

Do not remove, cut, deface, injure, or destroy existing landscape features without the Contracting Officer's permission. Do not fasten or attach ropes, cables, or guys to existing nearby trees for anchorages, unless authorized by the Contracting Officer. Where such use of attach ropes, cables, or guys is authorized, the Contractor shall be responsible for any resultant damage.

2.1.1.1 Protection

Protect existing landscape features which are to remain and which may be injured, bruised, defaced, or otherwise damaged by construction operations. By approved excavation, remove trees with 30 percent or more of their root systems destroyed.

2.1.1.2 Replacement

Remove existing landscape features scarred or damaged by equipment operations, and replace with equivalent, undamaged features. Obtain the Contracting Officer's approval before replacement.

2.1.2 Water Resources

2.1.2.1 Oily and Hazardous Substances

Prevent oily or other hazardous substances from entering the ground, drainage areas, or local bodies of water. Take precautions to ensure that no oil or other hazardous material is released to the water, land, sanitary sewer system, or storm sewer system. Environmental requirements for the prevention of oil spills are contained in 40 CFR 112. For oil and hazardous substance spills which may be large enough to violate Federal, State or Local Regulations, the Contracting Officer shall be verbally notified immediately. Clean up any spills of oil or hazardous substances which result from the Contractor's operations. If the Contractor can not clean up any spill in a timely manner, the Government will perform the cleanup at the Contractor's expense.

2.1.3 Fish and Wildlife Resources

Do not disturb fish and wildlife. Do not alter water flows or otherwise disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as indicated or specified.

2.2 HISTORICAL AND ARCHEOLOGICAL RESOURCES

Carefully preserve and report immediately to the Contracting Officer items having possible historical or archeological interest which are discovered in the course of work. Protect monuments, markers, and works of art.

## 2.3 EROSION AND SEDIMENT CONTROL

### 2.3.1 Burnoff

Burnoff of the ground cover is prohibited.

### 2.3.2 Protection of Erodible Soils

Immediately finish the earthwork brought to a final grade, as indicated or specified. Immediately protect the side slopes and back slopes upon completion of rough grading. Plan and conduct earthwork to minimize the duration of exposure of unprotected soils.

### 2.3.3 Temporary Protection of Erodible Soils

Provide the following methods to prevent erosion and control sedimentation.

#### 2.3.3.1 Mechanical Retardation and Control of Runoff

Mechanically retard and control the rate of runoff from the construction site. Provide diversion ditches, benches, and berms to retard and divert runoff to protected drainage courses.

#### 2.3.3.2 Seeding

Provide seeding where ground is disturbed. Include topsoil or nutriment during the seeding operation necessary to establish or re-establish a suitable stand of grass.

## 2.4 SOLID AND SANITARY WASTES

Pick up solid wastes, and place in containers which are regularly emptied. Prevent contamination of the site and other areas when handling and disposing wastes. On completion, leave the areas clean. Control and dispose of wastes.

### 2.4.1 Disposal of Solid Wastes

Dispose of solid wastes in accordance with the requirements specified.

#### 2.4.1.1 Sub Removal From Government Property

Remove solid wastes from Government property.

### 2.4.2 Garbage Disposal

Place garbage in approved containers, and move to a pickup point or disposal area, where directed.

### 2.4.3 Sewage, Odor, and Pest Control

Dispose of sewage through connection to a municipal or district sanitary sewage system, where such a system is not available, use chemical toilets. Empty wastes into a municipal, district, or station sanitary sewage system. Provide pest control and elimination of odors. No substances shall be disposed of which will interfere with treatment plant operations in accordance with 40 CFR 403.

## 2.5 HAZARDOUS WASTES

### 2.5.1 General

Handle generated hazardous wastes in accordance with 40 CFR 262. Store hazardous waste in approved containers in accordance with 40 CFR 152 - 186. Label to identify the type of waste and the date the container was filled. Remove the containers from the project site, and store and dispose of hazardous waste in accordance with 40 CFR 263 and 40 CFR 264. For hazardous wastes spills, verbally notify the Contracting Officer immediately.

### 2.5.2 Petroleum Products

Conduct the fueling and lubricating of equipment and motor vehicles to protect against spills and evaporation. Dispose of lubricants and excess oils.

## 2.6 DUST

Keep dust down at all times, including during nonworking periods. Sprinkle or treat, with dust suppressants, the soil at the site, haul roads, and other areas disturbed by operations. Dry power brooming will not be permitted. Instead, use vacuuming, wet mopping, wet sweeping, or wet power brooming. Air blowing will be permitted only for cleaning nonparticulate debris such as steel reinforcing bars.

## 2.7 ABRASIVE BLASTING

Provide tarpaulin drop cloths and windscreens to enclose abrasive blasting operations to confine and collect dust, abrasive agent, paint chips, and other debris. Dispose of abrasive blasting debris.

## 2.8 NOISE

Make the maximum use of low noise emission products, as certified by the EPA. Blasting or use of explosives will not be permitted without written permission from the Contracting Officer, and then only during the designated times.

## 2.9 FOOD

Preparation, cooking and dispensing of food products will not be permitted.

## 2.10 TEMPORARY CONSTRUCTION

Remove temporary construction facilities including haul roads, work areas, structures, foundations of temporary structures, and stockpiles of excess or waste materials. Grade temporary roads, parking areas, and temporarily used areas to conform with surrounding contours.

# PART 3 EXECUTION

Not used.

-- End of Section --

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SECTION 01 78 23

OPERATION AND MAINTENANCE DATA
**08/15**

PART 1 GENERAL

1.1 SUMMARY

Provide operation and maintenance data for products, equipment, and systems incorporated into the Work. Compile the information into a concise, searchable electronic manual suitable for operation, inspection, maintenance, repair, and replacement of the completed cattle-handling improvements.

Coordinate this section with Section 01 33 00 SUBMITTAL PROCEDURES, Section 01 45 00 QUALITY CONTROL, and the applicable technical specification sections.

1.2 O&M MANUAL FORMAT

Submit one bookmarked, text-searchable PDF file. Organize bookmarks by specification section and equipment or system. Provide native electronic files for schedules, spreadsheets, and record drawings when those files are part of the closeout requirements. Do not submit product literature for materials not installed.

Identify the project title, contract number, Contractor, date, and “Operation and Maintenance Manual” on the cover. Include a table of contents with links or bookmarks to each section.

1.3 O&M MANUAL CONTENT

1.3.1 Project and Contractor Information

Project title, location, contract number, and date of completion.

Names, addresses, telephone numbers, and email addresses for the Contractor, major subcontractors, fabricators, suppliers, and manufacturers.

Emergency and routine service contacts for installed equipment.

1.3.2 Approved Product and Fabrication Data

Include final approved product data, shop drawings, fabrication drawings, certifications, and manufacturer instructions applicable to the installed Work, including:

Concrete mix data, reinforcing information, curing materials, joint materials, and applicable test reports.

Steel pipe rails, posts, gates, blackout panels, gate hardware, anchors, and miscellaneous fabricated-metal items.

Welding procedures, welder qualifications, weld inspection records, and repair documentation.

Hot-dip galvanizing certificates, coating product data,

surface-preparation requirements, touch-up materials, and repair instructions.

Water piping, valves, hose bibbs, float valves, backflow-prevention components, and associated accessories.

Lighting fixtures, controls, 120-volt receptacles, boxes, covers, panels, disconnects, and other installed electrical equipment

1.3.3 Operating and Maintenance Instructions

For each maintainable product or item of equipment, provide only information applicable to the installed model and configuration. Include, as applicable:

Safe operation and applicable hazards.

Normal startup, operation, shutdown, isolation, and emergency procedures.

Inspection frequency, cleaning, lubrication, adjustment, and preventive maintenance.

Troubleshooting, repair, removal, and replacement instructions.

Recommended spare parts, special tools, and sources of replacement components.

Wiring diagrams, control diagrams, and valve or equipment identification

1.3.4 Cattle-Railing and Gate Maintenance

Provide project-specific inspection and maintenance instructions addressing weld cracking, impact damage, bent rails or gates, loose anchors and connections, gate alignment, hinge and latch operation, sharp projections, damaged galvanizing or coatings, corrosion, and approved repair procedures. Identify compatible galvanizing-repair or coating touch-up materials.

1.3.5 Warranties

Provide executed warranties, warranty start and expiration dates, conditions that could void a warranty, and procedures and contacts for obtaining warranty service.

1.3.6 Record Drawings and Utility Information

Provide record drawings showing the installed location and configuration of pens, gates, rails, posts, concrete foundations, slabs, water piping, shutoff valves, hose bibbs, float valves, backflow-prevention components, lighting, receptacles, panels, circuits, underground utilities, and concealed work. Clearly identify field changes and actual routing and dimensions needed for future maintenance.

1.3.7 Test and Inspection Records

Include applicable concrete test reports; reinforcing and embedment inspections; welding and galvanizing/coating certificates; water-system pressure, flushing, disinfection, and backflow test records where required; electrical test records; and final functional checks for gates,

latches, valves, lighting, and receptacles.

1.4 O&M DATA PACKAGES

Provide the O&M data package required by each technical section. Where no package is stated, provide the following minimum data:

1.4.1 Data Package 1 - Materials and Fabricated Items

Cleaning and inspection recommendations.

Maintenance and repair procedures.

Warranty information.

Contractor, fabricator, and supplier contacts.

Replacement materials and touch-up products.

Use Data Package 1 for concrete finishes, fabricated rails and gates, galvanizing/coatings, sealants, and similar non-operating items

1.4.2 Data Package 2 - Operating Equipment

Safety precautions and normal operation.

Preventive maintenance, cleaning, lubrication, and adjustment.

Troubleshooting, repair, removal, and replacement instructions.

Parts identification, spare parts, warranties, and service contacts.

Wiring or control diagrams where applicable.

Use Data Package 2 for float-valve assemblies, backflow-prevention devices, lighting controls, and other maintainable operating equipment unless a technical section requires more extensive data.

1.5 CHANGES TO APPROVED DATA

Submit manufacturer-originated changes or revisions affecting previously approved O&M data. Submit corrections, additions, or revisions requested by the Contracting Officer within 15 calendar days after notification and before final acceptance, whichever occurs first.

1.6 SUBMITTALS

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-10 Operation and Maintenance Data

Operation and Maintenance Manual, Prefinal Submittal; G

Operation and Maintenance Manual, Final Submittal G

SD-11 Closeout Submittals

Record Drawings And Utility Information; G

Training Outline; G

Completion Record; G

PART 2 PRODUCTS

Not Used.

PART 3 EXECUTION

3.1 TRAINING

Before final acceptance, provide field training for Government personnel on installed equipment and maintainable systems. Training must be conducted by personnel familiar with the installed products and must use the approved O&M information.

3.1.1 Training Content

Safe operation of gates, latches, water-system components, lighting controls, and other operating equipment.

Location and operation of water shutoff valves, electrical disconnects, panels, and circuits serving the Work.

Routine inspection and maintenance of rails, gates, anchors, welds, galvanizing/coatings, valves, fixtures, lighting, and receptacles.

Identification of unsafe damage, sharp edges, loose components, corrosion, leaks, and electrical defects requiring removal from service or repair.

Warranty procedures and service contacts.

3.1.2 Training Outline and Completion Record

Submit a concise training outline at least 7 calendar days before training. Coordinate the date and time with the Contracting Officer. Record the date, location, subjects covered, instructor, and names of attendees. Submit the completed attendance record with the Final O&M Manual. Video recording is not required.

3.2 SUBMITTAL SCHEDULING

3.2.1 Prefinal O&M Manual

Submit the complete Prefinal O&M Manual not later than 30 calendar days before final inspection. Include available warranties, record information, approved data, test records, and proposed training materials. Correct Government review comments before the Final submittal.

3.2.2 Final O&M Manual

Submit the Final O&M Manual before final acceptance. Incorporate review comments, final test and inspection records, executed warranties, completed training records, and final record drawings. The final manual must accurately represent the products and conditions installed.

-- End of Section --

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SECTION 02 41 00

DEMOLITION
**08/22**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

| | |
|---|---|
| AASHTO M 145 | (1991; R 2025) Standard Specification for Classification of Soils and Soil-Aggregate Mixtures for Highway Construction Purposes |

ASTM INTERNATIONAL (ASTM)

| | |
|---|---|
| ASTM D2487 | (2017; R 2025) Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System) |

U.S. ARMY CORPS OF ENGINEERS (USACE)

| | |
|---|---|
| EM 385-1-1 | (2024) Safety -- Safety and Occupational Health (SOH) Requirements |

1.2 PROJECT DESCRIPTION

1.2.1 Demolition/Deconstruction Plan

Prepare a Demolition Plan/Deconstruction Plan and submit proposed demolition/deconstruction, and removal procedures for approval before work is started. Include in the plan procedures for coordination with other work in progress, a disconnection schedule of utility services, a detailed description of methods and equipment to be used for each operation and of the sequence of operations. Append tracking forms for all removed materials indicating type, quantities, condition, destination, and end use. Provide procedures for safe conduct of the work in accordance with EM 385-1-1. Plan must be approved by Contracting Officer prior to work beginning.

1.2.2 General Requirements

Do not begin demolition or deconstruction until authorization is received from the Contracting Officer. The work of this section is to be performed in a manner that maximizes the value derived from the salvage and recycling of materials. Remove rubbish and debris from the station daily; do not allow accumulations. The work includes demolition/deconstruction, and removal of resulting rubbish and debris. Remove rubbish and debris from Government property daily, unless otherwise directed. Store materials that cannot be removed daily in areas specified by the Contracting Officer. In the interest of occupational safety and health,

perform the work in accordance with EM 385-1-1, Section 17, Demolition, and other applicable Sections.

## 1.3 ITEMS TO REMAIN IN PLACE

Comply with FAR 52.236-9 to protect existing vegetation, structures, equipment, utilities, and improvements. Coordinate the work of this section with all other work indicated. Construct and maintain shoring, bracing, and supports as required. Ensure that structural elements are not overloaded. Increase structural supports or add new supports as may be required as a result of any cutting, removal, deconstruction, or demolition work performed under this contract. Do not overload structural elements. Provide new supports and reinforcement for existing construction weakened by demolition, deconstruction, or removal work. Repairs, reinforcement, or structural replacement require approval by the Contracting Officer prior to performing such work.

### 1.3.1 Existing Construction Limits and Protection

Do not disturb existing construction beyond the extent indicated or necessary for installation of new construction. Provide temporary shoring and bracing for support of building components to prevent settlement or other movement. Provide protective measures to control accumulation and migration of dust and dirt in all work areas. Remove debris from work areas daily.

### 1.3.2 Utility Service

Maintain existing utilities indicated to stay in service and protect against damage during demolition and deconstruction operations.

### 1.3.3 Facilities

Verify existing conditions, dimensions, connections, utilities, and structural relationships before demolition. Notify the Contracting Officer of concealed or differing conditions that affect the work. Do not proceed in the affected area until direction is received.

Where suspect hazardous materials are encountered, stop work in the immediate area, prevent disturbance, secure the area, and notify the Contracting Officer. Hazardous-material abatement is not included unless specifically identified elsewhere in the Contract Documents.

## 1.4 BURNING

The use of burning at the project site for the disposal of refuse and debris will not be permitted.

## 1.5 AVAILABILITY OF WORK AREAS

Areas in which the work is to be accomplished will be available at contract award.

## 1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Demolition Plan; G

Deconstruction Plan; G

Existing Conditions

SD-07 Certificates

Notification; G

## 1.7 QUALITY ASSURANCE

Submit timely notification of demolition/deconstruction projects to Federal, State, regional, and local authorities in accordance with 40 CFR 61, Subpart M. Notify the Regional Office of the United States Environmental Protection Agency (USEPA) State's environmental protection agency local air pollution control district/agency and the Contracting Officer in writing 10 working days prior to the commencement of work in accordance with 40 CFR 61, Subpart M. Comply with federal, state, and local hauling and disposal regulations. In addition to the requirements of the "Contract Clauses," conform to the safety requirements contained in ASSP A10.6. Comply with the Environmental Protection Agency requirements specified. Use of explosives will not be permitted.

### 1.7.1 Debris Control

Prevent the spread of debris and avoid the creation of a nuisance in the surrounding area. Do not use water if it results in hazardous or objectionable conditions such as, but not limited to, ice, flooding, or pollution.

## 1.8 PROTECTION

### 1.8.1 Traffic Control Signs

a. Where pedestrian and driver safety is endangered in the area of removal work, use traffic barricades with flashing lights. Notify the Contracting Officer prior to beginning such work.

### 1.8.2 Protection of Personnel

Before, during and after the demolition and deconstruction work continuously evaluate the condition of the structure being demolished and deconstructed and take immediate action to protect all personnel working in and around the project site. No area, section, or component of floors, roofs, walls, columns, pilasters, or other structural element will be allowed to be left standing without sufficient bracing, shoring, or lateral support to prevent collapse or failure while workmen remove debris or perform other work in the immediate area.

## 1.9 EXISTING CONDITIONS

Before beginning any demolition or deconstruction work, survey the site and examine the drawings and specifications to determine the extent of the work. Record existing conditions in the presence of the Contracting Officer or the Contracting Officer's Representative showing the condition

of structures and other facilities adjacent to areas of alteration or removal. Photographs or electronic images with a minimum resolution of 3072 x 2304 pixels, capable of a print resolution of 300 dpi, will be acceptable as a record of existing conditions. Include in the record the elevation of the top of foundation walls, finish floor elevations, possible conflicting electrical conduits, plumbing lines, alarms systems, the location and extent of existing cracks and other damage and description of surface conditions that exist prior to starting work. It is the Contractor's responsibility to verify and document all required outages which will be required during the course of work, and to note these outages on the record document. Submit survey results to the Contracting Officer or the Contracting Officer's Representative.

## PART 2 PRODUCTS

### 2.1 FILL MATERIAL

a. Comply with excavating, backfilling, and compacting procedures for soils used as backfill material to fill basements, voids, depressions or excavations resulting from demolition or deconstruction of structures. Provide fill material consisting of waste products from demolition or deconstruction until all waste appropriate for this purpose is consumed.

b. Provide fill material conforming to the definition of satisfactory soil material as defined in ASTM D2487. In addition, fill material must be free from roots and other organic matter, trash, debris, frozen materials, and stones larger than 2 inches in any dimension.

c. Proposed fill material must be sampled and tested by an approved soil testing laboratory, as follows:

| | |
|---|---|
| Soil classification | AASHTO M 145 |
| Moisture-density relations | AASHTO T 180, Method B or D |

## PART 3 EXECUTION

### 3.1 GENERAL DEMOLITION REQUIREMENTS

Perform demolition in a controlled sequence. Use tools and equipment appropriate to the work. Prevent uncontrolled collapse, excessive vibration, flying debris, and damage beyond the demolition limits. Remove debris regularly; do not stockpile debris on retained framing or purlins.

### 3.2 SELECTIVE DEMOLITION

#### 3.2.1 Existing Metal Roof

Remove existing metal roof panels, ridge or edge closures, flashings, fasteners, sealants, and accessories indicated for removal. Dismantle in manageable sections and lower materials to grade. Do not drop panels or debris. Retain and protect the existing rigid steel frame. Remove residual fastener projections and seal or repair abandoned holes only as required for the new roofing system.

### 3.2.2 CMU Enclosure

Remove the CMU block enclosure indicated. Remove masonry in small, controlled sections from the top downward. Do not impose lateral or impact loads on retained framing, slabs, or utilities.

### 3.2.3 Sump Pit

Remove the existing sump-pit top, covers, curbs, reinforcing, and associated construction indicated. Verify that piping or electrical components have been disconnected or are designated to remain before infill. Remove loose material and debris from the pit. Where the pit structure is to remain below grade, provide drainage or relief measures indicated and fill the entire void.

Place approved fill in uniform lifts not exceeding 8 inches loose thickness and compact each lift to at least 95 percent of the maximum dry density determined by the specified laboratory compaction method. Do not use demolished concrete or CMU as infill. Provide compaction testing and complete the surface to the elevation and construction indicated.

### 3.2.4 Railings and Post Foundations

Remove existing railings, gates, posts, attachments, anchors, and concrete post foundations indicated. Extract foundations completely where they conflict with new work. Do not break off posts or foundations and conceal them below new slabs unless specifically shown. Backfill and compact resulting holes in accordance with the sump-pit infill requirements or prepare them for new foundations as indicated.

### 3.2.5 Canopy Structure

Dismantle and remove the existing canopy roof, framing, columns, braces, connections, anchors, and foundations indicated. Sequence removal to maintain stability. Provide temporary bracing as required.

### 3.2.6 Miscellaneous Concrete Slabs

Sawcut concrete at neat limits before removal. Make full-depth cuts at the nearest joint where practical. Remove slabs, pads, foundations, and embedded items indicated without undermining or spalling adjacent concrete to remain. Where removed concrete adjoins retained concrete, leave sound, straight edges suitable for patching or new construction.

## 3.3 STRUCTURAL STEEL TO REMAIN

After roof demolition, clean debris from retained rigid frames and purlins. Notify the Contracting Officer before repairing any member or connection. Do not straighten, weld, drill, cut, reinforce, or coat structural steel without approved repair details. Replace or repair components damaged by demolition as directed and at no additional cost to the Government.

## 3.4 DISPOSITION OF MATERIAL

Except for items identified for Government retention or reuse, removed materials become the Contractor’s property upon approval and must be removed from Government property. Recycle metal and other recoverable material where practical. Dispose of waste legally in accordance with

federal, state, and local requirements. On-site burning and burial are prohibited.

## 3.5 PATCHING AND CLEANUP

Patch surfaces exposed or damaged by demolition to provide a complete, safe, weather-resistant condition compatible with new work. Remove abandoned fasteners, sharp projections, loose masonry, concrete fragments, and debris. Clean the project area and retained framing before final inspection.

-- End of Section --

SECTION 03 30 53

MISCELLANEOUS CAST-IN-PLACE CONCRETE
**05/14**

PART 1 GENERAL

1.1 SUMMARY

Perform all work in accordance with ACI 318.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

| | |
|---|---|
| ACI 117 | (2010; R 2015) Specifications for Tolerances for Concrete Construction and Materials and Commentary |
| ACI 301 | (2020) Specifications for Structural Concrete |
| ACI 304R | (2000; R 2009) Guide for Measuring, Mixing, Transporting, and Placing Concrete |
| ACI 305R | (2020) Guide to Hot Weather Concreting |
| ACI 306R | (2016) Guide to Cold Weather Concreting |
| ACI 318 | (2019; R 2022) Building Code Requirements for Structural Concrete (ACI 318-19) and Commentary (ACI 318R-19) |
| ACI SP-66 | (2004) ACI Detailing Manual |

ASTM INTERNATIONAL (ASTM)

| | |
|---|---|
| ASTM A615/A615M | (2026) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement |
| ASTM A1064/A1064M | (2024) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete |
| ASTM C31/C31M | (2026a) Standard Practice for Making and Curing Concrete Test Specimens in the Field |
| ASTM C33/C33M | (2024a) Standard Specification for Concrete Aggregates |
| ASTM C39/C39M | (2026) Standard Test Method for Compressive Strength of Cylindrical |

| | |
|---|---|
| | Concrete Specimens |
| ASTM C94/C94M | (2026) Standard Specification for Ready-Mixed Concrete |
| ASTM C143/C143M | (2026a) Standard Test Method for Slump of Concrete |
| ASTM C150/C150M | (2024) Standard Specification for Portland Cement |
| ASTM C172/C172M | (2017) Standard Practice for Sampling Freshly Mixed Concrete |
| ASTM C173/C173M | (2026) Standard Test Method for Air Content of Freshly Mixed Concrete by the Volumetric Method |
| ASTM C231/C231M | (2026) Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method |
| ASTM C260/C260M | (2024) Standard Specification for Air-Entraining Admixtures for Concrete |
| ASTM C618 | (2025a) Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete |
| ASTM C685/C685M | (2025a) Standard Specification for Concrete Made by Volumetric Batching and Continuous Mixing |
| ASTM C920 | (2018; R 2024) Standard Specification for Elastomeric Joint Sealants |
| ASTM C989/C989M | (2025) Standard Specification for Slag Cement for Use in Concrete and Mortars |
| ASTM C1064/C1064M | (2023) Standard Test Method for Temperature of Freshly Mixed Hydraulic-Cement Concrete |
| ASTM C1602/C1602M | (2022) Standard Specification for Mixing Water Used in Production of Hydraulic Cement Concrete |
| ASTM D75/D75M | (2019) Standard Practice for Sampling Aggregates |
| ASTM D1752 | (2018) Standard Specification for Preformed Sponge Rubber, Cork and Recycled PVC Expansion Joint Fillers for Concrete Paving and Structural Construction |

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

| | |
|---|---|
| 40 CFR 247 | Comprehensive Procurement Guideline for Products Containing Recovered Materials |

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Installation Drawings; G

SD-03 Product Data

Air-Entraining Admixture

Expansion Joint Filler Strips, Premolded

Joint Sealants - Field Molded Sealants

Conveying and Placing Concrete

Mix Design Data; G

Ready-Mix Concrete

Mechanical Reinforcing Bar Connectors

SD-06 Test Reports

Aggregates

Concrete Mixture Proportions; G

Compressive Strength Testing; G

Slump; G

Air Content

Water

SD-07 Certificates

Cementitious Materials

Aggregates

## 1.4 QUALITY ASSURANCE

Indicate specific locations of Concrete Placement Forms Steel Reinforcement Expansion Joints Construction Joints Control Joints on installation drawings and include, but not be limited to, square feet of concrete placements, thicknesses and widths, plan dimensions, and arrangement of cast-in-place concrete section.

### 1.4.1 Regulatory Requirements

The state statutory and regulatory requirements: TX DOT form a part of

this specification to the extent referenced.

## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

The Government retains the option to sample and test aggregates and concrete to determine compliance with the specifications. Provide facilities and labor as may be necessary to assist the Government in procurement of representative test samples. Obtain samples of aggregates at the point of batching in accordance with ASTM D75/D75M. Sample concrete in accordance with ASTM C172/C172M. Determine slump and air content in accordance with ASTM C143/C143M and ASTM C231/C231M, respectively, when cylinders are molded. Prepare, cure, and transport compression test specimens in accordance with ASTM C31/C31M. Test compression test specimens in accordance with ASTM C39/C39M. Take samples for strength tests not less than once each shift in which concrete is produced. Provide a minimum of four 6 x 12 inch or six 4 x 8 inch specimens from each sample; two 6 x 12 inch or three 4 x 8 inch to be tested at 28 days (90 days if pozzolan or slag cement is used) for acceptance. Two 6 x 12 inch or three 4 x 8 inch will be tested at 7 days for information.

#### 2.1.1 Strength

Acceptance test results are the average strengths of two 6 x 12 inch or three 4 x 8 inch specimens tested at 28 days (90 days if pozzolan or slag cement is used). The strength of the concrete is considered satisfactory so long as the average of all three consecutive acceptance test results equal or exceed the specified compressive strength, f'c, and no individual acceptance test result falls below f'c by more than 500 psi.

#### 2.1.2 Construction Tolerances

Apply a Class "C" finish to all surfaces except those specified to receive a Class "D" finish. Apply a Class "D" finish to all post-construction surfaces which will be permanently concealed. Surface requirements for the classes of finish required are as specified in ACI 117.

#### 2.1.3 Concrete Mixture Proportions

Concrete mixture proportions are the responsibility of the Contractor. Mixture proportions must include the dry weights of cementitious material(s); the nominal maximum size of the coarse aggregate; the specific gravities, absorptions, and saturated surface-dry weights of fine and coarse aggregates; the quantities, types, and names of admixtures; and quantity of water per yard of concrete. Provide materials included in the mixture proportions of the same type and from the same source as will be used on the project. The specified compressive strength f'c is 3,000 psi at 28 days (90 days if pozzolan is used). The maximum nominal size coarse aggregate is 3/4 inch, in accordance with ACI 304R. The air content must be between 4.5 and 7.5 percent with a slump between 2 and 5 inches. The maximum water-cementitious material ratio is 0.50. Submit the applicable test reports and mixture proportions that will produce concrete of the quality required, ten days prior to placement of concrete.

### 2.2 MATERIALS

Submit manufacturer's literature from suppliers which demonstrates

compliance with applicable specifications for the specified materials.

2.2.1 Cementitious Materials

Submit Manufacturer's certificates of compliance, accompanied by mill test reports, attesting that the concrete materials meet the requirements of the specifications in accordance with the Special Clause "CERTIFICATES OF COMPLIANCE". Also, certificates for all material conforming to EPA's Comprehensive Procurement Guidelines (CPG), in accordance with 40 CFR 247. Provide cementitious materials that conform to the appropriate specifications listed:

2.2.1.1 Portland Cement

ASTM C150/C150M, Type II, with tri-calcium aluminates (C3A) content less than 10 percent and a maximum cement-alkali content of 0.80 percent Na2Oe (sodium oxide) equivalent.

2.2.1.2 Pozzolan

Provide pozzolan that conforms to ASTM C618, Class F, including requirements of Tables 1A and 2A.

2.2.1.3 Slag Cement

Slag Cement must conform to ASTM C989/C989M, Grade 80.

2.2.2 Aggregates

For fine and coarse aggregates meet the quality and grading requirements of ASTM C33/C33M. Submit certificates of compliance and test reports for aggregates showing the material(s) meets the quality and grading requirements of the specifications under which it is furnished.

2.2.3 Admixtures

Provide admixtures, when required or approved, in compliance with the appropriate specification listed. Retest chemical admixtures that have been in storage at the project site, for longer than 6 months or that have been subjected to freezing, at the expense of the Contractor at the request of the Contracting Officer and will be rejected if test results are not satisfactory.

2.2.3.1 Air-Entraining Admixture

Provide air-entraining admixture that meets the requirements of ASTM C260/C260M.

2.2.4 Water

Mixing and curing water in compliance with the requirements of ASTM C1602/C1602M; potable, and free of injurious amounts of oil, acid, salt, or alkali. Submit test report showing water complies with ASTM C1602/C1602M.

2.2.5 Reinforcing Steel

Provide reinforcing bars conforming to the requirements of ASTM A615/A615M, Grade 60, deformed. Provide welded steel wire reinforcement conforming to

the requirements of ASTM A1064/A1064M. Detail reinforcement not indicated in accordance with ACI 301 and ACI SP-66. Provide mechanical reinforcing bar connectors in accordance with ACI 301 and provide 125 percent minimum yield strength of the reinforcement bar.

2.2.6 Expansion Joint Filler Strips, Premolded

Expansion joint filler strips, premolded of sponge rubber conforming to ASTM D1752, Type I.

2.2.7 Joint Sealants - Field Molded Sealants

Conform to ASTM C920, Type M, Grade NS, Class 25, use NT for vertical joints and Type M, Grade P, Class 25, use T for horizontal joints. Provide polyethylene tape, coated paper, metal foil, or similar type bond breaker materials. The backup material needs to be compressible, nonshrink, nonreactive with the sealant, and a nonabsorptive material such as extruded butyl or polychloroprene foam rubber. Immediately prior to installation of field-molded sealants, clean the joint of all debris and further cleaned using water, chemical solvents, or other means as recommended by the sealant manufacturer or directed.

2.2.8 Form Coatings

Provide form coating in accordance with ACI 301.

2.3 READY-MIX CONCRETE

Provide ready-mix concrete with mix design data conforming to ACI 301 Part 4. Include the following additional information: .

a. Type and brand cement

b. Cement content in 94-pound bags per cubic yard of concrete

c. Maximum size of aggregate

d. Amount and brand name of admixture

e. Total water content expressed by water cementitious material ratio

PART 3 EXECUTION

3.1 PREPARATION

Prepare construction joints to expose coarse aggregate. The surface must be clean, damp, and free of laitance. Construct ramps and walkways, as necessary, to allow safe and expeditious access for concrete and workmen. Remove snow, ice, standing or flowing water, loose particles, debris, and foreign matter. Satisfactorily compact earth foundations. Make spare vibrators available. Placement cannot begin until the entire preparation has been accepted by the Government.

3.1.1 Embedded Items

Secure reinforcement in place after joints, anchors, and other embedded items have been positioned. Arrange internal ties so that when the forms are removed the metal part of the tie is not less than 2 inches from concrete surfaces permanently exposed to view or exposed to water on the

finished structures. Prepare embedded items so they are be free of oil and other foreign matters such as loose coatings or rust, paint, and scale. The embedding of wood in concrete is permitted only when specifically authorized or directed. Provide all equipment needed to place, consolidate, protect, and cure the concrete at the placement site and in good operating condition.

3.1.2 Formwork Installation

Forms must be properly aligned, adequately supported, and mortar-tight. Provide smooth form surfaces, free from irregularities, dents, sags, or holes when used for permanently exposed faces. Chamfer all exposed joints and edges , unless otherwise indicated.

3.1.3 Production of Concrete

3.1.3.1 Ready-Mixed Concrete

Provide ready-mixed concrete conforming to ASTM C94/C94M except as otherwise specified.

3.1.3.2 Concrete Made by Volumetric Batching and Continuous Mixing

Conform to ASTM C685/C685M.

3.2 CONVEYING AND PLACING CONCRETE

Convey and place concrete in accordance with ACI 301, Section 5.

3.2.1 Cold-Weather Requirements

Place concrete in cold weather in accordance with ACI 306R

3.2.2 Hot-Weather Requirements

Place concrete in hot weather in accordance with ACI 305R

3.3 FINISHING

3.3.1 Temperature Requirement

Do not finish or repair concrete when either the concrete or the ambient temperature is below 50 degrees F.

3.3.2 Finishing Formed Surfaces

Remove all fins and loose materials , and surface defects including filling of tie holes. Repair all honeycomb areas and other defects. Remove all unsound concrete from areas to be repaired. Ream or chip surface defects greater than 1/2 inch in diameter and holes left by removal of tie rods in all surfaces not to receive additional concrete and fill with dry-pack mortar. Brush-coat the prepared area with an approved epoxy resin or latex bonding compound or with a neat cement grout after dampening and filling with mortar or concrete. Use a blend of portland cement and white cement in mortar or concrete for repairs to all surfaces permanently exposed to view so that the final color when cured is the same as adjacent concrete.

3.3.3 Finishing Unformed Surfaces

Finish unformed surfaces in accordance with ACI 301, Section 5.

| FINISH | LOCATION |
|---|---|
| Float | Exposed Concrete |
| Trowel | N/A |
| Broom or Belt | Concrete Walks |

3.3.3.1 Expansion and Contraction Joints

Make expansion and contraction joints in accordance with the details shown or as otherwise specified. Provide 1/2 inch thick transverse expansion joints where new work abuts an existing concrete. Provide expansion joints at a maximum spacing of 30 feet on center in sidewalks, unless otherwise indicated. Provide contraction joints at a maximum spacing of 10 linear feet in sidewalks, unless otherwise indicated. Cut contraction joints at a minimum of 1 inch(es) deep with a jointing tool after the surface has been finished.

3.4 CURING AND PROTECTION

Cure and protect in accordance with ACI 301, Section 5.

3.5 FORM WORK

Provide form work in accordance with ACI 301, Section 2 and Section 5.

3.5.1 Removal of Forms

Remove forms in accordance with ACI 301, Section 2.

3.6 STEEL REINFORCING

Reinforcement must be free from loose, flaky rust and scale, and free from oil, grease, or other coating which might destroy or reduce the reinforcement's bond with the concrete.

3.6.1 Fabrication

Shop fabricate steel reinforcement in accordance with ACI 318 and ACI SP-66. Provide shop details and bending in accordance with ACI 318 and ACI SP-66.

3.6.2 Splicing

Perform splices in accordance with ACI 318 and ACI SP-66.

3.6.3 Supports

Secure reinforcement in place by the use of metal or concrete supports, spacers, or ties.

3.7 EMBEDDED ITEMS

Before placing concrete, take care to determine that all embedded items

are firmly and securely fastened in place. Provide embedded items free of oil and other foreign matter, such as loose coatings of rust, paint and scale. Embedding of wood in concrete is permitted only when specifically authorized or directed.

## 3.8 TESTING AND INSPECTING

Report the results of all tests and inspections conducted at the project site informally at the end of each shift. Submit written reports weekly. Deliver within three days after the end of each weekly reporting period. See Section 01 45 00 QUALITY CONTROL.

### 3.8.1 Field Testing Technicians

The individuals who sample and test concrete must have demonstrated a knowledge and ability to perform the necessary test procedures equivalent to the ACI minimum guidelines for certification of Concrete Field Testing Technicians, Grade I.

### 3.8.2 Preparations for Placing

Inspect foundation or construction joints, forms, and embedded items in sufficient time prior to each concrete placement to certify that it is ready to receive concrete.

### 3.8.3 Sampling and Testing

a. Obtain samples and test concrete for quality control during placement. Sample fresh concrete for testing in accordance with ASTM C172/C172M. Make six test cylinders.

b. Test concrete for compressive strength at 7 and 28 days for each design mix and for every 100 cubic yards of concrete. Test two cylinders at 7 days; two cylinders at 28 days; and hold two cylinders in reserve. Conform test specimens to ASTM C31/C31M. Perform compressive strength testing conforming to ASTM C39/C39M.

c. Test slump at the site of discharge for each design mix in accordance with ASTM C143/C143M. Check slump once during each shift that concrete is produced.

d. Test air content for air-entrained concrete in accordance with ASTM C231/C231M. Test concrete using lightweight or extremely porous aggregates in accordance with ASTM C173/C173M. Check air content at least once during each shift that concrete is placed.

e. Determine temperature of concrete at time of placement in accordance with ASTM C1064/C1064M. Check concrete temperature at least once during each shift that concrete is placed.

### 3.8.4 Action Required

#### 3.8.4.1 Placing

Do not begin placement until the availability of an adequate number of acceptable vibrators, which are in working order and have competent operators, has been verified. Discontinue placing if any lift is inadequately consolidated.

3.8.4.2 Air Content

Whenever an air content test result is outside the specification limits, adjust the dosage of the air-entrainment admixture prior to delivery of concrete to forms.

3.8.4.3 Slump

Whenever a slump test result is outside the specification limits, adjust the batch weights of water and fine aggregate prior to delivery of concrete to the forms. Make the adjustments so that the water-cementitious material ratio does not exceed that specified in the submitted concrete mixture proportion and the required concrete strength is still met.

-- End of Section --

SECTION 05 50 13

MISCELLANEOUS METAL FABRICATIONS
**08/24, CHG 1: 08/25**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WELDING SOCIETY (AWS)

| | |
|---|---|
| AWS D1.1/D1.1M | (2025) Structural Welding Code - Steel |

ASTM INTERNATIONAL (ASTM)

| | |
|---|---|
| ASTM A36/A36M | (2019) Standard Specification for Carbon Structural Steel |
| ASTM A53/A53M | (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless |
| ASTM A123/A123M | (2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products |
| ASTM A500/A500M | (2023) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes |
| ASTM A780/A780M | (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings |
| ASTM F593 | (2024) Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs |

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

CORRAL RAILINGS; G

GATES; G

BOLLARDS; G

Anchors; G

MISCELLANEOUS STEEL FABRICATIONS; G

SD-03 Product Data

Pipe; G

Hinges; G

Latches; G

Galvanizing-Repair Products; G

Proprietary Hardware; G

SD-07 Certificates

GALVANIZING CERTIFICATES; G

WELDERS Qualifications; G

Welding procedures; G

Mill certificates; G

1.3 QUALIFICATION OF WELDERS

Qualify procedures and personnel according to the following:

a. AWS D1.1/D1.1M Structural Welding Code - Steel.

1.4 DELIVERY, STORAGE, AND PROTECTION

Protect from corrosion, deformation, and other types of damage. Store items in an enclosed area free from contact with soil and weather. Remove and replace damaged items with new items.

1.5 MISCELLANEOUS REQUIREMENTS

1.5.1 Fabrication Drawings

Show member sizes and schedules; post spacing; gate dimensions and swing; panel lengths; rail spacing; welds; vent and drain holes for galvanizing; hinges, latches, chains, stops, and keepers; anchors and embedment; foundations; field connections; and erection sequence. Coordinate openings and gate clearances with cattle movement and operating equipment.

PART 2 PRODUCTS

2.1 STEEL MATERIALS

2.1.1 Cattle-Corral Pipe

Provide ASTM A53/A53M, Type E or Type S, Grade B, Schedule 80 steel pipe for cattle-corral rails, posts, and gate frames, as follows:

Horizontal rails and primary gate-frame rails: Nominal 2-inch pipe, 2.375-inch outside diameter, minimum 0.218-inch wall thickness (XS2).

Corral posts and heavy gate hinge or latch posts: Nominal 3-inch pipe, 3.500-inch outside diameter, minimum 0.300-inch wall thickness (XS3).

Provide six horizontal rails and post spacing approximately 5 feet on center unless otherwise indicated on the Drawings

Pipe must be new, straight, and free from laminations, splits, excessive mill scale, dents, or other defects. Do not substitute light-gage tubing, fence-framework tubing, water-well casing, used pipe, or pipe with wall thickness less than specified

2.1.2 Structural Tubing

Where rectangular or square HSS is specifically indicated for gate frames, brackets, or supports, provide ASTM A500/A500M, Grade C, with minimum dimensions and wall thicknesses indicated. HSS is not a substitute for specified Schedule 80 round pipe without approval.

2.1.3 Plates and Shapes

Provide angles, channels, bars, plates, base plates, connection plates, tabs, gussets, stops, and miscellaneous rolled shapes complying with ASTM A36/A36M. Provide minimum 1/4-inch thickness unless a greater thickness is indicated or required by design. Provide ASTM A500/A500M, Grade C, for miscellaneous HSS shapes.

2.1.4 Anchor Rods and Fasteners

Provide anchor rods complying with ASTM F593, Grade 36, unless a higher grade is indicated. Provide bolts, nuts, and washers complying with ASTM A307 and applicable ASME standards. Hot-dip galvanize exposed anchors, bolts, nuts, washers, and fasteners in accordance with ASTM A153/A153M.

2.2 PIPE BOLLARDS

Provide nominal 6-inch, Schedule 40 steel pipe complying with ASTM A53/A53M, Type E or S, Grade B; 6.625-inch outside diameter and minimum 0.280-inch wall thickness. Hot-dip galvanize after fabrication. Set bollards in concrete foundations as indicated and fill the pipe solidly with minimum 3,000 psi concrete. Crown the top to shed water unless a welded steel cap is indicated.

2.3 CATTLE-CORRAL RAILINGS

Fabricate railing panels with six continuous horizontal XS2 rails welded to XS3 posts or connected to individually installed XS3 posts as indicated. Provide finished top height and rail spacing shown on the Drawings. Maintain smooth cattle-contact surfaces without sharp edges, protruding fasteners, open pipe ends, burrs, weld spatter, or projections capable of injuring animals or personnel

Shop-fabricate panel sections in practical lengths of 10, 15, or 20 feet; do not exceed 20 feet. Hot-dip galvanize panels and posts separately where needed for handling and galvanizing. Use bolted or sleeved field connections where indicated. Field welding of galvanized assemblies is permitted only where specifically detailed and approved

## 2.4 CATTLE GATES

Fabricate gates as rigid welded assemblies using ASTM A53/A53M Grade B Schedule 80 pipe. Provide nominal 2-inch, 2.375-inch-OD, 0.218-inch-wall pipe for horizontal rails and perimeter framing unless heavier members are indicated. Provide nominal 3-inch, 3.500-inch-OD, 0.300-inch-wall pipe for hinge or latch posts and heavily loaded vertical members where indicated.

Provide diagonal bracing, gussets, or internal reinforcement as required to prevent sag, twist, and permanent deformation. Provide heavy-duty galvanized hinges, adjustable where indicated, and positive cattle-resistant latches operable from the designated side. Provide gate stops, hold-open devices, chains, and keepers where indicated. Arrange hardware to eliminate pinch points and sharp projections.

## 2.5 FABRICATION

Verify field dimensions before fabrication. Make members true to line and free from warp. Cope and miter joints accurately. Provide continuous welds at cattle-contact and weather-exposed joints. Seal exposed pipe ends with welded steel caps. Provide vent and drain holes required for safe, complete hot-dip galvanizing; locate holes to minimize water retention and animal contact.

Grind welds and cut edges smooth where exposed to cattle or personnel, without reducing required weld size. Fabricate hinges, latches, brackets, stops, and connections for repeated impact and operation in a cattle-handling environment.

## 2.6 RED-OXIDE PRIMER (BASE BID AND ADDITIVE ALTERNATE #1)

Provide SSPC-Paint 15, Type I, red-oxide primer on all fencing, gates, and bollards. Bollards shall receive two coats industrial urethane alkyd enamel matching ANSI safety Yellow or federal standard 595 color 13591.

## 2.7 GALVANIZING (ADDITIVE ALTERNATE #2)

Hot-dip galvanize corral railings, gates, posts, bollards, hardware, anchors, and exterior miscellaneous steel after fabrication in accordance with ASTM A123/A123M. Galvanize fasteners and small hardware in accordance with ASTM A153/A153M. Do not rely on pregalvanized pipe where fabrication welds or cuts would interrupt the coating.

Design fabrications for proper venting, drainage, handling, and galvanizing without distortion. Remove spikes, runs, ash, and rough deposits from cattle-contact surfaces while maintaining the required zinc coating.

# PART 3 EXECUTION

## 3.1 INSTALLATION

Install fabrications plumb, level, aligned, rigid, and at locations indicated. Verify gate swing, latch engagement, clearances, and alignment before final anchorage. Set embedded posts and bollards in concrete foundations as indicated. Brace posts until concrete has attained sufficient strength. Do not damage galvanized coatings during erection.

## 3.2 ANCHORAGE AND CONNECTIONS

Provide anchorage required for a complete installation. Use templates to position anchor rods. Tighten connections without crushing pipe or HSS. Conceal or orient bolts and projections away from cattle-contact surfaces. Do not use powder-actuated fasteners for corral railings, gates, bollards, or primary supports.

## 3.3 FIELD WELDING

Perform field welding only where indicated or approved and in accordance with AWS D1.1/D1.1M. Remove zinc from the weld zone using safe methods, provide ventilation and respiratory protection required for welding galvanized steel, and restore the coating after inspection. Grind exposed welds smooth where required for safe cattle contact.

## 3.4 GALVANIZING REPAIR

Clean field welds, cut edges, abraded areas, and damaged coatings and repair in accordance with ASTM A780/A780M using an approved zinc-rich repair system. Apply the repair coating to the thickness required by ASTM A780/A780M. Repair areas must be continuous, adherent, and free of sharp edges or runs.

## 3.5 ADJUSTMENT AND FINAL INSPECTION

Operate each gate through its full range. Adjust hinges, latches, stops, chains, and keepers for smooth operation without sagging, binding, excessive clearance, or unintended release. Inspect all cattle-contact surfaces and remove sharp edges, projections, weld spatter, and coating defects. Replace members that are bent, cracked, improperly welded, or otherwise defective.

## 3.6 REPAIRS

a. Galvanized Surfaces:

(1) Clean field welds, bolted connections, and abraded areas and repair galvanizing to comply with ASTM A780/A780M.

-- End of Section --

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SECTION 13 34 19

METAL BUILDING SYSTEMS
**08/20, CHG 1: 02/21**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

| | |
|---|---|
| AISC 360 | (2016) Specification for Structural Steel Buildings |

AMERICAN WELDING SOCIETY (AWS)

| | |
|---|---|
| AWS D1.1/D1.1M | (2025) Structural Welding Code - Steel |

ASTM INTERNATIONAL (ASTM)

| | |
|---|---|
| ASTM A36/A36M | (2019) Standard Specification for Carbon Structural Steel |
| ASTM A572/A572M | (2025) Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel |
| ASTM A653/A653M | (2025a) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process |
| ASTM A755/A755M | (2018; R 2024) Standard Specification for Steel Sheet, Metallic Coated by the Hot-Dip Process and Prepainted by the Coil-Coating Process for Exterior Exposed Building Products |
| ASTM A792/A792M | (2025a) Standard Specification for Steel Sheet, 55% Aluminum-Zinc Alloy-Coated by the Hot-Dip Process |
| ASTM A992/A992M | (2022) Standard Specification for Structural Steel Shapes |
| ASTM A1011/A1011M | (2025) Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High-Strength |
| ASTM F593 | (2024) Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs |

METAL BUILDING MANUFACTURERS ASSOCIATION (MBMA)

MBMA MBSM (2018) Metal Building Systems Manual

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Existing-Condition Survey; G

Verification Report; G

Erection Plan; G

SD-02 Shop Drawings

Existing-To-New Interfaces; G

Rigid Frames; G

Columns; G

Rafters

Purlin Extensions; G

Splices; G

Bracing; G

Roof Panels; G

Fasteners; G

Flashings; G

Trim; G

Closures; G

Insulation Layout; G

SD-03 Product Data

Structural Materials; G

Red-Oxide Primer; G

Exposed-Fastener Roof Panels; G

Sealants; G

Prodex Total 5M Plus Insulation; G

SD-05 Design Data

Signed And Sealed Structural Calculations; G

SD-07 Certificates

Welding Qualifications; G

Mill Certificates; G

Manufacturer/Erector Qualifications; G

SD-11 Closeout Submittals

Record Drawings; G

Finish Repair Records; G

Warranties; G

Final Inspection Report; G

## 1.3 SUMMARY

Provide engineering, fabrication, finishing, delivery, and erection required to expand the existing open-sided metal structure. Work includes investigation of the existing structure; extension and integration of roof purlins; new gabled rigid frames with straight columns; permanent bracing; exposed-fastener metal roof panels; Prodex Total 5M Plus insulation; flashings, trim, closures, sealants, and accessories required for a complete weather-resistant roof. No wall system, wall panels, gutters, or downspouts are included. Provide only framing required for structural stability, roof support, edge conditions, equipment support, and connections shown or required by the engineered system.

## 1.4 DESIGN RESPONSIBILITY

Provide a complete, integrated, engineered metal-building expansion designed by a professional engineer licensed in Texas. Design the new work and connections to the existing structure for the loads and criteria indicated in the Contract Documents and UFC 3-301-01. Comply with MBMA MBSM, AISC 360, and applicable cold-formed-steel standards.

Field-verify the existing rigid frames, purlins, bracing, elevations, spacing, slopes, member sizes, material thicknesses, connections, and condition before final design. Do not assume the existing roof panels provide diaphragm strength. Provide new or supplemental roof-plane and longitudinal bracing where required for a complete load path and for stability of the combined existing and new structure.

Design purlin extensions, splices, laps, bridging, flange braces, clips, bolts, and connections to transfer required gravity, uplift, diaphragm, and lateral forces without overstressing existing members. Submit calculations demonstrating adequacy of affected existing members and connections or identify required strengthening.

1.5 SYSTEM DESCRIPTION

a. Primary framing: Rigid clear-span, solid-member frames.

b. Frame configuration: Single gable.

c. Exterior columns: Straight, uniform-depth columns; tapered columns are not permitted.

d. Roof slope: 2 inches per 12 inches.

e. Secondary framing: Existing and new C- or Z-purlins, flange bracing, eave struts where required, clips, bridging, and permanent bracing.

f. Roof system: Field-assembled, exposed-fastener metal roof panels over Prodex Total 5M Plus insulation.

g. Exterior walls: None; facility remains open-sided.
Roof drainage: Free drainage from eaves; no gutters or downspouts.

1.6 SHOP DRAWINGS AND CALCULATIONS

Show existing members and connections; demolition or alteration limits; new framing; purlin extensions and splice locations; bracing; frame elevations; 2:12 slope; column profile; base plates and anchor rods; roof-panel layout; sidelaps and end laps; fastener patterns; insulation placement; closures; flashings; trim; and erection sequence. Clearly distinguish existing, reused, strengthened, and new construction.

Roof fastener calculations must account for the thickness and compressibility of the Prodex insulation between panels and purlins. Provide fastener type, length, diameter, washer, spacing, minimum penetration, pullout, pullover, shear, and uplift capacities. Do not rely on panel-to-purlin friction through the insulation for diaphragm resistance.

1.7 QUALITY ASSURANCE

Use a metal-building-system manufacturer regularly engaged in engineering and manufacturing comparable rigid-frame systems. Use an experienced erection contractor approved by the manufacturer. Perform welding in accordance with AWS D1.1/D1.1MAWS D1.1/D1.1M. Conduct a pre-erection conference addressing the existing structure, purlin extensions, temporary bracing, roof installation, insulation, weather protection, and safety.

1.8 WARRANTIES

Provide manufacturer's standard structural system warranty and roof-panel finish warranty. Provide a minimum 5-year contractor weather-tightness warranty for roof panels, sidelaps, end laps, penetrations, fasteners, flashings, trim, closures, and sealants.

1.9 SUBMITTAL DESCRIPTION

Preconstruction

Existing-condition survey and verification report; erection plan.

Shop Drawings

Existing-to-new interfaces, rigid frames, columns, rafters, purlin extensions and splices, bracing, roof panels, fasteners, flashings, trim, closures, and insulation layout.

Product Data

Structural materials, red-oxide primer, exposed-fastener roof panels, fasteners, closures, sealants, and Prodex Total 5M Plus insulation.

Design Data

Signed and sealed structural calculations for the expansion and affected existing structure

Certificates

Welding qualifications, mill certificates, and manufacturer/erector qualifications.

Closeout Submittals

Record drawings, finish repair records, warranties, and final inspection report

## PART 2 PRODUCTS

### 2.1 STRUCTURAL FRAMING MATERIALS

Provide structural steel shapes and plates complying with ASTM A992/A992M, ASTM A572/A572M, or ASTM A36/A36M, as applicable. Provide cold-formed purlins and secondary framing complying with ASTM A1011/A1011M or ASTM A653/A653M in grades selected by the system engineer. Provide high-strength bolts complying with ASTM F3125/F3125M and anchor rods complying with ASTM F593.

### 2.2 PRIMARY FRAMING

Provide shop-fabricated rigid clear-span frames consisting of straight, uniform-depth columns and gabled rafters. Fabricate from welded built-up plate sections or structural shapes engineered for required loads. Provide base plates, splice plates, stiffeners, connection plates, bolts, and attachments required for bolted field erection. Interior columns and tapered exterior columns are not permitted unless specifically shown.

### 2.3 SECONDARY FRAMING AND PURLIN EXTENSIONS

Provide new purlins matching the existing purlin depth and profile where practicable and engineered for the required loads and connections. Extend, lap, sleeve, or splice purlins only at locations shown on approved shop drawings. Provide bridging, sag rods, flange braces, clips, eave struts, and roof-plane bracing required to stabilize both existing and new framing. Do not field cut, drill, weld, heat-straighten, or otherwise alter existing rigid frames or purlins without written approval and sealed details from the metal-building-system engineer. Replace or reinforce damaged or inadequate members as indicated by the approved design.

## 2.4 RED-OXIDE FINISH

Provide SSPC-Paint 15, Type I, red-oxide primer on new and existing exposed primary and secondary structural framing. Shop-clean new framing to the primer manufacturer's requirements and apply one complete coat to not less than 1.5 mils dry-film thickness unless a greater thickness is recommended. Prepare existing framing by removing dirt, grease, loose rust, loose mill scale, failed coating, and contaminants using SSPC SP 2 and SSPC SP 3, with additional preparation where required by the coating manufacturer. Feather sound existing coating edges. Spot-prime bare steel, then apply a complete field coat of compatible red-oxide primer to produce a uniform red-oxide appearance. Do not coat faying surfaces of slip-critical connections unless the coating is qualified for that use. Touch up welds, bolted connections, abrasions, and damaged primer after erection using the same compatible red-oxide primer. Existing and new framing must have a consistent finish at completion.

## 2.5 EXPOSED-FASTENER ROOF PANELS

Provide manufacturer's standard structural, through-fastened, lap-seam steel roof panels designed for the 2:12 slope and specified loads. Provide minimum 24-gage metallic-coated steel sheet complying with ASTM A653/A653M, G90, or ASTM A792/A792M, AZ50, prefinished by the coil-coating process in accordance with ASTM A755/A755M. Provide profile, depth, coverage, and color as indicated or selected from the manufacturer's standard offerings. Provide panels in full lengths from eave to ridge where shipping and erection permit. Where end laps are approved, locate laps over purlins, provide minimum lap required by the panel manufacturer and not less than 6 inches, and seal with continuous tape sealant. Provide minimum one full major rib side lap or greater overlap required by the tested assembly.

## 2.6 ROOF FASTENERS

Provide corrosion-resistant self-drilling or self-tapping exposed fasteners recommended by the roof-panel manufacturer and verified by design calculations. Provide integral metal washers matching the roof-panel finish and bonded EPDM sealing washers. Fasteners must penetrate supporting steel not less than the panel manufacturer's tested minimum after accounting for the Prodex insulation thickness.

Use stitch screws at sidelaps and trim as required by the tested assembly. Do not overdrive or underdrive fasteners. Provide oversized or repair fasteners only where approved by the roof-panel manufacturer.

## 2.7 ROOF INSULATION

Basis of Design: Prodex Total 5M Plus by Prodex/Insulation4Less, nominal 5 mm thick, closed-cell polyethylene foam core with reflective facers, furnished with manufacturer's compatible joining tape and accessories. Provide the nonperforated product where condensation and vapor control are intended. Equivalent products require approval and must match the basis-of-design composition, thickness, durability, vapor-control characteristics, flame/smoke performance, and installation method.

Do not assign an assembly R-value based solely on the reflective

insulation product. Thermal performance depends on the installed assembly and adjacent air spaces. Submit manufacturer's current technical data, code evaluation, flame/smoke test data, installation instructions, and proposed roof assembly.

## 2.8 FLASHING, TRIM, CLOSURES, AND SEALANTS

Provide ridge caps, rake trim, eave trim, transition flashing, closures, sealants, and accessories compatible with the roof panels. Provide factory-fabricated trim of the same base metal, coating, and color as the roof panels. Provide formed closures matching the panel profile and continuous tape sealant at weather-exposed laps. Do not provide gutters or downspouts.

# PART 3 EXECUTION

## 3.1 EXAMINATION AND EXISTING CONDITIONS

Before fabrication and erection, verify existing frame and purlin geometry, elevations, alignment, condition, connections, and anchor locations. Submit discrepancies and proposed corrective work. Proceed only after the engineered design has incorporated verified conditions and detrimental conditions have been corrected.

## 3.2 ERECTION OF STRUCTURAL FRAMING

Erect framing in accordance with the manufacturer's instructions, approved erection plan, MBMA MBSM, and AISC requirements. Maintain stability with temporary shoring, guys, and bracing until permanent framing and bracing are complete. Align and plumb new straight columns and gabled frames before final bolt tightening. Do not alter structural members without sealed approval from the system engineer.

## 3.3 PURLIN EXTENSIONS AND EXISTING-TO-NEW CONNECTIONS

Install purlin extensions, splices, bridging, clips, and bracing at the approved locations. Maintain purlin alignment and roof plane across the existing-to-new interface. Use bolted field connections unless approved details require welding. Protect existing framing during work and repair damage in accordance with approved procedures.

## 3.4 INSULATION INSTALLATION

Install Prodex Total 5M Plus concurrently with roof panels in accordance with the manufacturer's approved instructions. Orient facers as recommended. Keep material continuous, taut, clean, and dry; avoid excessive stretching. Lap or butt joints as required by the manufacturer and seal continuously with compatible tape. Seal cuts, penetrations, tears, and discontinuities. Replace wet, torn, punctured, delaminated, or contaminated material.

Coordinate insulation placement with roof fasteners so washers seat uniformly and panels bear as required by the tested assembly. Do not allow insulation buildup, wrinkles, or multiple layers at purlins to prevent proper fastener seating or panel alignment.

## 3.5 ROOF PANEL INSTALLATION

Install exposed-fastener roof panels true to line and plane with ribs parallel to the roof slope. Secure panels to each supporting purlin using the approved fastener pattern. Install sidelaps, end laps, ridge caps, flashings, trim, closures, and sealants to form a complete weather-resistant roof. Field cutting by torch is prohibited. Remove metal shavings and drilling debris daily.

## 3.6 FREE-DRAINING EAVES

Terminate roof panels and eave trim to discharge water freely from the open eaves at locations indicated. Provide uniform panel overhang, closures, and drip edges. No gutters, downspouts, collector heads, or enclosed roof-drainage system are required.

## 3.7 CLEANUP AND FINAL INSPECTION

Clean existing and new red-oxide framing and repair finish damage. Clean roof panels, remove protective films, sealant smears, filings, and fasteners, and replace damaged panels. Verify roof fasteners are properly seated, laps and closures are sealed, insulation joints are taped, bracing is complete, and the roof drains freely from the eaves.

-- End of Section --

SECTION 22 00 00

PLUMBING, GENERAL PURPOSE
**05/24, CHG 1: 11/24**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

| | |
|---|---|
| ASME A112.6.1M | (1997; R 2017) Floor Affixed Supports for Off-the-Floor Plumbing Fixtures for Public Use |
| ASME A112.36.2M | (2022) Cleanouts |
| ASME B1.20.1 | (2013; R 2018) Pipe Threads, General Purpose (Inch) |
| ASME B40.100 | (2022) Pressure Gauges and Gauge Attachments |

AMERICAN WATER WORKS ASSOCIATION (AWWA)

| | |
|---|---|
| AWWA B300 | (2024) Hypochlorites |
| AWWA B301 | (2024) Liquid Chlorine |
| AWWA C203 | (2025) Coal-Tar Protective Coatings and Linings for Steel Water Pipe and Fittings |
| AWWA C651 | (2023) Standard for Disinfecting Water Mains |
| AWWA C652 | (2019) Disinfection of Water-Storage Facilities |

ASTM INTERNATIONAL (ASTM)

| | |
|---|---|
| ASTM B370 | (2022) Standard Specification for Copper Sheet and Strip for Building Construction |
| ASTM D4586/D4586M | (2025) Standard Specification for Asphalt Roof Cement |
| ASTM E1 | (2014; R 2025) Standard Specification for ASTM Liquid-in-Glass Thermometers |
| ASTM F877 | (2025) Standard Specification for Crosslinked Polyethylene (PEX) Plastic Hot- and Cold-Water Distribution Systems |
| ASTM F2389 | (2024a) Standard Specification for |

| | |
|---|---|
| | Pressure-rated Polypropylene (PP) Piping Systems |

INTERNATIONAL CODE COUNCIL (ICC)

| | |
|---|---|
| ICC IPC | (2024) International Plumbing Code |

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

| | |
|---|---|
| MSS SP-58 | (2025) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation |

NSF INTERNATIONAL (NSF)

| | |
|---|---|
| NSF 372 | (2022) Drinking Water System Components - Lead Content |
| NSF/ANSI 14 | (2024) Plastics Piping System Components and Related Materials |
| NSF/ANSI/CAN 61 | (2024) Drinking Water System Components - Health Effects |

PLASTIC PIPE AND FITTINGS ASSOCIATION (PPFA)

| | |
|---|---|
| PPFA Fire Man | (2016) Firestopping: Plastic Pipe in Fire Resistive Construction |

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

| | |
|---|---|
| SAE J1508 | (2023) Hose Clamp Specifications |

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

| | |
|---|---|
| EPA SM 9223 | (2004) Enzyme Substrate Coliform Test |
| PL 93-523 | (1974; A 1999) Safe Drinking Water Act |

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Plumbing System; G

SD-03 Product Data

Recycled Content for Steel Pipe; S

Recycled Content for Cast Iron Pipe; S

Press Fittings; G

Plumbing System

SD-06 Test Reports

Tests, Flushing and Disinfection

SD-07 Certificates

Materials and Equipment

Bolts

SD-10 Operation and Maintenance Data

Plumbing System; G

## 1.3 SHOP DRAWINGS

Submit Plumbing System Detail drawings consisting of schedules, performance charts, instructions, diagrams, and other information to illustrate the requirements and operations of systems that are not covered by the Plumbing Code. Submit Plumbing System Detail drawings for the complete plumbing system including piping layouts and locations of connections; dimensions for roughing-in, foundation, and support points; schematic diagrams and wiring diagrams or connection and interconnection diagrams. Detail drawings must indicate clearances required for maintenance and operation. Where piping and equipment are to be supported other than as indicated, details must include loadings and proposed support methods. Mechanical drawing plans, elevations, views, and details, must be drawn to scale.

## 1.4 STANDARD PRODUCTS

Specified materials and equipment must be standard products of a manufacturer regularly engaged in the manufacture of such products. Specified equipment must essentially duplicate equipment that has performed satisfactorily at least 2 years prior to bid opening. Standard products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year use must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

Where equipment is specified to conform to requirements of the ASME Boiler and Pressure Vessel Code, the design, fabrication, and installation must conform to the code and a certificate must be submitted for the equipment.

### 1.4.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

### 1.4.2 Service Support

The equipment items must be supported by service organizations. Submit a certified list of qualified permanent service organizations for support of

the equipment which includes their addresses and qualifications. These service organizations must be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.4.3 Manufacturer's Nameplate

Each item of equipment must have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

1.4.4 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction", or words of similar meaning, to mean the Contracting Officer.

1.4.4.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions must be considered mandatory, the word "should" must be interpreted as "must." Reference to the "code official" must be interpreted to mean the "Contracting Officer." For Navy owned property, references to the "owner" must be interpreted to mean the "Contracting Officer." For leased facilities, references to the "owner" must be interpreted to mean the "lessor." References to the "permit holder" must be interpreted to mean the "Contractor."

1.4.4.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, must be applied appropriately by the Contracting Officer as authorized by his administrative cognizance and the FAR.

1.5 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

1.6 REGULATORY REQUIREMENTS

Unless otherwise required herein, plumbing work must be in accordance with ICC IPC.

1.7 PROJECT/SITE CONDITIONS

The Contractor must become familiar with details of the work, verify dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

## 1.8 INSTRUCTION TO GOVERNMENT PERSONNEL

Furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system. Instructors must be thoroughly familiar with all parts of the installation and must be trained in operating theory as well as practical operation and maintenance work. Submit Plumbing System Operation and Maintenance Data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

Instruction must be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished must be as specified in the individual section. When more than 4 man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system. When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

## 1.9 ACCESSIBILITY OF EQUIPMENT

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

# PART 2 PRODUCTS

## 2.1 MATERIALS

Materials for various services must be in accordance with TABLES I and II. Cement pipe must contain recycled content as specified in Section 03 30 00 CAST-IN-PLACE CONCRETE. Steel pipe must contain a minimum of 25 percent recycled content, with a minimum of 16 percent post-consumer recycled content. Provide data identifying percentage of recycled content for steel pipe. Pipe schedules must be selected based on service requirements. Pipe fittings must be compatible with the applicable pipe materials. Plastic pipe, fittings, and solvent cement must meet NSF/ANSI 14 and must be NSF listed for the service intended. Plastic pipe, fittings, and solvent cement used for potable hot and cold water service must bear the NSF seal "NSF-PW." Polypropylene pipe and fittings must conform to dimensional requirements of Schedule 40, Iron Pipe size and must comply with NSF/ANSI 14, NSF/ANSI/CAN 61 and ASTM F2389. Polypropylene piping that will be exposed to UV light must be provided with a Factory applied UV resistant coating. Pipe threads (except dry seal) must conform to ASME B1.20.1. Grooved pipe couplings and fittings must be from the same manufacturer. Material or used in any potable water system intended for human consumption be certified in accordance with NSF 372 with the exception of brazing materials. In line devices such as water meters, building valves, check valves, meter stops, valves, fittings and back flow preventers must comply with PL 93-523 and NSF/ANSI/CAN 61, Section 8. End point devices such as drinking water fountains, lavatory faucets, kitchen and bar faucets, residential ice makers, supply stops and end point control valves used to dispense water for drinking must meet the requirements of NSF/ANSI/CAN 61, Section 9. Hubless cast-iron soil pipe

must not be installed underground, under concrete floor slabs, or in crawl spaces below kitchen floors (for Air Force/Army/Navy projects). Cast-iron pipe must contain a minimum of 95 percent recycled content. Provide data identifying percentage of recycled content for cast iron pipe. Plastic pipe must not be installed in air plenums. Plastic pipe must not be installed in a pressure piping system in buildings greater than three stories including any basement levels.

2.1.1 Miscellaneous Materials

Miscellaneous materials must conform to the following:

a. Copper, Sheet and Strip for Building Construction: ASTM B370.

b. Asphalt Roof Cement: ASTM D4586/D4586M.

c. Hose Clamps: SAE J1508.

d. Supports for Off-The-Floor Plumbing Fixtures: ASME A112.6.1M.

e. Metallic Cleanouts: ASME A112.36.2M.

f. Plumbing Fixture Setting Compound: A preformed flexible ring seal molded from hydrocarbon wax material. The seal material must be nonvolatile nonasphaltic and contain germicide and provide watertight, gastight, odorproof and verminproof properties.

g. Coal-Tar Protective Coatings and Linings for Steel Water Pipelines: AWWA C203.

h. Hypochlorites: AWWA B300.

i. Liquid Chlorine: AWWA B301.

j. Gauges - Pressure and Vacuum Indicating Dial Type - Elastic Element: ASME B40.100.

k. Thermometers: ASTM E1. Mercury must not be used in thermometers.

2.2 PIPE HANGERS, INSERTS, AND SUPPORTS

Pipe hangers, inserts, and supports must conform to MSS SP-58.

2.3 VALVES

2.3.1 Wall Faucets

Wall faucets with vacuum-breaker backflow preventer must be brass with 3/4 inch male inlet threads, hexagon shoulder, and 3/4 inch hose connection. Faucet handle must be securely attached to stem.

PART 3 EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

Installation of plastic pipe where in compliance with NFPA may be installed in accordance with PPFA Fire Man. The plumbing system must be installed complete with necessary fixtures, fittings, traps, valves, and accessories. Submit diagrams, instructions, and other sheets proposed for

posting for Plumbing System. A full port ball valve and drain must be installed on the water service line inside the cast-in-place concrete valve access box. Piping must be connected to the exterior service lines or capped or plugged if the exterior service is not in place. Exterior underground utilities must be at least 12 inches below the average local frost depth or as indicated on the drawings. If trenches are closed or the pipes are otherwise covered before being connected to the service lines, the location of the end of each plumbing utility must be marked with a stake or other acceptable means. Valves must be installed with control no lower than the valve body. Avoid the use of combination double wye and one-eighth bend fittings in the horizontal orientation below grade or below slab.

3.1.1 Water Pipe, Fittings, and Connections

3.1.1.1 Utilities

The piping must be extended to fixtures, outlets, and equipment. The hot-water and cold-water piping system must be arranged and installed to permit draining. The supply line to each item of equipment or fixture, except faucets, flush valves, or other control valves which are supplied with integral stops, must be equipped with a shutoff valve to enable isolation of the item for repair and maintenance without interfering with operation of other equipment or fixtures. Supply piping to fixtures, faucets, hydrants, shower heads, and flushing devices must be anchored to prevent movement.

3.1.1.2 Cutting and Repairing

The work must be carefully laid out in advance, and unnecessary cutting of construction must be avoided. Damage to building, piping, wiring, or equipment as a result of cutting must be repaired by mechanics skilled in the trade involved.

3.1.1.3 Protection of Fixtures, Materials, and Equipment

Pipe openings must be closed with caps or plugs during installation. Fixtures and equipment must be tightly covered and protected against dirt, water, chemicals, and mechanical injury. Upon completion of the work, the fixtures, materials, and equipment must be thoroughly cleaned, adjusted, and operated. Safety guards must be provided for exposed rotating equipment.

3.1.1.4 Mains, Branches, and Runouts

Piping must be installed as indicated. Pipe must be accurately cut and worked into place without springing or forcing. Structural portions of the building must not be weakened. Aboveground piping must run parallel with the lines of the building, unless otherwise indicated. Branch pipes from service lines may be taken from top, bottom, or side of main, using crossover fittings required by structural or installation conditions. Supply pipes, valves, and fittings must be kept a sufficient distance from other work and other services to permit not less than 1/2 inch between finished covering on the different services. Bare and insulated water lines must not bear directly against building structural elements so as to transmit sound to the structure or to prevent flexible movement of the lines. Water pipe must not be buried in or under floors unless specifically indicated or approved. Buried domestic waste vent piping must not be less than 2 inch diameter. Changes in pipe sizes must be made

with reducing fittings. Use of bushings will not be permitted except for use in situations in which standard factory fabricated components are furnished to accommodate specific accepted installation practice. Change in direction must be made with fittings, except that bending of pipe 4 inches and smaller will be permitted, provided a pipe bender is used and wide sweep bends are formed. The center-line radius of bends must be not less than six diameters of the pipe. Bent pipe showing kinks, wrinkles, flattening, or other malformations will not be acceptable.

3.1.1.5 Pipe Drains

Pipe drains indicated must consist of 3/4 inch hose bibb with renewable seat and full port ball valve ahead of hose bibb. At other low points, 3/4 inch brass plugs or caps must be provided. Disconnection of the supply piping at the fixture is an acceptable drain.

3.1.2 Joints

3.1.2.1 Plastic Pipe

Acrylonitrile-Butadiene-Styrene (ABS) pipe must have joints made with solvent cement. PVC and CPVC pipe must have joints made with solvent cement elastomeric, threading, (threading of Schedule 80 Pipe is allowed only where required for disconnection and inspection; threading of Schedule 40 Pipe is not allowed), or mated flanged.

3.1.3 Dissimilar Pipe Materials

Connections between ferrous and non-ferrous copper water pipe must be made with dielectric unions or flange waterways. Dielectric waterways must have temperature and pressure rating equal to or greater than that specified for the connecting piping. Waterways must have metal connections on both ends suited to match connecting piping. Dielectric waterways must be internally lined with an insulator specifically designed to prevent current flow between dissimilar metals. Dielectric flanges must meet the performance requirements described herein for dielectric waterways. Connecting joints between plastic and metallic pipe must be made with transition fitting for the specific purpose.

3.1.4 Pipe Sleeves and Flashing

Pipe sleeves must be furnished and set in their proper and permanent location.

3.1.4.1 Pipe Penetrations

Provide sealants for all pipe penetrations. All pipe penetrations must be sealed to prevent infiltration of air, insects, and vermin.

3.1.5 Supports

3.1.5.1 General

Hangers used to support piping 2 inches and larger must be fabricated to permit adequate adjustment after erection while still supporting the load. Pipe guides and anchors must be installed to keep pipes in accurate alignment, to direct the expansion movement, and to prevent buckling, swaying, and undue strain. Piping subjected to vertical movement when operating temperatures exceed ambient temperatures must be supported by

variable spring hangers and supports or by constant support hangers. In the support of multiple pipe runs on a common base member, a clip or clamp must be used where each pipe crosses the base support member. Spacing of the base support members must not exceed the hanger and support spacing required for an individual pipe in the multiple pipe run. Threaded sections of rods must not be formed or bent.

3.1.5.2 Structural Attachments

Attachment to building structure concrete and masonry must be by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Inserts and anchors must be applied with a safety factor not less than 5. Supports must not be attached to metal decking. Supports must not be attached to the underside of concrete filled floor or concrete roof decks unless approved by the Contracting Officer. Masonry anchors for overhead applications must be constructed of ferrous materials only.

3.2 IDENTIFICATION SYSTEMS

3.2.1 Identification Tags

Identification tags made of brass, engraved laminated plastic, or engraved anodized aluminum, indicating service and valve number must be installed on valves, except those valves installed on supplies at plumbing fixtures. Tags must be 1-3/8 inch minimum diameter, and marking must be stamped or engraved. Indentations must be black, for reading clarity. Tags must be attached to valves with No. 12 AWG, copper wire, chrome-plated beaded chain, or plastic straps designed for that purpose.

3.3 ESCUTCHEONS

Escutcheons must be provided at finished surfaces where bare or insulated piping, exposed to view, passes through floors, walls, or ceilings, except in boiler, utility, or equipment rooms. Escutcheons must be fastened securely to pipe or pipe covering and must be satin-finish, corrosion-resisting steel, polished chromium-plated zinc alloy, or polished chromium-plated copper alloy. Escutcheons must be either one-piece or split-pattern, held in place by internal spring tension or setscrew.

3.4 TESTS, FLUSHING AND DISINFECTION

Submit test reports in booklet form showing all field tests performed to adjust each component and all field tests performed to prove compliance with the specified performance criteria, completion and testing of the installed system. Each test report must indicate the final position of controls.

3.4.1 Plumbing System

The following tests must be performed on the plumbing system in accordance with ICC IPC.

a. Water Supply Systems Tests.

3.4.2 Defective Work

If inspection or test shows defects, such defective work or material must be replaced or repaired as necessary and inspection and tests must be

repeated. Repairs to piping must be made with new materials. Caulking of screwed joints or holes will not be acceptable.

3.4.3 Operational Test

Upon completion of flushing and prior to disinfection procedures, the Contractor must subject the plumbing system to operating tests to demonstrate satisfactory installation, connections, adjustments, and functional and operational efficiency. Such operating tests must cover a period of not less than 8 hours for each system and must include the following information in a report with conclusion as to the adequacy of the system:

a. Time, date, and duration of test.

b. Water pressures at the most remote and the highest fixtures.

c. Operation of each fixture and fixture trim.

d. Operation of each valve, hydrant, and faucet.

e. Operation of each vacuum breaker and backflow preventer.

f. Compressed air readings at each compressor and at each outlet. Each indicating instrument must be read at 1/2 hour intervals. The report of the test must be submitted in quadruplicate. The Contractor must furnish instruments, equipment, and personnel required for the tests; the Government will furnish the necessary water and electricity.

3.4.4 Disinfection

After all system components are provided and operational tests are complete, the entire domestic water distribution system must be disinfected. Before introducing disinfecting chlorination material, entire system must be flushed with potable water until any entrained dirt and other foreign materials have been removed.

Water chlorination procedure must be in accordance with AWWA C651 and AWWA C652 as modified and supplemented by this specification. The chlorinating material must be hypochlorites or liquid chlorine. The chlorinating material must be fed into the water piping system at a constant rate at a concentration of at least 50 parts per million (ppm). Feed a properly adjusted hypochlorite solution injected into the system with a hypochlorinator, or inject liquid chlorine into the system through a solution-feed chlorinator and booster pump until the entire system is completely filled.

3.4.4.1 Sample Testing

Take additional samples of water in disinfected containers, for bacterial examination, at locations specified by the Contracting Officer. Test these samples for total coliform organisms (coliform bacteria, fecal coliform, streptococcal, and other bacteria) in accordance with EPA SM 9223. The testing method used must be EPA approved for drinking water systems and must comply with applicable local and state requirements.

3.4.4.2 Disinfection Acceptance

Disinfection must be repeated until bacterial tests indicate the absence of coliform organisms (zero mean coliform density per 100 milliliters) in the samples for at least 2 full days. The system will not be accepted until satisfactory bacteriological results have been obtained.

3.4.5 OPTIONAL DISINFECTION METHOD

Disinfect new potable water piping and affected portions of existing potable water piping with geothermal water. Geothermal water must be not less than 194 degrees F and contact time must be not less than 30 minutes. After disinfection, thoroughly flush new portable water piping and affected portions of existing potable water piping with the chlorinated base water supply for a minimum of 2 hours.

3.5 POSTED INSTRUCTIONS

Framed instructions under glass or in laminated plastic, including wiring and control diagrams showing the complete layout of the entire system, must be posted where directed. Condensed operating instructions explaining preventive maintenance procedures, methods of checking the system for normal safe operation, and procedures for safely starting and stopping the system must be prepared in typed form, framed as specified above for the wiring and control diagrams and posted beside the diagrams. The framed instructions must be posted before acceptance testing of the systems.

3.6 TABLES

TABLE II

PIPE AND FITTING MATERIALS FOR PRESSURE PIPING SYSTEMS

| Item # | Pipe and Fitting Materials | SERVICE A | SERVICE B | SERVICE C | SERVICE D |
|---|---|---|---|---|---|
| 1 | Crosslinked Polyethylene (PEX) Plastic Pipe ASTM F877 | X | X | | X |
| | SERVICE:<br>A - Cold Water Service Aboveground<br>B - Hot and Cold Water Distribution<br>180 degrees F Maximum Aboveground<br>C - Compressed Air Lubricated<br>D - Cold Water Service Belowground<br>Indicated types are minimum wall thicknesses.<br>** - Type L - Hard<br>*** - Type K - Hard temper with brazed joints only or type K-soft temper without joints in or under floors<br>**** - In or under slab floors only brazed joints | | | | |

-- End of Section --

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SECTION 26 20 00

INTERIOR DISTRIBUTION SYSTEM
**05/25, CHG 2: 02/26**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

| | |
|---|---|
| ASTM B1 | (2013) Standard Specification for Hard-Drawn Copper Wire |
| ASTM B8 | (2023) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft |
| ASTM D709 | (2017) Standard Specification for Laminated Thermosetting Materials |

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

| | |
|---|---|
| IEEE 81 | (2025) IEEE Guide for Measuring Earth Resistivity, Ground Impedance, and Earth |
| IEEE C2 | (2023) National Electrical Safety Code |
| IEEE Stds Dictionary | (2009) IEEE Standards Dictionary: Glossary of Terms & Definitions |

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

| | |
|---|---|
| NETA ATS | (2025) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems |

NATIONAL ELECTRICAL CONTRACTORS ASSOCIATION (NECA)

| | |
|---|---|
| NECA NEIS 1 | (2015) Standard for Good Workmanship in Electrical Construction |

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

| | |
|---|---|
| ANSI C80.1 | (2020) American National Standard for Electrical Rigid Steel Conduit (ERSC) |
| ANSI C80.3 | (2020) American National Standard for Electrical Metallic Tubing (EMT) |
| ANSI C80.5 | (2020) American National Standard for Electrical Rigid Aluminum Conduit |
| NEMA 250 | (2020) Enclosures for Electrical Equipment |

| | |
|---|---|
| | (1000 Volts Maximum) |
| NEMA FU 1 | (2012) Low Voltage Cartridge Fuses |
| NEMA ICS 1 | (2022) Standard for Industrial Control and Systems: General Requirements |
| NEMA ICS 6 | (1993; R 2016) Industrial Control and Systems: Enclosures |
| NEMA KS 1 | (2013) Enclosed and Miscellaneous Distribution Equipment Switches (600 V Maximum) |
| NEMA RN 1 | (2018) Polyvinyl-Chloride (PVC) Externally Coated Galvanized Rigid Steel Conduit and Intermediate Metal Conduit |
| NEMA ST 20 | (2014) Dry-Type Transformers for General Applications |
| NEMA TC 2 | (2020) Standard for Electrical Polyvinyl Chloride (PVC) Conduit |
| NEMA TC 3 | (2021) Polyvinyl Chloride (PVC) Fittings for Use With Rigid PVC Conduit and Tubing |
| NEMA WD 1 | (1999; R 2020) Standard for General Color Requirements for Wiring Devices |
| NEMA WD 6 | (2021) Wiring Devices Dimensions Specifications |
| NEMA Z535.4 | (2023) Product Safety Signs and Labels |
| NATIONAL FIRE PROTECTION ASSOCIATION (NFPA) | |
| NFPA 70 | (2026; TIA 26-1; ERTA 26-1; TIA 26-2; TIA 26-3; TIA 26-4; TIA 26-5; TIA 26-6; TIA 26-7; ERTA 26-2; ERTA 26-3) National Electrical Code |
| NFPA 70E | (2024) Standard for Electrical Safety in the Workplace |
| U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA) | |
| 29 CFR 1910.147 | The Control of Hazardous Energy (Lock Out/Tag Out) |
| 29 CFR 1910.303 | Electrical, General |
| UL SOLUTIONS (UL) | |
| UL 1 | (2005; Reprint Jan 2022) UL Standard for Safety Flexible Metal Conduit |
| UL 6 | (2022) UL Standard for Safety Electrical Rigid Metal Conduit-Steel |

| | |
|---|---|
| UL 6A | (2008; Reprint Nov 2024) UL Standard for Safety Electrical Rigid Metal Conduit - Aluminum, Red Brass, and Stainless Steel |
| UL 20 | (2025) UL Standard for Safety General-Use Snap Switches |
| UL 44 | (2018; Reprint Oct 2025 ) UL Standard for Safety Thermoset-Insulated Wires and Cables |
| UL 50 | (2024) UL Standard for Safety Enclosures for Electrical Equipment, Non-Environmental Considerations |
| UL 67 | (2018; Reprint Aug 2025) UL Standard for Safety Panelboards |
| UL 83 | (2017; Reprint Oct 2025) UL Standard for Safety Thermoplastic-Insulated Wires and Cables |
| UL 248-4 | (2010; Reprint Apr 2019) Low-Voltage Fuses - Part 4: Class CC Fuses |
| UL 248-8 | (2011; Reprint Aug 2020) Low-Voltage Fuses - Part 8: Class J Fuses |
| UL 248-10 | (2011; Reprint Aug 2020) Low-Voltage Fuses - Part 10: Class L Fuses |
| UL 248-12 | (2011; Reprint Aug 2020) Low Voltage Fuses - Part 12: Class R Fuses |
| UL 248-15 | (2018) Low-Voltage Fuses - Part 15: Class T Fuses |
| UL 360 | (2025) UL Standard for Safety Liquid-Tight Flexible Metal Conduit |
| UL 467 | (2022) UL Standard for Safety Grounding and Bonding Equipment |
| UL 489 | (2025) UL Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures |
| UL 498 | (2017; Reprint Jun 2024) UL Standard for Safety Attachment Plugs and Receptacles |
| UL 514A | (2024) UL Standard for Safety Metallic Outlet Boxes |
| UL 514B | (2012; Reprint Mar 2024) UL Standard for Safety Conduit, Tubing and Cable Fittings |
| UL 514C | (2014; Reprint Apr 2024) UL Standard for Safety Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers |

| | |
|---|---|
| UL 651 | (2011; Reprint Jan 2026) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings |
| UL 797 | (2007; Reprint Apr 2023) UL Standard for Safety Electrical Metallic Tubing -- Steel |
| UL 817 | (2025) UL Standard for Safety Cord Sets and Power-Supply Cords |
| UL 869A | (2006; Reprint Jun 2020) Reference Standard for Service Equipment |
| UL 870 | (2016; Reprint Nov 2023) UL Standard for Safety Wireways, Auxiliary Gutters, and Associated Fittings |
| UL 943 | (2016; Reprint Aug 2025) UL Standard for Safety Ground-Fault Circuit-Interrupters |
| UL 1242 | (2025) UL Standard for Safety for Electrical Intermediate Metal Conduit – Steel |
| UL 1449 | (2021; Reprint Oct 2025) UL Standard for Safety Surge Protective Devices |
| UL 1660 | (2019; Reprint May 2024) Liquid-Tight Flexible Nonmetallic Conduit |
| UL 4248-1 | (2022; Reprint Jun 2025) UL Standard for Safety Fuseholders - Part 1: General Requirements |
| UL 4248-12 | (2025) UL Standard for Safety Fuseholders - Part 12: Class R |

## 1.2 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in IEEE Stds Dictionary.

## 1.3 RELATED REQUIREMENTS

Section 25 05 11 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS applies to this section, with the additions and modifications specified herein.

## 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Panelboards; G

Wireways; G

Marking Strips Drawings; G

SD-03 Product Data

Switches; G

Receptacles; G

Circuit Breakers; G

Fuses; G

Enclosed Circuit Breakers; G

Surge Protective Devices; G

SD-06 Test Reports

600-volt Wiring Test; G

Grounding System Test; G

Ground-fault Receptacle Test; G

SD-07 Certificates

Fuses; G

SD-09 Manufacturer's Field Reports

Transformer Factory Tests

## 1.5 QUALITY ASSURANCE

### 1.5.1 Electrician Qualifications

Electrical work must be performed by qualified persons with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed.

### 1.5.2 Fuses

Submit coordination data as specified in paragraph, FUSES of this section.

### 1.5.3 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Provide equipment, materials, installation, and workmanship in accordance with NFPA 70 unless more stringent requirements are specified or indicated. NECA NEIS 1 is

the minimum standard for workmanship.

1.5.4 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and:

a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.

b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.

c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.4.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 consecutive hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.5.4.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable.

1.6 MAINTENANCE

1.7 WARRANTY

Provide equipment items supported by service organizations that are located within 250 miles of the equipment installation in order to render service to the equipment on a regular and emergency basis during the warranty period of the contract.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

As a minimum, meet requirements of UL, where UL standards are established for those items, and requirements of NFPA 70 for all materials, equipment, and devices.

2.2 CONDUIT AND FITTINGS

Conform to the following:

2.2.1 Rigid Metallic Conduit

2.2.1.1 Rigid, Threaded Zinc-Coated Steel Conduit

ANSI C80.1, UL 6.

2.2.1.2 Rigid Aluminum Conduit

ANSI C80.5, UL 6A.

2.2.2 Rigid Nonmetallic Conduit

PVC Type EPC-40 in accordance with NEMA TC 2,UL 651.

2.2.3 Intermediate Metal Conduit (IMC)

UL 1242, zinc-coated steel only.

2.2.4 Electrical, Zinc-Coated Steel Metallic Tubing (EMT)

UL 797, ANSI C80.3.

2.2.5 Plastic-Coated Rigid Steel and IMC Conduit

NEMA RN 1, Type 40 (40 mils thick).

2.2.6 Flexible Metal Conduit

UL 1, limited to 6 feet in areas where not accessible or below a raised floor installation.

2.2.6.1 Liquid-Tight Flexible Metal Conduit, Steel

UL 360, limited to 6 feet in areas where not accessible or below a raised floor installation. The use of flexible metal conduit for bends or transitions in RMC, IMC, or EMT is prohibited.

2.2.7 Liquid-Tight Flexible Nonmetallic Conduit

UL 1660. The use of liquid-tight flexible metal conduit for bends or transitions in RMC, IMC, or EMT is prohibited.

2.2.8 Fittings for Metal Conduit, EMT, and Flexible Metal Conduit

UL 514B. Ferrous fittings: cadmium- or zinc-coated in accordance with UL 514B.

2.2.8.1 Fittings for Rigid Metal Conduit and IMC

Threaded-type. Split couplings unacceptable.

2.2.8.2 Fittings for EMT

Steel compression type.

2.2.9 Fittings for Rigid Nonmetallic Conduit

NEMA TC 3 for PVC, and UL 514B.

2.3 OUTLET BOXES AND COVERS

UL 514A, cadmium- or zinc-coated, if ferrous metal. UL 514C, if nonmetallic.

## 2.4 CABINETS, JUNCTION BOXES, AND PULL BOXES

UL 50; volume greater than 100 cubic inches, NEMA Type 1 enclosure; sheet steel, hot-dip, zinc-coated. Where exposed to wet, damp, or corrosive environments, NEMA Type 4X.

## 2.5 WIRES AND CABLES

Provide wires and cables in accordance with applicable requirements of NFPA 70 and UL for type of insulation, jacket, and conductor specified or indicated. Do not use wires and cables manufactured more than 24 months prior to date of delivery to site.

### 2.5.1 Conductors

Provide the following:

a. Conductor sizes and capacities shown are based on copper, unless indicated otherwise.

b. Conductors No. 10 AWG and larger diameter: stranded.

c. Conductors No. 12 AWG and smaller diameter: solid.

d. Conductors for remote control, alarm, and signal circuits, classes 1, 2, and 3: stranded unless specifically indicated otherwise.

e. All conductors: copper.

#### 2.5.1.1 Minimum Conductor Sizes

Provide minimum conductor size in accordance with the following:

a. Branch circuits: No. 12 AWG.

b. Class 1 remote-control and signal circuits: No. 14 AWG.

c. Class 2 low-energy, remote-control and signal circuits: No. 16 AWG.

d. Class 3 low-energy, remote-control, alarm and signal circuits: No. 22 AWG.

e. Digital low voltage lighting control (DLVLC) system at 24 Volts or less: Category 5 UTP cables in accordance with DLVLC system manufacturer requirements.

### 2.5.2 Color Coding

Provide color coding for service, feeder, branch, control, and signaling circuit conductors.

#### 2.5.2.1 Ground and Neutral Conductors

Provide color coding of ground and neutral conductors as follows:

a. Grounding conductors: Green.

b. Neutral conductors: White.

c. Exception, where neutrals of more than one system are installed in same raceway or box, other neutrals color coding: white with a different colored (not green) stripe for each.

2.5.2.2 Ungrounded Conductors

Provide color coding of ungrounded conductors in different voltage systems as follows:

a. 208/120 volt, three-phase

(1) Phase A - black

(2) Phase B - red

(3) Phase C - blue

2.5.3 Insulation

Unless specified or indicated otherwise or required by NFPA 70, provide power and lighting wires rated for 600-volts, Type THWN/THHN conforming to UL 83 conforming to UL 44, except that grounding wire may be type TW conforming to UL 83; remote-control and signal circuits: Type TW or TF, conforming to UL 83. Where equipment or devices require 90-degree Centigrade (C) conductors, provide only conductors with 90-degree C insulation or better.

2.5.4 Bonding Conductors

ASTM B1, solid bare copper wire for sizes No. 8 AWG and smaller diameter; ASTM B8, Class B, stranded bare copper wire for sizes No. 6 AWG and larger diameter.

2.5.5 Cord Sets and Power-Supply Cords

UL 817.

2.6 DEVICE PLATES

Provide the following:

a. UL listed, one-piece device plates for outlets to suit the devices installed.

b. For metal outlet boxes, plates on unfinished walls: zinc-coated sheet steel or cast metal having round or beveled edges.

c. For nonmetallic boxes and fittings, other suitable plates may be provided.

d. Plates on finished walls: satin finish stainless steel or brushed-finish aluminum, minimum 0.03 inch thick.

e. Screws: machine-type with countersunk heads in color to match finish of plate.

f. Sectional type device plates are not be permitted.

g. Plates installed in wet locations: gasketed and UL listed for "wet

locations."

## 2.7 SWITCHES

### 2.7.1 Toggle Switches

NEMA WD 1, UL 20, single pole, three-way, totally enclosed with bodies of thermoplastic or thermoset plastic and mounting strap with grounding screw. Include the following:

a. Handles: white thermoplastic.

b. Wiring terminals: screw-type, side-wired. Do not use back-stabbed conductor terminals.

c. Contacts: silver-cadmium and contact arm - one-piece copper alloy.

d. Switches: rated quiet-type ac only, 120/277 volts, with current rating and number of poles indicated.

### 2.7.2 Switch with Red Pilot Handle

NEMA WD 1. Provide the following:

a. Pilot lights that are integrally constructed as a part of the switch's handle.

b. Pilot light color: red and illuminate whenever the switch is closed or "on".

c. Pilot lighted switch: rated 20 amps and 120 volts or 277 volts as indicated.

d. The circuit's neutral conductor to each switch with a pilot light.

### 2.7.3 Breakers Used as Switches

For 120- and 277-Volt fluorescent fixtures, mark breakers "SWD" in accordance with UL 489.

### 2.7.4 Disconnect Switches

NEMA KS 1. Provide heavy duty-type switches where indicated, where switches are rated higher than 240 volts, and for double-throw switches. Utilize Class R fuseholders and fuses for fused switches, unless indicated otherwise. Provide horsepower rated for switches serving as the motor-disconnect means. Provide switches in NEMA stainless steel, enclosure as indicated in accordance with NEMA ICS 6.

## 2.8 FUSES

NEMA FU 1. Provide complete set of fuses for each fusible switch. Coordinate time-current characteristics curves of fuses serving motors or connected in series with circuit breakers for proper operation. Submit coordination data for approval. Provide fuses with a voltage rating not less than circuit voltage.

2.8.1 Fuseholders

Provide in accordance with UL 4248-1.

2.8.2 Cartridge Fuses, Current Limiting Type (Class R)

UL 248-12, Class RK-1. Provide only Class R associated fuseholders in accordance with UL 4248-12.

2.8.3 Cartridge Fuses, High-Interrupting Capacity, Current Limiting Type (Classes J, L, and CC)

UL 248-8, UL 248-10, UL 248-4, Class J for zero to 600 amperes, Class L for 601 to 6,000 amperes, and Class CC for zero to 30 amperes.

2.8.4 Cartridge Fuses, Current Limiting Type (Class T)

UL 248-15, Class T for zero to 1,200 amperes, 300 volts; and zero to 800 amperes, 600 volts.

2.9 RECEPTACLES

Provide the following:

a. UL 498, general purpose specification grade, grounding-type. Residential grade receptacles are not acceptable.

b. Ratings and configurations: as indicated.

c. Bodies: white in accordance with NEMA WD 1.

d. Face and body: thermoplastic supported on a metal mounting strap.

e. Dimensional requirements: in accordance with NEMA WD 6.

f. Screw-type, side-wired wiring terminals. Do not use back-stabbed conductor terminals.

g. Grounding pole connected to mounting strap.

h. The receptacle: containing triple-wipe power contacts and double or triple-wipe ground contacts.

2.9.1 Split Duplex Receptacles

Provide separate terminals for each ungrounded pole. One receptacle must be controlled separately.

2.9.2 Weather Resistant Receptacles

Provide receptacles, UL listed for use in "wet locations" with integral GFCI protection. Include cast metal box with gasketed, hinged, lockable and weather resistant while-in-use, die-cast metal/aluminum extra-duty rated hood.

2.9.3 Ground-Fault Circuit Interrupter Receptacles

UL 943, duplex type for mounting in standard outlet box. Provide device capable of detecting current leak when the current to ground is 6

milliamperes or higher, and tripping in accordance with requirements of UL 943 for Class A ground-fault circuit interrupter devices. Provide screw-type, side-wired wiring terminals or pre-wired (pigtail) leads.

## 2.10 PANELBOARDS

Provide panelboards in accordance with the following:

a. UL 67 and UL 50 having a short-circuit current rating as indicated.

b. Panelboards for use as service disconnecting means: additionally conform to UL 869A.

c. Panelboards: circuit breaker-equipped.

d. Designed such that individual breakers can be removed without disturbing adjacent units or without loosening or removing supplemental insulation supplied as means of obtaining clearances as required by UL.

e. "Specific breaker placement" is required in panelboards to match the breaker placement indicated in the panelboard schedule on the design drawings. If it is not possible to match "specific breaker placement" during construction, obtain Government approval prior to device installation.

f. Use of "Subfeed Breakers" is not acceptable.

g. Main breaker: "separately" mounted "above" branch breakers.

h. Where "space only" is indicated, make provisions for future installation of breakers.

i. Directories: indicate load served by each circuit in panelboard.

j. Directories: indicate source of service to panelboard (e.g., Panel PA served from Panel MDP).

k. Provide new directories for existing panels modified by this project as indicated.

l. Type directories and mount in holder behind transparent protective covering.

m. Panelboards: listed and labeled for their intended use.

n. Panelboard nameplates: provided in accordance with paragraph FIELD FABRICATED NAMEPLATES.

### 2.10.1 Enclosure

Provide panelboard enclosure in accordance with the following:

a. UL 50.

b. Cabinets mounted outdoors or flush-mounted: hot-dipped galvanized after fabrication.

c. Cabinets: painted in accordance with paragraph FIELD APPLIED PAINTING.

d. Outdoor cabinets: NEMA 3R raintight a removable steel plate 1/4 inch thick in the bottom for field drilling for conduit connections.

e. Front edges of cabinets: form-flanged or fitted with structural shapes welded or riveted to the sheet steel, for supporting the panelboard front.

f. All cabinets: fabricated such that no part of any surface on the finished cabinet deviates from a true plane by more than 1/8 inch.

g. Holes: provided in the back of indoor surface-mounted cabinets, with outside spacers and inside stiffeners, for mounting the cabinets with a 1/2 inch clear space between the back of the cabinet and the wall surface.

h. Flush doors: mounted on hinges that expose only the hinge roll to view when the door is closed.

i. Each door: fitted with a combined catch and lock latch.

j. Keys: two provided with each lock, with all locks keyed alike.

k. Finished-head cap screws: provided for mounting the panelboard fronts on the cabinets.

2.10.2 Panelboard Buses

Support bus bars on bases independent of circuit breakers. Design main buses and back pans so that breakers may be changed without machining, drilling, or tapping. Provide isolated neutral bus in each panel for connection of circuit neutral conductors. Provide separate ground bus identified as equipment grounding bus in accordance with UL 67 for connecting grounding conductors; bond to steel cabinet.

2.10.2.1 Panelboard Neutrals for Non-Linear Loads

Provide in accordance with the following:.

a. UL listed, with panelboard type specifically UL heat rise tested for use on non-linear loads.

b. Panelboard: heat rise tested in accordance with UL 67, except with the neutral assembly installed and carrying 200 percent of the phase bus current during testing.

c. Verification of the testing procedure: provided upon request.

d. Two neutral assemblies paralleled together with cable is not acceptable.

e. Nameplates for panelboard rated for use on non-linear loads: marked "SUITABLE FOR NON-LINEAR LOADS" and in accordance with paragraph FIELD FABRICATED NAMEPLATES.

f. Provide a neutral label with instructions for wiring the neutral of panelboards rated for use on non-linear loads.

2.10.3 Circuit Breakers

UL 489, thermal magnetic-type having a minimum short-circuit current rating equal to the short-circuit current rating of the panelboard in which the circuit breaker will be mounted. Breaker terminals: UL listed as suitable for type of conductor provided. Series rated circuit breakers and plug-in circuit breakers that do not use a retaining bolt are unacceptable.

2.10.3.1 Multipole Breakers

Provide common trip-type with single operating handle. Design breaker such that overload in one pole automatically causes all poles to open. Maintain phase sequence throughout each panel so that any three adjacent breaker poles are connected to Phases A, B, and C, respectively.

2.10.3.2 Circuit Breaker With Ground-Fault Circuit Interrupter

UL 943 and NFPA 70. Provide with auto-monitoring (self-test) and lockout features, "push-to-test" button, visible indication of tripped condition, and ability to detect and trip when current imbalance is 6 milliamperes or higher in accordance with requirements of UL 943 for Class A ground-fault circuit interrupter devices.

2.10.3.3 LSIG Type Circuit Breakers

Provide circuit breaker rated for the appropriate current and voltage level as indicated by the system design and a short circuit current capacity as indicated for the system's fault levels. The circuit breaker must have a reliable operating mechanism for manual and automatic operation. Provide circuit breaker with a suitable enclosure for the installation environment (e.g., indoor, outdoor, hazardous locations) and be suitable for the specified environmental conditions (e.g., temperature, humidity, altitude).

2.10.3.3.1 Long-Time Protection (L)

Provide adjustable long-time current protection with a time delay measured in seconds to safeguard against sustained, moderate levels of overcurrent.

2.10.3.3.2 Short-Time Protection (S)

Provide adjustable short-time current protection with a time delay measured in seconds to respond to temporary overcurrent conditions.

2.10.3.3.3 Instantaneous Protection (I)

Provide adjustable instantaneous current protection with no intentional delay to interrupt severe short circuits.

2.10.3.3.4 Ground Fault Protection (G)

Provide ground fault protection with adjustable settings to detect and interrupt ground fault conditions.

2.11 ENCLOSED CIRCUIT BREAKERS

UL 489. Individual molded case circuit breakers with voltage and continuous current ratings, number of poles, overload trip setting, and

short circuit current interrupting rating as indicated. Enclosure type as indicated. Provide solid neutral.

## 2.12 LOCKOUT REQUIREMENTS

Provide circuit breakers, disconnecting means, and other devices that are electrical energy-isolating capable of being locked out for machines and other equipment to prevent unexpected startup or release of stored energy in accordance with 29 CFR 1910.147, NFPA 70E and 29 CFR 1910.303. Comply with requirements of Division 23, "Heating, Ventilating, and Air Conditioning (HVAC)" for mechanical isolation of machines and other equipment.

## 2.13 GROUNDING AND BONDING EQUIPMENT

### 2.13.1 Ground Rods

UL 467. Ground rods: cone pointed copper-clad steel, with minimum diameter of 3/4 inch and minimum length 10 feet. Sectional type rods may be used for rods 20 feet or longer.

## 2.14 NAMEPLATES AND SIGNS

### 2.14.1 Manufacturer's Nameplate

Provide on each item of equipment a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

### 2.14.2 Field Fabricated Nameplates

Provide field fabricated nameplates in accordance with the following:

a. ASTM D709.

b. Provide laminated plastic nameplates for each equipment enclosure, relay, switch, and device; as specified or as indicated on the drawings.

c. Each nameplate inscription: identify the function and, when applicable, the position.

d. Nameplates: melamine plastic, 0.125 inch thick, white with black center core.

e. Provide red laminated plastic label with white center core where indicated.

f. Surface: matte finish. Corners: square. Accurately align lettering and engrave into the core.

g. Minimum size of nameplates: one by 2.5 inches.

h. Lettering size and style: a minimum of 0.25 inch high normal block style.

2.14.3 Warning Signs

Provide warning signs for flash protection in accordance with NFPA 70E and NEMA Z535.4 for switchboards, panelboards, industrial control panels, and motor control centers that are in other than dwelling occupancies and are likely to require examination, adjustment, servicing, or maintenance while energized. Provide field installed signs to warn qualified persons of potential electric arc flash hazards when warning signs are not provided by the manufacturer. Provide marking that is clearly visible to qualified persons before examination, adjustment, servicing, or maintenance of the equipment.

2.15 FIRESTOPPING MATERIALS

Provide firestopping around electrical penetrations.

2.16 WIREWAYS

UL 870. Material: steel galvanized 16 gauge for heights and depths up to 6 by 6 inches, and 14 gauge for heights and depths up to 12 by 12 inches. Provide in length required for the application with screw-cover NEMA 1 3R enclosure in accordance with NEMA ICS 6.

2.17 SURGE PROTECTIVE DEVICES

Provide parallel type surge protective devices (SPD) which comply with UL 1449 at the service entrance, panelboards. Provide surge protectors in a NEMA 3R enclosure in accordance with NEMA ICS 6. Do not install SPD inside a panelboard or switchboard enclosure. However, SPD may be installed in a separate compartment of a switchgear provided that it is supplied by a dedicated circuit breaker. SPD must have the same short-circuit current rating as the protected equipment and must not be installed at a point of system where the available fault current is in excess of that rating. Use Type 1 or Type 2 SPD and connect on the load side of a dedicated circuit breaker. Submit performance and characteristic curves.

Provide the following modes of protection:

FOR SINGLE PHASE AND THREE PHASE WYE CONNECTED SYSTEMS-

Phase to phase ( L-L )

Each phase to neutral ( L-N )

Neutral to ground ( N-G )

Phase to ground ( L-G )

FOR DELTA CONNECTIONS-

Phase to phase ( L-L )

Phase to ground ( L-G )

The minimum MCOV (Maximum Continuous Operating Voltage) rating for L-N and L-G modes of operation: 120 percent of nominal voltage for 240 volts and below; 115 percent of nominal voltage above 240 volts to 480 volts.

2.17.1 Warranty

Provide not less than a 5-year warranty and include unlimited free replacements of the unit if destroyed by lightning or other transients during the warranty period.

2.18 FACTORY APPLIED FINISH

Provide factory-applied finish on electrical equipment in accordance with the following:

a. NEMA 250 corrosion-resistance test and the additional requirements as specified herein.

b. Interior and exterior steel surfaces of equipment enclosures: thoroughly cleaned followed by a rust-inhibitive phosphatizing or equivalent treatment prior to painting.

c. Exterior surfaces: free from holes, seams, dents, weld marks, loose scale or other imperfections.

d. Interior surfaces: receive not less than one coat of corrosion-resisting paint in accordance with the manufacturer's standard practice.

e. Exterior surfaces: primed, filled where necessary, and given not less than two coats baked enamel with semigloss finish.

f. Equipment located indoors: ANSI Light Gray, and equipment located outdoors: ANSI Light Gray Dark Gray.

g. Provide manufacturer's coatings for touch-up work and as specified in paragraph FIELD APPLIED PAINTING.

2.19 SOURCE QUALITY CONTROL

2.19.1 Transformer Factory Tests

Submittal: include routine NEMA ST 20 transformer test results on each transformer and also provide the results of NEMA "design" and "prototype" tests that were made on transformers electrically and mechanically equal to those specified.

PART 3 EXECUTION

3.1 INSTALLATION

Electrical installations, including weatherproof and hazardous locations and ducts, plenums and other air-handling spaces: conform to requirements of NFPA 70 and IEEE C2 and to requirements specified herein.

3.1.1 Wiring Methods

Provide insulated conductors installed in rigid steel conduit, IMC, rigid nonmetallic conduit, or EMT, except where specifically indicated or specified otherwise or required by NFPA 70 to be installed otherwise. Grounding conductor: separate from electrical system neutral conductor. Provide insulated green equipment grounding conductor for circuit(s) installed in conduit and raceways. Shared neutral, or multi-wire branch

circuits, are not permitted with arc-fault circuit interrupters. Minimum conduit size: 1/2 inch in diameter for low voltage lighting and power circuits. Vertical distribution in multiple story buildings: made with metal conduit in fire-rated shafts, with metal conduit extending through shafts for minimum distance of 6 inches. Firestop conduit which penetrates fire-rated walls, fire-rated partitions, or fire-rated floors in accordance with Section 07 84 00 FIRESTOPPING.

3.1.1.1 Pull Wire

Install pull wires in empty conduits. Pull wire: plastic having minimum 200-pound force tensile strength. Leave minimum 36 inches of slack at each end of pull wire.

3.1.1.2 Metal-Clad Cable

Install in accordance with NFPA 70, Type MC cable. MC Cable is also allowed for branch circuits in the following locations:

a. New construction in exposed locations.

b. Renovations in exposed locations.

c. Concealed in renovations in existing areas where walls and ceilings are not disturbed.

d. Cable trays.

MC Cable is prohibited in the following locations:

a. Feeder circuits other than feeder circuits for aerial messengers between buildings.

b. Embedded or direct buried locations.

c. Concealed locations, except as allowed above.

3.1.1.3 Armored Cable

Install in accordance with NFPA 70, Type AC cable.

3.1.2 Conduit Installation

Unless indicated otherwise, conceal conduit under floor slabs and within finished walls, ceilings, and floors. Keep conduit minimum 6 inches away from parallel runs of flues and steam or hot water pipes. Install conduit parallel with or at right angles to ceilings, walls, and structural members where located above accessible ceilings and where conduit will be visible after completion of project.

3.1.2.1 Restrictions Applicable to Aluminum Conduit

a. Do not install underground or encase in concrete or masonry.

b. Do not use brass or bronze fittings.

c. Do not use when the enclosed conductors must be shielded from the effects of High-altitude Electromagnetic Pulse (HEMP).

3.1.2.2 Restrictions Applicable to EMT

a. Do not install underground.

b. Do not encase in concrete, mortar, grout, or other cementitious materials.

c. Do not use in areas subject to physical damage including but not limited to equipment rooms where moving or replacing equipment could physically damage the EMT.

d. Do not use in hazardous areas.

e. Do not use outdoors.

f. Do not use in fire pump rooms.

g. Do not use when the enclosed conductors must be shielded from the effects of High-altitude Electromagnetic Pulse (HEMP).

3.1.2.3 Restrictions Applicable to Nonmetallic Conduit

a. PVC Schedule 40.

(1) Do not use where subject to physical damage, including but not limited to, mechanical equipment rooms, electrical equipment rooms, fire pump rooms, and where restrictions are applying to both PVC Schedule 40 and PVC Schedule 80.

(2) Do not use above grade, except where allowed in this section for rising through floor slab or indicated otherwise.

b. PVC Schedule 80.

(1) Do not use where subject to physical damage, including but not limited to, hospitals, power plant, missile magazines, and other such areas.

(2) Do not use in hazardous (classified) areas.

(3) Do not use in penetrating fire-rated walls or partitions, or fire-rated floors.

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3.1.2.4 Restrictions Applicable to Flexible Conduit

Use only as specified in paragraph FLEXIBLE CONNECTIONS. Do not use when the enclosed conductors must be shielded from the effects of High-altitude Electromagnetic Pulse (HEMP).

3.1.2.5 Conduit Support

Support conduit by pipe straps, wall brackets, threaded rod conduit hangers, or ceiling trapeze. Plastic cable ties are not acceptable. Fasten by wood screws to wood; by toggle bolts on hollow masonry units; by concrete inserts or expansion bolts on concrete or brick; and by machine screws, welded threaded studs, or spring-tension clamps on steel work. Threaded C-clamps may be used on rigid steel conduit only. Do not weld conduits or pipe straps to steel structures. Do not exceed one-fourth proof test load for load applied to fasteners. Provide vibration

resistant and shock-resistant fasteners attached to concrete ceiling. Do not cut main reinforcing bars for any holes cut to depth of more than 1 1/2 inches in reinforced concrete beams or to depth of more than 3/4 inch in concrete joints. Fill unused holes. In partitions of light steel construction, use sheet metal screws. In suspended-ceiling construction, run conduit above ceiling. Do not support conduit by ceiling support system. Conduit and box systems: supported independently of both (a) tie wires supporting ceiling grid system, and (b) ceiling grid system into which ceiling panels are placed. Do not share supporting means between electrical raceways and mechanical piping or ducts. Identify independent conduit support in both fire and non-fire rated assemblies in accordance with NFPA 70. Coordinate installation with above-ceiling mechanical systems to assure maximum accessibility to all systems. Spring-steel fasteners may be used for lighting branch circuit conduit supports in suspended ceilings in dry locations. Where conduit crosses building expansion joints, provide suitable watertight expansion fitting that maintains conduit electrical continuity by bonding jumpers or other means. For conduits greater than 2 1/2 inches inside diameter, provide supports to resist forces of 0.5 times the equipment weight in any direction and 1.5 times the equipment weight in the downward direction.

3.1.2.6 Directional Changes in Conduit Runs

Make changes in direction of runs with symmetrical bends or cast-metal fittings. Make field-made bends and offsets with hickey or conduit-bending machine. Do not install crushed or deformed conduits. Avoid trapped conduits. Prevent plaster, dirt, or trash from lodging in conduits, boxes, fittings, and equipment during construction. Free clogged conduits of obstructions.

3.1.2.7 Locknuts and Bushings

Fasten conduits to sheet metal boxes and cabinets with two locknuts where required by NFPA 70, where insulated bushings are used, and where bushings cannot be brought into firm contact with the box; otherwise, use at least minimum single locknut and bushing. Provide locknuts with sharp edges for digging into wall of metal enclosures. Install bushings on ends of conduits, and provide insulating type where required by NFPA 70.

3.1.2.8 Flexible Connections

Provide flexible steel conduit between 3 and 6 feet in length for recessed and semirecessed lighting fixtures; for equipment subject to vibration, noise transmission, or movement; and for motors. Install flexible conduit to allow 20 percent slack. Minimum flexible steel conduit size: 1/2 inch diameter. Provide liquid tight flexible conduit in wet and damp locations for equipment subject to vibration, noise transmission, movement or motors. Provide separate ground conductor across flexible connections. Plastic cable ties are not acceptable as a support method.

3.1.3 Boxes, Outlets, and Supports

Provide boxes in wiring and raceway systems wherever required for pulling of wires, making connections, and mounting of devices or fixtures. Boxes for metallic raceways: cast-metal, hub-type when located in wet locations, when surface mounted on outside of exterior surfaces, when surface mounted on interior walls exposed up to 7 feet above floors and walkways, and when specifically indicated. Boxes in other locations: sheet steel, except that aluminum boxes may be used with aluminum conduit,

and nonmetallic boxes may be used with nonmetallic conduit system. Provide each box with volume required by NFPA 70 for number of conductors enclosed in box. Boxes for mounting lighting fixtures: minimum 4 inches square, or octagonal, except that smaller boxes may be installed as required by fixture configurations, as approved. Boxes for use in masonry-block or tile walls: square-cornered, tile-type, or standard boxes having square-cornered, tile-type covers. Provide gaskets for cast-metal boxes installed in wet locations and boxes installed flush with outside of exterior surfaces. Provide separate boxes for flush or recessed fixtures when required by fixture terminal operating temperature; provide readily removable fixtures for access to boxes unless ceiling access panels are provided. Support boxes and pendants for surface-mounted fixtures on suspended ceilings independently of ceiling supports. Fasten boxes and supports with wood screws on wood, with bolts and expansion shields on concrete or brick, with toggle bolts on hollow masonry units, and with machine screws or welded studs on steel. In open overhead spaces, cast boxes threaded to raceways need not be separately supported except where used for fixture support; support sheet metal boxes directly from building structure or by bar hangers. Where bar hangers are used, attach bar to raceways on opposite sides of box, and support raceway with approved-type fastener maximum 24 inches from box. When penetrating reinforced concrete members, avoid cutting reinforcing steel.

3.1.3.1 Boxes

Boxes for use with raceway systems: minimum 1 1/2 inches deep, except where shallower boxes required by structural conditions are approved. Boxes for other than lighting fixture outlets: minimum 4 inches square, except that 4 by 2 inch boxes may be used where only one raceway enters outlet.

3.1.3.2 Pull Boxes

Construct of at least minimum size required by NFPA 70 of code-gauge aluminum or galvanized sheet steel, compatible with nonmetallic raceway systems, except where cast-metal boxes are required in locations specified herein. Provide boxes with screw-fastened covers. Where several feeders pass through common pull box, tag feeders to indicate clearly electrical characteristics, circuit number, and panel designation.

3.1.4 Mounting Heights

Mount panelboards, circuit breakers, motor controller and disconnecting switches so height of center of grip of the operating handle of the switch or circuit breaker at its highest position is maximum 79 inches above floor or working platform or as allowed in Section 404.8 in accordance with NFPA 70. Mount lighting switches 48 inches above finished floor. Mount receptacles and telecommunications outlets 18 inches above finished floor, unless otherwise indicated. Measure mounting heights of wiring devices and outlets to center of device or outlet.

3.1.5 Conductor Identification

Provide conductor identification within each enclosure where tap, splice, or termination is made. For conductors No. 6 AWG and smaller diameter, provide color coding by factory-applied, color-impregnated insulation. For conductors No. 4 AWG and larger diameter, provide color coding by plastic-coated, self-sticking markers; colored nylon cable ties and plates; or heat shrink-type sleeves.

3.1.5.1 Marking Strips

Provide marking strips for identification of power distribution, control, data, and communications cables in accordance with the following:

a. Provide white or other light-colored plastic marking strips, fastened by screws to each terminal block, for wire designations.

b. Use permanent ink for the wire numbers

c. Provide reversible marking strips to permit marking both sides, or provide two marking strips with each block.

d. Size marking strips to accommodate the two sets of wire numbers.

e. Assign a device designation in accordance with NEMA ICS 1 to each device to which a connection is made. Mark each device terminal to which a connection is made with a distinct terminal marking corresponding to the wire designation used on the Contractor's schematic and connection diagrams.

f. The wire (terminal point) designations used on the Contractor's wiring diagrams and printed on terminal block marking strips may be according to the Contractor's standard practice; however, provide additional wire and cable designations for identification of remote (external) circuits for the Government's wire designations.

g. Prints of the marking strips drawings submitted for approval will be so marked and returned to the Contractor for addition of the designations to the terminal strips and tracings, along with any rearrangement of points required.

3.1.6 Splices

Make splices in accessible locations. Make splices in conductors No. 10 AWG and smaller diameter with insulated, pressure-type connector. Make splices in conductors No. 8 AWG and larger diameter with solderless connector, and cover with insulation material equivalent to conductor insulation.

3.1.7 Covers and Device Plates

Install with edges in continuous contact with finished wall surfaces without use of mats or similar devices. Plaster fillings are not permitted. Install plates with alignment tolerance of 1/16 inch. Use of sectional-type device plates are not permitted. Provide gasket for plates installed in wet locations.

3.1.7.1 Circuit Identification

Indicate the electrical source(s) for the conductors supplying the receptacles or light switches at a given cover/device plate using a 'Panel-Circuit Number' naming convention (e.g., L1-23 for a circuit supplied from Panel L1, Pole Number 23). Labels must be black on a white field. Labels must be digitally printed; handwritten lables are unacceptable.

### 3.1.8 Electrical Penetrations

Seal openings around electrical penetrations through fire resistance-rated walls, partitions, floors, or ceilings.

### 3.1.9 Grounding and Bonding

Provide in accordance with NFPA 70. Ground exposed, non-current-carrying metallic parts of electrical equipment, metallic raceway systems, grounding conductor in metallic and nonmetallic raceways, telecommunications system grounds, grounding conductor of nonmetallic sheathed cables, and neutral conductor of wiring systems. Make ground connection to driven ground rods on exterior of building. Bond additional driven rods together with a minimum of 4 AWG soft bare copper wire buried to a depth of at least 12 inches. Interconnect all grounding media in or on the structure to provide a common ground potential. This includes lightning protection, electrical service, telecommunications system grounds, as well as underground metallic piping systems. Make interconnection to the gas line on the customer's side of the meter. Use main size lightning conductors for interconnecting these grounding systems to the lightning protection system. Where ground fault protection is employed, ensure that connection of ground and neutral does not interfere with correct operation of fault protection.

#### 3.1.9.1 Ground Rods

Provide ground rods and measure the resistance to ground using the fall-of-potential method described in IEEE 81. Do not exceed 25 ohms under normally dry conditions for the maximum resistance of a driven ground. If this resistance cannot be obtained, provide 2 additional rods, spaced on center. Spacing for additional rods must be a minimum of 10 feet. If the resultant resistance exceeds 25 ohms measured not less than 48 hours after rainfall, notify the Contracting Officer who will decide on the number of ground rods to add.

#### 3.1.9.2 Grounding Connections

Make grounding connections which are buried or otherwise normally inaccessible, by exothermic weld or high compression connector.

a. Make exothermic welds strictly in accordance with the weld manufacturer's written recommendations. Welds which are "puffed up" or which show convex surfaces indicating improper cleaning are not acceptable. Mechanical connectors are not required at exothermic welds.

b. Make high compression connections using a hydraulic or electric compression tool to provide the correct circumferential pressure. Provide tools and dies as recommended by the manufacturer. Use an embossing die code or other standard method to provide visible indication that a connector has been adequately compressed on the ground wire.

#### 3.1.9.3 Resistance

Maximum resistance-to-ground of grounding system: do not exceed 5 ohms under dry conditions. Where resistance obtained exceeds 5 ohms, contact Contracting Officer for further instructions.

3.1.10 Equipment Connections

Provide power wiring and connections to motors and equipment. Except as otherwise specifically noted or specified, automatic control wiring, control devices, and protective devices within the control circuitry are not included in this section of the specifications and are provided under the section specifying the associated equipment.

3.1.11 Government-Furnished Equipment

Contractor rough-in for Government-furnished equipment to make equipment operate as intended, including providing miscellaneous items such as plugs, receptacles, wire, cable, conduit, flexible conduit, and outlet boxes or fittings.

3.1.12 Repair of Existing Work

Perform repair of existing work, demolition, and modification of existing electrical distribution systems as follows:

3.1.12.1 Workmanship

Lay out work in advance. Exercise care where cutting, channeling, chasing, or drilling of floors, walls, partitions, ceilings, or other surfaces is necessary for proper installation, support, or anchorage of conduit, raceways, or other electrical work. Repair damage to buildings, piping, and equipment using skilled craftsmen of trades involved.

3.1.12.2 Existing Concealed Wiring to be Removed

Disconnect existing concealed wiring to be removed from its source. Remove conductors; cut conduit flush with floor, underside of floor, and through walls; and seal openings.

3.1.12.3 Continuation of Service

Maintain continuity of existing circuits of equipment to remain. Maintain existing circuits of equipment energized. Restore circuits wiring and power which are to remain but were disturbed during demolition back to original condition.

3.1.13 Surge Protective Devices

Connect the surge protective devices in parallel to the power source, keeping the conductors as short and straight as practically possible. Maximum allowed lead length is 3 feet avoiding 90 degree bends. Do not locate surge protective devices inside a panelboard or switchboard enclosure.

3.2 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets. Provide nameplate on all equipment in access controlled spaces and areas.

3.3 WARNING SIGN MOUNTING

Provide the number of signs required to be readable from each accessible

side. Space the signs in accordance with NFPA 70E.

## 3.4 FIELD APPLIED PAINTING

Paint electrical equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Where field painting of enclosures for panelboards, load centers or the like is specified to match adjacent surfaces, to correct damage to the manufacturer's factory applied coatings, or to meet the indicated or specified safety criteria, provide manufacturer's recommended coatings and apply in accordance to manufacturer's instructions.

## 3.5 FIELD QUALITY CONTROL

Furnish test equipment and personnel and submit written copies of test results. Give Contracting Officer 5 working days notice prior to each test s. Where applicable, test electrical equipment in accordance with NETA ATS.

### 3.5.1 Devices Subject to Manual Operation

Operate each device subject to manual operation at least five times, demonstrating satisfactory operation each time.

### 3.5.2 600-Volt Wiring Test

Test wiring rated 600 volt and less to verify that no short circuits or accidental grounds exist. Perform insulation resistance tests on wiring No. 6 AWG and larger diameter using instrument which applies voltage of 1,000 volts DC for 600 volt rated wiring and 500 volts DC for 300 volt rated wiring in accordance with NETA ATS to provide direct reading of resistance. All existing wiring to be reused must also be tested.

### 3.5.3 Ground-Fault Receptacle Test

Test ground-fault receptacles with a "load" (such as a plug in light) to verify that the "line" and "load" leads are not reversed. Press the TEST button and then the RESET button to verify by LED status that the device is a self-test model as specified in UL 943.

### 3.5.4 Grounding System Test

Test grounding system to ensure continuity, and that resistance to ground is not excessive. Test each ground rod for resistance to ground before making connections to rod; tie grounding system together and test for resistance to ground. Make resistance measurements in dry weather, not earlier than 48 hours after rainfall. Submit written results of each test to Contracting Officer, and indicate location of rods as well as resistance and soil conditions at time measurements were made.

### 3.5.5 Phase Rotation Test

Perform phase rotation test to ensure proper rotation of service power prior to operation of new or reinstalled equipment using a phase rotation meter. Follow the meter manual directions performing the test.

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SECTION 26 51 00

INTERIOR LIGHTING
**05/20, CHG 2: 11/21**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

| | |
|---|---|
| ASTM A580/A580M | (2023) Standard Specification for Stainless Steel Wire |
| ASTM A641/A641M | (2019; R 2025) Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire |
| ASTM A653/A653M | (2025a) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process |
| ASTM A1008/A1008M | (2025) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Required Hardness, Solution Hardened, and Bake Hardenable |
| ASTM B164 | (2003; R 2014) Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire |
| ASTM B633 | (2023) Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel |

EUROPEAN UNION (EU)

| | |
|---|---|
| Directive 2011/65/EU | (2011) Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment |

ILLUMINATING ENGINEERING SOCIETY (IES)

| | |
|---|---|
| ANSI/IES LM-79 | (2019) Approved Method: Electrical and Photometric Measurements of Solid State Lighting Products |
| ANSI/IES LM-80 | (2020) Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays and Modules |
| ANSI/IES LS-1 | (2020) Lighting Science: Nomenclature and Definitions for Illuminating Engineering |

| | |
|---|---|
| ANSI/IES TM-21 | (2021) Technical Memorandum: Projecting Long-TermLuminous, Photon, and Radiant Flux Maintenance of LED Light Sources |
| ANSI/IES TM-30 | (2020) Technical Memorandum: IES Method for Evaluating Light Source Color Rendition |
| IES Lighting Library | IES Lighting Library |

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

| | |
|---|---|
| IEEE 100 | (2000; Archived) The Authoritative Dictionary of IEEE Standards Terms |
| IEEE C2 | (2023) National Electrical Safety Code |

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

| | |
|---|---|
| ANSI C78.54 | (2019) Specification Sheet for Tubular Fluorescent Replacement and Retrofit LED Lamps |
| NEMA 250 | (2020) Enclosures for Electrical Equipment (1000 Volts Maximum) |
| NEMA ANSLG C78.377 | (2024) Electric Lamps— Specifications for the Chromaticity of Solid State Lighting Products |
| NEMA C82.77-10 | (2020) Harmonic Emission Limits - Related Power Quality Requirements |
| NEMA ICS 6 | (1993; R 2016) Industrial Control and Systems: Enclosures |
| NEMA SSL 1 | (2016) Electronic Drivers for LED Devices, Arrays, or Systems |
| NEMA SSL 3 | (2011) High-Power White LED Binning for General Illumination |

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

| | |
|---|---|
| NFPA 70 | (2026; TIA 26-1; ERTA 26-1; TIA 26-2; TIA 26-3; TIA 26-4; TIA 26-5; TIA 26-6; TIA 26-7; ERTA 26-2; ERTA 26-3) National Electrical Code |
| NFPA 101 | (2024) Life Safety Code |

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

| | |
|---|---|
| 47 CFR 15 | Radio Frequency Devices |

UL SOLUTIONS (UL)

| | |
|---|---|
| UL 917 | (2006; Reprint Aug 2013) UL Standard for Safety Clock-Operated Switches |

| | |
|---|---|
| UL 1598 | (2021; Reprint Jan 2024) UL Standard for Safety Luminaires |
| UL 1993 | (2017) Self-Ballasted Lamps and Lamp Adapters |
| UL 8750 | (2020; Reprint Aug 2024) UL Standard for Safety Light Emitting Diode (LED) Equipment for Use in Lighting Products |

1.2 RELATED REQUIREMENTS

Materials not considered to be luminaires, luminaire accessories, or lighting equipment are specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Luminaires and accessories that are mounted in exterior environments and not attached to the exterior of the building are specified in Section 26 56 00 EXTERIOR LIGHTING.

1.3 DEFINITIONS

a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications and on the drawings, must be as defined in IEEE 100 and ANSI/IES LS-1.

b. For LED luminaire light sources, "Useful Life" is the operating hours before reaching 70 percent of the initial rated lumen output (L70) with no catastrophic failures under normal operating conditions. This is also known as 70 percent "Rated Lumen Maintenance Life" as defined in ANSI/IES LM-80.

c. For LED luminaires, "Luminaire Efficacy" (LE) is the appropriate measure of energy efficiency, measured in lumens/watt. This is gathered from LM-79 data for the luminaire, in which absolute photometry is used to measure the lumen output of the luminaire as one entity, not the source separately and then the source and housing together.

d. Total harmonic distortion (THD) is the root mean square (RMS) of all the harmonic components divided by the total fundamental current.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Luminaire Drawings; G

Lighting Control System One-Line Diagram; G

Sequence of Operation for Lighting Control System; G

SD-03 Product Data

Luminaires; G

Light Sources; G

LED Drivers; G

Luminaire Warranty; G

Lighting Controls Warranty; G

Local Area Controller; G

Switches; G

Time Clocks; G

Power Hook Luminaire Hangers; G

SD-05 Design Data

Luminaire Design Data; G

Photometric Plan; G

SD-06 Test Reports

ANSI/IES LM-79 Test Report; G

ANSI/IES LM-80 Test Report; G

ANSI/IES TM-21 Test Report; G

ANSI/IES TM-30 Test Report; G

Occupancy/Vacancy Sensor Verification Test; G

SD-07 Certificates

LED Driver and Dimming Switch Compatibility Certificate; G

SD-10 Operation and Maintenance Data

Lighting System, Data Package 5; G

Lighting Control System, Data Package 5; G

Maintenance Staff Training Plan; G

End-User Training Plan; G

## 1.5 QUALITY ASSURANCE

Data, drawings, and reports must employ the terminology, classifications and methods prescribed by the IES Lighting Library as applicable, for the lighting system specified.

### 1.5.1 Luminaire Drawings

Include dimensions, accessories installation details, and construction details. Photometric data, including CRI, CCT, LED driver type, aiming

diagram, zonal lumen data, and candlepower distribution data must accompany shop drawings.

1.5.2 Luminaire Design Data

a. Provide safety certification and file number for the luminaire family that must be listed, labeled, or identified in accordance with the NFPA 70. Applicable testing bodies are determined by the US Occupational Safety Health Administration (OSHA) as Nationally Recognized Testing Laboratories (NRTL) and include: CSA (Canadian Standards Association), ETL (Edison Testing Laboratory), and UL (Underwriters Laboratories).

b. Provide long term lumen maintenance projections for each LED luminaire in accordance with ANSI/IES TM-21. Data used for projections must be obtained from testing in accordance with ANSI/IES LM-80.

1.5.3 ANSI/IES LM-79 Test Report

Submit test report on manufacturer's standard production model of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data in IES format as outlined under "14.0 Test Report" in ANSI/IES LM-79.

1.5.4 ANSI/IES LM-80 Test Report

Submit report on manufacturer's standard production LED light source (package, array, or module) of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data as outlined under "8.0 Test Report" in ANSI/IES LM-80.

1.5.5 ANSI/IES TM-21 Test Report

Submit test report on manufacturer's standard production LED light source (package, array, or module) of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data, as well as required interpolation information as outlined under "7.0 Report" in ANSI/IES TM-21.

1.5.6 ANSI/IES TM-30 Test Report

Submit color vector graphic in accordance with ANSI/IES TM-30 on manufacturer's standard production LED light source (package, array, or module) of specified luminaire. Include spectral distribution of test LED light source.

1.5.7 LED Driver and Dimming Switch Compatibility Certificate

Submit certification from the luminaire, driver, or dimmer switch manufacturer that ensures compatibility and operability between devices without flickering and to specified dimming levels.

1.5.8 Photometric Plan

1.5.8.1 Computer-generated Photometric Plans

Computer-generated photometric plans for each space are required to verify proposed luminaires and locations meet the required performance criteria

of the design using the applicable light loss factor (LLF).

Target illumination levels are provided for each Interior Application. Depending on the application and the recommendations provided by the IES, values are given as one of the following:

a. Minimum: No values anywhere on the calculation grid may be less than this value, within a 10 percent margin of error.

b. Minimum Average: An average, taken over the entire task area for the application, may not be less than this value, within a 10 percent margin of error.

c. Maximum: No values anywhere on the calculation grid may be greater than this value, within a 10 percent margin of error.

d. Maximum Average: An average, taken over the entire task area for the application, may not be greater than this value, within a 10 percent margin of error.

e. Uniformity: Unless otherwise noted, uniformity is calculated as a ratio of the average calculated illuminance over the minimum calculated illuminance of the calculation grid.

1.5.8.2 Schematic Photometric Plan Calculations

Schematic photometric plan calculations must include:

a. Horizontal illuminance measurements at workplane or other designated height above finished floor, taken at a maximum of every one foot across the task area.

b. Average maintained illuminance level.

c. Minimum and maximum maintained illuminance levels.

d. Lighting power density (Watts per square foot).

e. LLF. Recommended LLF is 0.81 for LED luminaires but LLF varies based on environment and application.

1.5.8.3 Final Photometric Plan Calculations

Final photometric plan calculations must include:

a. Horizontal illuminance measurements at workplane or other designated height above finished floor, taken at a maximum of every one foot across the task area.

b. Where applicable, vertical illuminance measurements at designated surface, taken at a maximum of every one foot across task area.

c. Minimum and maximum maintained illuminance levels.

d. Average maintained illuminance level.

e. Average to minimum and maximum to minimum ratios for horizontal illuminance.

f. Lighting power density (Watts per square foot).

g. LLF. Recommended LLF is 0.81 for LED luminaires but LLF varies based on environment and application.

1.5.9 Regulatory Requirements

Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions of NFPA 70, unless more stringent requirements are specified or indicated. Provide luminaires and assembled components that are approved by and bear the label of UL for the applicable location and conditions unless otherwise specified.

1.5.10 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design, and workmanship. Products must have been in satisfactory commercial or industrial use for six months prior to bid opening. The six-month period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the six-month period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.10.1 Alternative Qualifications

Products having less than a six-month field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.5.10.2 Material and Equipment Manufacturing Date

Do not use products manufactured more than six months prior to date of delivery to site, unless specified otherwise.

1.6 WARRANTY

Support all equipment items by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.6.1 Luminaire Warranty

Provide and transfer to the government the original LED luminaire manufacturers standard commercial warranty for each different luminaire manufacturer used in the project.

a. Provide a written five year minimum replacement warranty for material, luminaire finish, and workmanship. Provide written warranty document that contains all warranty processing information needed, including customer service point of contact, whether or not a return authorization number is required, return shipping information, and closest return location to the luminaire location.

(1) Finish warranty must include failure and substantial deterioration such as blistering, cracking, peeling, chalking, or fading.

(2) Material warranty must include:

(a) All LED drivers and integral control equipment.

(b) Replacement when more than 15 percent of LED sources in any lightbar or subassembly(s) are defective, non-starting, or operating below 70 percent of specified lumen output.

b. Warranty period must begin in accordance with the manufacturer's standard warranty starting date.

c. Provide replacements that are promptly shipped, without charge, to the using Government facility point of contact and that are identical to or an improvement upon the original equipment. All replacements must include testing of new components and assembly.

1.6.2 Lighting Controls Warranty

Provide and transfer to the government the original lighting controls manufacturers standard commercial warranty for each different lighting controls manufacturer used in the project. Warranty coverage must begin from date of final system commissioning or three months from date of delivery, whichever is the earliest. Warranty service must be performed by a factory-trained engineer or technician.

a. Unless otherwise noted, provide a written five year minimum warranty on the complete system for all systems with factory commissioning. Provide warranty that covers 100 percent of the cost of any replacement parts and services required over the five years which are directly attributable to the product failure. Failures include, but are not limited to, the following:

(1) Software: Failure of input/output to execute switching or dimming commands.

(2) Damage of electronic components due to transient voltage surges.

(3) Failure of control devices, including but not limited to occupancy sensors, photosensors, and manual wall station control devices.

b. Provide a written five year minimum warranty on all input devices against defect in workmanship or materials provided by device manufacturer.

c. Provide a written five year minimum warranty on all control components attached to luminaires against defect in workmanship or materials.

1.7 OPERATION AND MAINTENANCE MANUALS

1.7.1 Lighting System

Provide operation and maintenance manuals for the lighting system in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA that

provide basic data relating to the design, operation, and maintenance of the lighting system for the building. Include the following:

a. Manufacturers' operating and maintenance manuals.

b. Luminaire shop drawings for modified and custom luminaires.

c. Luminaire Manufacturers' standard commercial warranty information as specified in paragraph LUMINAIRE WARRANTY.

1.7.2 Lighting Control System

Provide operation and maintenance manuals for the lighting control system in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA that provide basic data relating to the design, operation, and maintenance of the lighting control system for the building. Include the following:

a. Lighting control system layout and wiring plan.

b. Lighting control system one-line diagram.

c. Product data for all devices, including installation and programming instructions.

d. Occupancy/vacancy sensor coverage layout.

e. Training materials, such as videos or in-depth manuals, that cover basic operation of the lighting control system and instructions on modifying the lighting control system. Training materials must include calibration, adjustment, troubleshooting, maintenance, repair, and replacement.

f. Sequence of operation descriptions for each typical room type, including final programming, schedules, and calibration settings.

PART 2 PRODUCTS

2.1 PRODUCT COORDINATION

2.2 LUMINAIRES

UL 1598, NEMA C82.77-10. Provide luminaires as indicated in the luminaire schedule and NL plates or details on project plans, complete with light source, wattage, and lumen output indicated. All luminaires of the same type must be provided by the same manufacturer. Luminaires must be specifically designed for use with the driver and light source provided.

2.2.1 Luminaires

UL 8750, ANSI/IES LM-79, ANSI/IES LM-80. For all luminaires, provide:

a. Complete system with LED drivers and light sources.

b. Housings constructed of non-corrosive materials. All new aluminum housings must be anodized or powder-coated. All new steel housings must be treated to be corrosion resistant.

c. ANSI/IES TM-21, ANSI/IES LM-80. Minimum L70 lumen maintenance value of 50,000 hours unless otherwise indicated in the luminaire schedule.

Luminaire drive current value must be identical to that provided by test data for luminaire in question.

d. Minimum efficacy as specified in the luminaire schedule. Theoretical models of initial lamp lumens per watt are not acceptable. If efficacy values are not listed in the luminaire schedule, provide luminaires that meet the following minimum values:

| Luminaire Style | Minimum Luminaire Efficacy |
|---|---|
| Recessed 1 by 4, 2 by 4, and 2 by 2 | 100 LPW |
| Recessed Downlight (fixed, adjustable, wallwash) | 80 LPW |
| Linear, Accent (undercabinet, cove) | 45 LPW |
| Linear, Ambient (indirect wall mount, linear pendent) | 100 LPW |
| High Bay, Low Bay, and Industrial Locations | 100 LPW |
| Food Service and Hazardous Locations | 60 LPW |
| Other (track, residential diffusers) | 50 LPW |
| Exterior Wall Sconce | 50 LPW |
| Steplight | 30 LPW |
| Parking Garage Luminaire | 100 LPW |

e. UL listed for dry or damp location typical of interior installations. Any luminaire mounted on the exterior of the building must be UL listed for wet location typical of exterior installations.

f. LED driver and light source package, array, or module are accessible for service or replacement without removal or destruction of luminaire.

g. Lenses constructed of heat tempered borosilicate glass, UV-resistant acrylic, or silicone. Provide polycarbonate vandal-resistant lenses as indicated. Sandblasting, etching and polishing must be performed as indicated in the luminaire description.

## 2.3 LIGHT SOURCES

NEMA ANSLG C78.377, NEMA SSL 3. Provide type, delivered lumen output, and wattage as indicated in the luminaire schedule on project plans.

### 2.3.1 LED Light Sources

Provide LED light sources that meet the following requirements:

a. NEMA ANSLG C78.377. Emit white light and have a nominal CCT of 3000 Kelvin.

b. Minimum Color Rendering Index (CRI) of 80.

c. Directive 2011/65/EU. Restriction of Hazardous Substances (RoHS) compliant.

d. Light source color consistency by utilizing a binning tolerance within a 3-step McAdam ellipse.

2.3.1.1 Linear LED Lamps

Provide linear LED Lamps that are compatible with existing instant-start or programmed-start ballasts, and meet the following additional requirements:

a. UL 1993 UL Type A linear LED lamp.

b. Power Factor greater than or equal to 0.90 at full input power and across specified dimming range.

c. Maximum Total Harmonic Distortion (THD) less than or equal to 20 percent at full input power and across specified dimming range.

d. Lumen per watt efficacy no less than 120.

e. Minimum beam angle of 180 degrees.

f. Lamp datasheet complies with ANSI C78.54. Manufacturer must provide list of all ballasts that are compatible for use with lamp.

2.4 LED DRIVERS

NEMA SSL 1, UL 8750. Provide LED drivers that are electronic, UL Class 1 or Class 2, constant-current type and that comply with the following requirements:

a. The combined driver and LED light source system does not exceed the minimum luminaire efficacy values as listed in the luminaire schedule provided.

b. Operates at a voltage of 120-277 volts at 50/60 hertz, with input voltage fluctuations of plus/minus 10 percent.

c. Power Factor (PF) greater than or equal to 0.90 at full input power and across specified dimming range.

d. Maximum Total Harmonic Distortion (THD) less than 20 percent at full input power and across specified dimming range.

e. Operates for at least 50,000 hours at maximum case temperature and 90 percent non-condensing relative humidity.

f. Withstands Category A surges of 2 kV without impairment of performance. Provide surge protection that is integral to the driver.

g. Integral thermal protection that reduces the output power to protect the driver and light source from damage if the case temperature approaches or exceeds the driver's maximum operating temperature.

h. 47 CFR 15. Complies with the requirements of the Federal Communications Commission (FCC) rules and regulations, Non-Consumer (Class A) for EMI/RFI (conducted and radiated).

i. Class A sound rating.

j. Directive 2011/65/EU. Restriction of Hazardous Substances (RoHS) compliant.

k. Provide dimming capability as indicated in the luminaire schedule on project plans. Dimmable drivers must dim down to 10 percent. Dimmable drivers must be controlled by a Class 2 low voltage 0-10VDC controller.

## 2.5 LIGHTING CONTROLS

Provide network certification for all networked lighting control systems and devices. Provide lighting control systems that do not switch off battery-operated or emergency backup luminaires or exit signs in path of egress. Provide system with override of lighting control devices controlling luminaires in path of egress with activation of fire alarm system.

### 2.5.1 System

Provide lighting control system that operates the lighting system as described in the lighting control strategies in the project plans. Submit Sequence of Operation for Lighting Control System describing the operation of the proposed lighting control system and devices. Sequence of Operation must provide the strategies identified in the lighting control strategies.

#### 2.5.1.1 Localized Control Systems

Provide room or area-wide lighting control system capable of manual control, time-based control, and receiving input from photosensors and occupancy/vacancy sensors.

##### 2.5.1.1.1 Local Area Controller

Provide controller designed for single area or room with the following requirements:

a. Operates at a voltage of 120-277 volts at 50/60 hertz.

b. 2 zone, with 1 relay rated 20 amps with one manual dimmer per zone.

c. Provide inputs for occupancy/vacancy sensors, photosensors, and low-voltage wall switches.

d. Capable of 0-10V dimming.

e. Provide override 'ON' function with input from Fire Alarm Control Panel for all emergency lighting. Controller must not turn off power to emergency batteries or exit signs.

### 2.5.2 Devices

#### 2.5.2.1 Switches

Provide line-voltage toggle switches as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. When used for non-digital loads, devices

must be rated at 20 Amps inductive load, and be compatible with the lighting control systems.

2.5.2.2 Time Clocks

UL 917, NEMA ICS 6. House time clock in a surface-mounted, lockable, NEMA 3R enclosure constructed of painted steel or plastic polymer. Provide electronic type time clock that meets the following criteria:

a. Astronomic programming function, providing a total of 56 on/off set points.

b. 12 hour AM/PM type digital clock display format.

c. Power outage back-up for time clock utilizing a capacitor which provides coverage for a minimum of four days.

d. Capable of controlling a minimum of 1 channels or loads.

e. Contacts are rated for 30 amps at 120-277 VAC resistive load in a SPST normally open (NO) configuration.

f. Contains function that allows automatic control to be skipped on certain selected days of the week, manual bypass or remote override control, daylight savings time automatic adjustment.

2.6 LUMINAIRE MOUNTING ACCESSORIES

2.6.1 Recess and Surface Mounted Luminaires

Provide access to light source and LED driver from bottom of luminaire. Provide trim and lenses for the exposed surface of flush-mounted luminaires as indicated on project drawings and specifications. Luminaires recessed in ceilings which have a fire resistive rating of one hour or more must be enclosed in a box which has a fire resistive rating equal to that of the ceiling. For surface mounted luminaires with brackets, provide flanged metal stem attached to outlet box, with threaded end suitable for supporting the luminaire rigidly in design position. Flanged part of luminaire stud must be of broad base type, secured to outlet box at not fewer than three points.

2.6.2 Luminaire Support Hardware

2.6.2.1 Wire

ASTM A641/A641M. Galvanized, soft tempered steel, minimum 0.11 inches in diameter, or galvanized, braided steel, minimum 0.08 inches in diameter.

2.6.2.2 Wire for Humid Spaces

ASTM A580/A580M. Composition 302 or 304, annealed stainless steel, minimum 0.11 inches in diameter.

ASTM B164. UNS NO4400, annealed nickel-copper alloy, minimum 0.11 inches in diameter.

2.6.2.3 Threaded Rods

Threaded steel rods, 3/16 inch diameter, zinc or cadmium coated.

2.6.2.4 Straps

Galvanized steel, one by 3/16 inch, conforming to ASTM A653/A653M, with a light commercial zinc coating or ASTM A1008/A1008M with an electrodeposited zinc coating conforming to ASTM B633, Type RS.

2.6.3 Power Hook Luminaire Hangers

UL 1598. Provide an assembly consisting of through-wired power hook housing, interlocking plug and receptacle, power cord, and luminaire support loop. Power hook housing must be cast aluminum having two 3/4 inch threaded hubs. Support hook must have safety screw. Luminaire support loop must be cast aluminum with provisions for accepting 3/4 inch threaded stems. Power cord must include 16 inches of 3 conductor No. 16 Type SO cord. Assembly must be rated 120 volts or 277 volts, 15 amperes.

2.7 EQUIPMENT IDENTIFICATION

2.7.1 Manufacturer's Nameplate

Each item of equipment must have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.7.2 Labels

UL 1598. All luminaires must be clearly marked for operation of specific light sources and LED drivers. The labels must be easy to read when standing next to the equipment, and durable to match the life of the equipment to which they are attached. Note the following light source characteristics in the format "Use Only _____":

a. Correlated Color Temperature (CCT) and Color Rendering Index (CRI) for all luminaires.

b. Driver and dimming protocol.

All markings related to light source type must be clear and located to be readily visible to service personnel, but unseen from normal viewing angles when light sources are in place. LED drivers must have clear markings indicating dimming type and indicate proper terminals for the various outputs.

2.8 FACTORY APPLIED FINISH

NEMA 250. Provide all luminaires and lighting equipment with factory-applied painting system that as a minimum, meets requirements of corrosion-resistance testing.

PART 3 EXECUTION

3.1 INSTALLATION

IEEE C2, NFPA 70.

### 3.1.1 Light Sources

When light sources are not provided as an integral part of the luminaire, deliver light sources of the type, wattage, lumen output, color temperature (CCT), color rendering index (CRI), and voltage rating indicated to the project site and install just prior to project completion, if not already installed in the luminaires from the factory.

### 3.1.2 Luminaires

Set luminaires plumb, square, and level with ceiling and walls, in alignment with adjacent luminaires and secure in accordance with manufacturers' directions and approved drawings. Provide accessories as required for ceiling construction type indicated on Finish Schedule. Luminaire catalog numbers do not necessarily denote specific mounting accessories for type of ceiling in which a luminaire may be installed. Provide wires, straps, or rods for luminaire support in this section. Install luminaires with vent holes free of air blocking obstacles.

#### 3.1.2.1 Recessed and Semi-Recessed Luminaires

a. Support recessed and semi-recessed luminaires independently from the building structure by a minimum of two wires, straps or rods per luminaire and located near opposite corners of the luminaire. Secure horizontal movement with clips provided by manufacturer. Ceiling grid clips are not allowed as an alternative to independently supported luminaires.

b. Support round luminaires or luminaires smaller in size than the ceiling grid independently from the building structure by a minimum of four wires, straps or rods per luminaire, spaced approximately equidistant around.

c. Do not support luminaires by acoustical tile ceiling panels.

d. Where luminaires of sizes less than the ceiling grid are indicated to be centered in the acoustical panel, support each independently and provide at least two 3/4 inch metal channels spanning, and secured to, the ceiling tees for centering and aligning the luminaire.

e. Adjust aperture rings on all applicable ceiling recessed luminaires to accommodate various ceiling material thickness. Coordinate cut-out size in ceiling to ensure aperture covers cut-out entirely. Install aperture rings such that the bottom of the ring is flush with finished ceiling or not more than 1/16 inch above. Do not install luminaires such that the aperture ring extends below the finished ceiling surface.

#### 3.1.2.2 Field Applied Painting

Paint lighting equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Provide painting as specified in Section 09 90 00 PAINTS AND COATINGS.

### 3.1.3 LED Drivers

Provide LED drivers integral to luminaire as constructed by the manufacturer.

3.1.3.1 Remote LED Drivers

Locate Remote LED Drivers within the maximum distance allowed to minimize voltage drop. Do not locate remote LED drivers further from the light source than specified by the manufacturer. Locate remote LED drivers in dry, well-ventilated, and accessible location, above accessible ceilings or behind a removable wall or ceiling panel. Mount housing or junction box so that it is rigidly and securely fastened in place. Install LED drivers such that components are not in contact with combustible materials unless listed for such condition. Remote LED drivers must be grounded in accordance with NFPA 70.

Provide separate compartments for Class 2 wiring connections and for Class 1 wiring connections. Separation must be barrier-type within the same box or separate boxes with close connector conduit fittings. Field connections must be inside housing or junction box or secured by a quick disconnect wire connector.

3.1.4 Lighting Controls

3.2 FIELD QUALITY CONTROL

3.2.1 Tests

3.2.1.1 Lighting Control Verification Tests

Verify lighting control system and devices operate according to approved sequence of operations. Verification tests are to be completed after commissioning.

a. Verify occupancy/vacancy sensors operate as described in sequence of operations. Provide testing of sensor coverage, sensitivity, and time-out settings in all spaces where sensors are placed. This is to be completed only after all furnishings have been installed. Submit occupancy/vacancy sensor verification test.

b. Verify photosensors operate as described in sequence of operations. Provide testing of sensor coverage, aiming, and calibration in all spaces where sensors are placed. This is to be completed only after all furnishings have been installed. Submit photosensor verification test.

c. Verify wall box dimmers and scene wallstations operate as described in sequence of operations.

3.3 CLOSEOUT ACTIVITIES

3.3.1 Commissioning

NFPA 101. Factory Trained Field Service Technician is responsible for calibration and programming sequences for input devices and systems in accordance with the requirements described in the sequence of operation.

3.3.2 Training

3.3.2.1 Maintenance Staff Training

Submit a Maintenance Staff Training Plan at least 30 calendar days prior to training session that describes training procedures for Owner's

personnel in the operation and maintenance of lighting and lighting control system. Provide on-site training which demonstrates full system functionality, assigning schedules, calibration adjustments for light levels and sensor sensitivity, integration procedures for connecting to third-party devices, and manual override including information on appropriate use. Provide protocols for troubleshooting, maintenance, repair, and replacement, and literature on available system updates and process for implementing updates.

3.3.2.2 End-User Training

Submit an End-User Training Plan at least 30 calendar days prior to training session that describes training procedures for end-users on the lighting control system. Provide users with a list of control devices located within user-occupied spaces, such as photosensors and occupancy and vacancy sensors, including information on the proper operation and schedule for each device. Provide demonstration for each type of interface. Provide users with the building schedule as currently commissioned, including conditional programming based on astronomic time clock functionality. Provide users with the correct contact information for maintenance personnel who will be available to address any lighting control issues.

Provide laminated instructions to the user at each scene wallstation. Provide only instructions relevant to the functionality of the specific scene wallstation. Provide a description of each labeled scene control button. If the room utilizes occupancy/vacancy sensors or photosensors, include a description of this functionality on the instruction sheet.

-- End of Section --

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